TRAINER SLIDES · NON-WSQ COURSE

AI Powered Quantitative Trading Fundamentals

Course code C728 · 2 days · 15 instructional hours

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version 2.0 · 24 October 2026

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About the Trainer



Complete this panel with your own details before delivery.

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

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YOUR TRAINER

About the Trainer



Dr Alfred Ang

Course Developer and Trainer for this course.

ROLE
Course Developer and Trainer

CERTIFICATION
ACTA · DACE · PhD

FINANCIAL SERVICES
Programmes in AI, algorithmic trading and fintech

DELIVERS
AI, machine learning and data programmes for financial institutions

FOUNDER
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ICE-BREAKER

Let's Know Each Other

- Your name, your institution and your role.
- Do you trade, invest, analyse or build systems for markets today?
- Have you used TradingView, a broker API or Python before?
- One trading rule you believe works — we will test it this week.

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HOUSEKEEPING

Ground Rules

Be on time
Sessions restart promptly after each break.

Phones on silent
Take calls outside the room.

Ask as we go
Questions are welcome at any point.

Hands on the keyboard
Twenty labs — follow along on your own laptop.

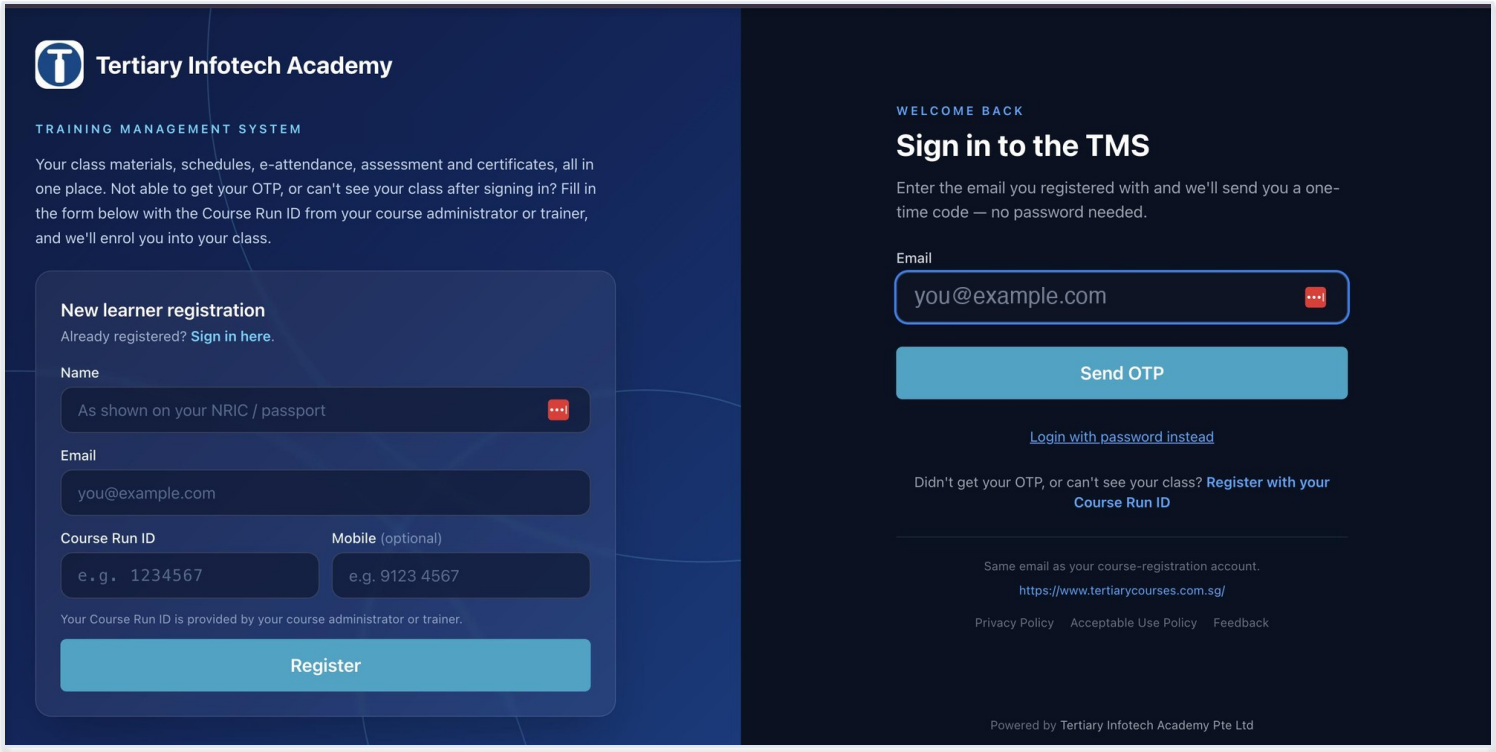
Paper trading only
No real money is used anywhere in this course.

Not investment advice
Lion City Capital and all scenarios are fictitious.

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COURSE PORTAL

Login to TMS and Download Course Material



Log in at lms-tms.tertiaryinfotech.com with your registered email + the OTP from your inbox, then download the slides, Learner Guide and labs.

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COURSE PORTAL · STEPS

Download Course Material

1. Go to lms-tms.tertiaryinfotech.com
2. Enter your registered email and click Send OTP
3. Key in the one-time password from your inbox
4. Open this course: download the Learner Slides + Learner Guide PDFs
5. Activities / Lab: download the labs folder (lab manuals)

Keep the Learner Guide open all course — every topic and lab step is in it.

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COURSE PORTAL · GITHUB

Download the Labs and Simulator (GitHub)

Course repository

github.com/tertiarycourses/C728-AI-Powered-Quantitative-Trading-Fundamentals

Trading simulator software

github.com/alfredang/algorithmic-trader-bot

The quant / algo trading simulator the labs run in: backtest, forward test, paper trading, analytics.

Install steps: next section and the Learner Guide.

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COURSE JOURNEY

The Course Journey

1

Topic 1 — Quant and Algo Trading Foundations

Day 1 · Labs 1–4 · Foundations of Quant and Algorithmic Trading: From Thorp to AI Agents

2

Topic 2 — Indicators, Strategies and Metrics

Day 1 · Labs 5–9 · Technical Indicators, Trading Strategies and Metrics with TradingView and Alpaca

3

Topic 3 — ML Models and A/B Testing

Day 2 · Labs 10–14 · Machine Learning Models, Statistical Modelling and A/B Testing for Trading Signals

4

Topic 4 — End-to-End Trading Pipeline

Day 2 · Labs 15–20 · Building an End-to-End Algorithmic Trading Pipeline: Backtest, Forward Test and Performance Analysis

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Low code. You own the algorithm.

This is a low-code course. Every lab runs in the Trading Simulator: you set the hypothesis, choose and configure the algorithm, run the tests and judge the results, while the software does the programming. What matters is the big concepts: the hypothesis, how each algorithm makes its decision, how it is tested without bias and with costs, how risk is controlled, and how the results are judged.

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THE PARADIGM SHIFT

Your Role versus the Software's Role

1

Your job: the hypothesis

What edge do you expect, why, and how will you know it is real?

2

Your job: the algorithm's logic

The rule or model, its inputs, and when it buys or sells.

3

Your job: testing and judgement

Out-of-sample, next-bar, net of costs; read the metrics.

4

Your job: risk and accountability

Limits, kill switch, approval - you answer for every trade.

⚙

The simulator's job: the code

It runs every stage - data, signals, backtests, paper orders and reports - from settings you choose.

⚙

The simulator's job: the drudgery

Data plumbing, charts, Pine Script export, reports - you review what it produces.

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COURSE SCHEDULE

Lesson Plan — Day 1 (1 of 2)

09:30 - 17:30 · 7.5 instructional hours · excluding a 30-minute lunch

09:30 - 10:00	Course Overview, Trainer Introduction and Ground Rules
10:00 - 11:10	T1.1-1.4 Quant Success Stories, the Evolution of Quant, the ML-to-LLM Shift, and Agentic AI versus AI Agents
11:10 - 11:40	T1.5-1.6 Forms of Financial Data, Big Data Applications and Trading Hypotheses
11:40 - 12:00	**Lab 1: Pipeline Skeleton, Configuration and the Alpaca Paper Account**
12:00 - 12:20	**Lab 2: Data-Validation Stage: Cross-Check Pipeline Bars against TradingView**
12:20 - 12:40	**Lab 3: Ingestion Stage: Multi-Timeframe Bars, Headlines and Data Quality**
12:40 - 13:10	**Lunch Break**
13:10 - 13:25	**Lab 4: Hypothesis Stage: from a Written Card to an Automated Statistical Test**
13:25 - 14:05	T2.1-2.2 Technical Indicators and Standard Algorithmic Trading Strategies

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COURSE SCHEDULE

Lesson Plan — Day 1 (2 of 2)

09:30 - 17:30 · 7.5 instructional hours · excluding a 30-minute lunch

14:05 - 14:30	**Lab 5: Indicator Stage: Pine Script v6 Indicators and their Python Twins**
14:30 - 14:55	**Lab 6: Strategy Prototype: TradingView Strategy Tester, Paper Trading from the Chart and Pipeline Reconciliation**
14:55 - 15:20	**Lab 7: Signal Stage: a Library of Trend, Mean-Reversion, Momentum and Breakout Signals**
15:20 - 16:00	T2.3-2.4 Analytics Platforms, Coding Languages and Order Placement
16:00 - 16:25	**Lab 8: Execution Stage: an Order Manager that Trades the Paper Account in Any Session**
16:25 - 16:50	T2.5 Performance and Risk Metric Definitions
16:50 - 17:15	**Lab 9: Evaluation Stage: Performance and Risk Metrics with a Benchmark Table**
17:15 - 17:30	Day 1 Recap and Q&A

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COURSE SCHEDULE

Lesson Plan — Day 2 (1 of 2)

09:30 - 17:30 · 7.5 instructional hours · excluding a 30-minute lunch

09:30 - 09:50	Day 1 Recap and Day 2 Overview
09:50 - 10:20	T3.1-3.2 Statistical Analyses, Algorithms and Supervised Learning Models
10:20 - 10:45	**Lab 10: Strategy-Search Stage: Grid Search, Out-of-Sample Ranking and Decay**
10:45 - 11:10	**Lab 11: ML-Signal Stage: a Walk-Forward Classifier from Probability to Position**
11:10 - 11:35	T3.3-3.4 Unsupervised Learning and Statistical Modelling Techniques
11:35 - 11:55	**Lab 12: Regime-Filter Stage: PCA and K-Means as a Gate on the Signals**
11:55 - 12:15	T3.5-3.6 A/B Testing, and Pairs Trading as a Hedging Algorithm
12:15 - 12:35	**Lab 13: A/B-Test Stage: a Statistical Decision before Capital Moves**
12:35 - 13:00	**Lab 14: Pairs Stage: a Hedged, Market-Neutral Statistical-Arbitrage Sleeve**
13:00 - 13:30	**Lunch Break**
13:30 - 14:10	T4.1-4.2 Pipeline Architecture and Backtesting with Trading Costs

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COURSE SCHEDULE

Lesson Plan — Day 2 (2 of 2)

09:30 - 17:30 · 7.5 instructional hours · excluding a 30-minute lunch

14:10 - 14:40	**Lab 15: Backtest-Engine Stage: Cost Models, Execution Timing and Break-Even**
14:40 - 14:55	T4.3 Forward Testing with Automated Paper Trading
14:55 - 15:25	**Lab 16: Automation Stage: a Scheduled, Risk-Checked Forward Test and Live Paper Trading**
15:25 - 15:50	**Lab 17: Event-Driven Automation: TradingView Webhook to Signal, Risk and Execution**
15:50 - 16:00	T4.4 Potential Reasons for Unintended Outcomes
16:00 - 16:25	**Lab 18: Diagnostics Stage: Automated Bias Checks before Go-Live**
16:25 - 16:35	T4.5 Performance Analysis, Insights and Reporting
16:35 - 17:05	**Lab 19: Monitoring and Reporting Stage: Automated Tearsheet and Backtest-vs-Forward Report**
17:05 - 17:30	**Lab 20: Capstone: One Command Runs the Pipeline End to End, with an Agentic Layer and Human Approval**

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GETTING STARTED

01

Set Up Alpaca and TradingView

LEARNING OUTCOME

Install the Trading Simulator and create the free accounts every lab uses. Everything runs on paper — no real money at any point.

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GETTING STARTED

Download, Install and Start the Trading Simulator Software

Full step-by-step guide: Learner Guide, 'Platform Guides'.

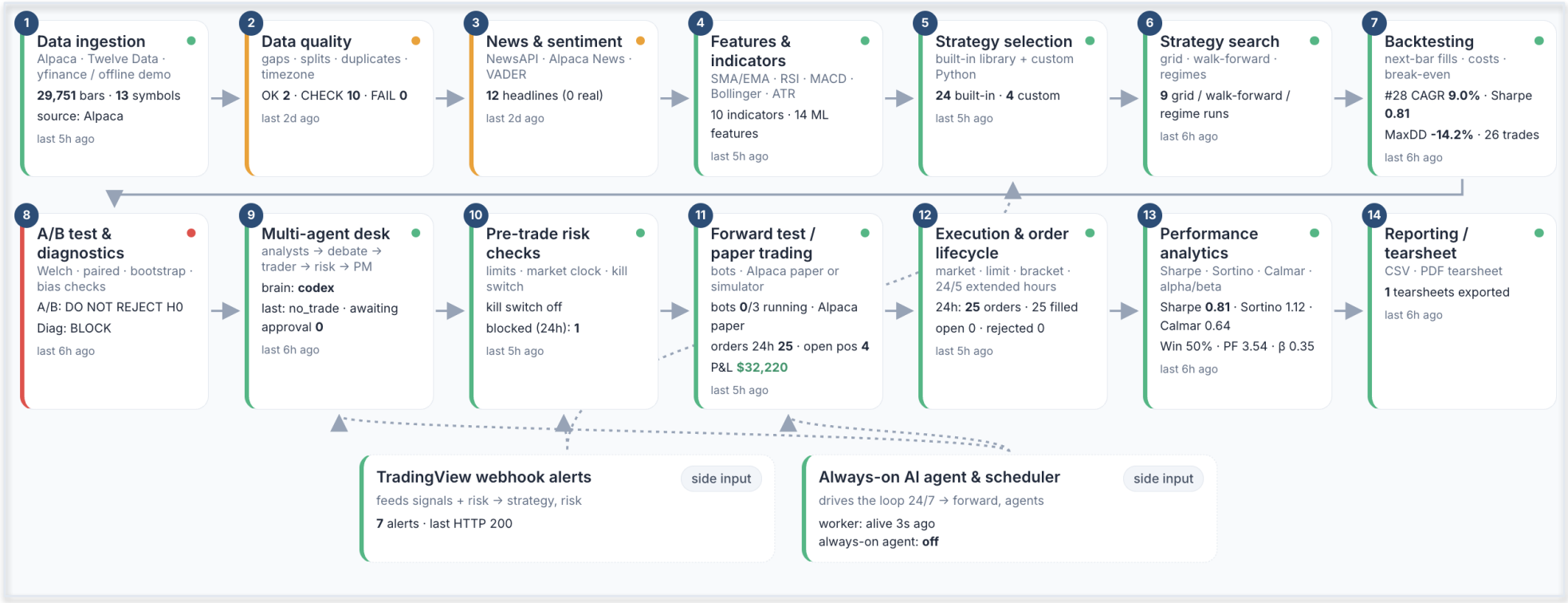
1. Install Docker Desktop - or Python 3.11+ and uv
2. Download the simulator: GitHub > Code > Download ZIP
3. Start it: docker compose up -d (or uv sync, then uv run trader serve --with-worker)
4. Load the demo data: Settings > Load demo data
5. Open http://localhost:8000 > Course Labs
6. Keys: Integrations page (no files to edit)

Simulator: github.com/alfredang/algorithmic-trader-bot - the README has the full start-up steps and troubleshooting.

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GETTING STARTED

The Trading Simulator — Pipeline Map



Pipeline Map shows the 14 stages of the automated pipeline, plus TradingView webhook alerts and the always-on AI agent as side inputs, each with its live state. Click a stage to see its activity.

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GETTING STARTED

Sign Up for Alpaca and Open the Paper-Trading Account

Full step-by-step guide: Learner Guide, 'Platform Guides'.

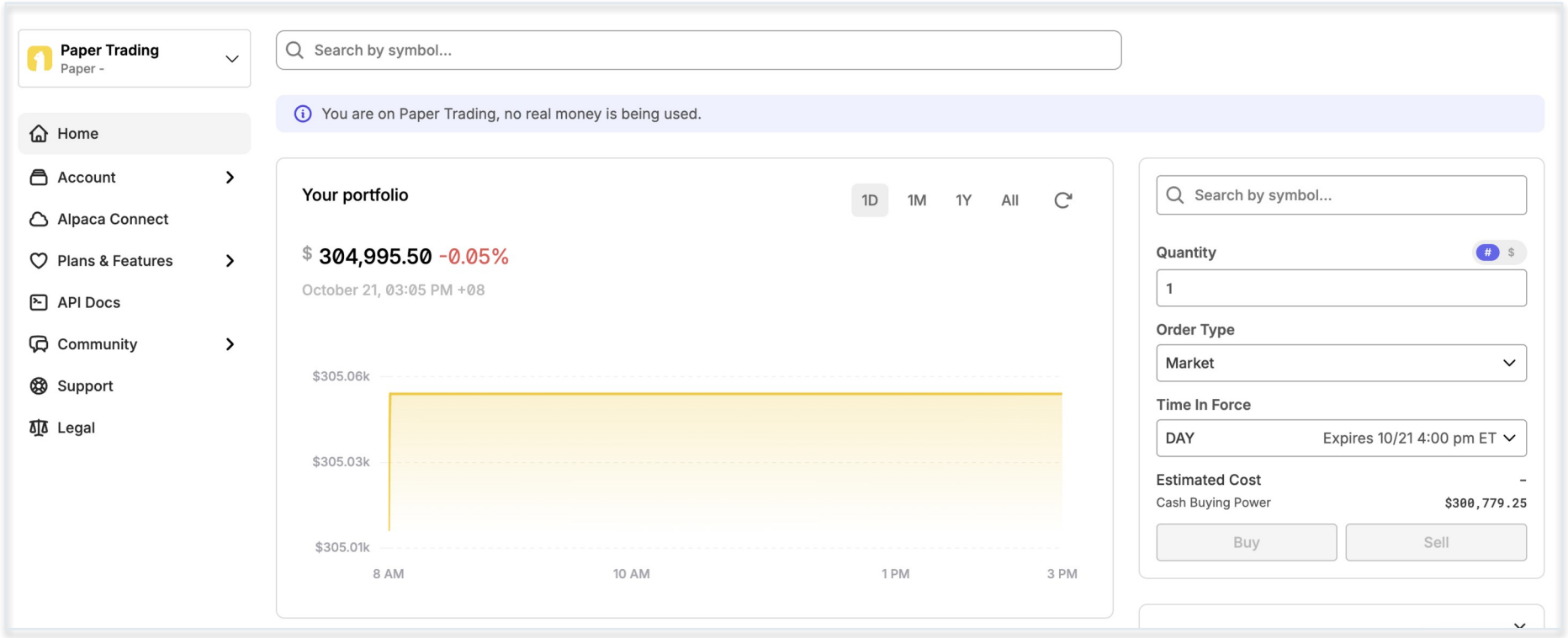
1. Sign up at alpaca.markets and verify your email
2. Log in; choose 'Paper Trading' (upper left)
3. API section: generate key - copy the secret once
4. Simulator: Integrations > Alpaca > paste > Save & test
5. Test connection: 'Paper account ACTIVE'

No deposit needed. Paper keys only work on paper-api.alpaca.markets. No account? Every lab also runs offline.

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GETTING STARTED

Your Alpaca Paper-Trading Dashboard

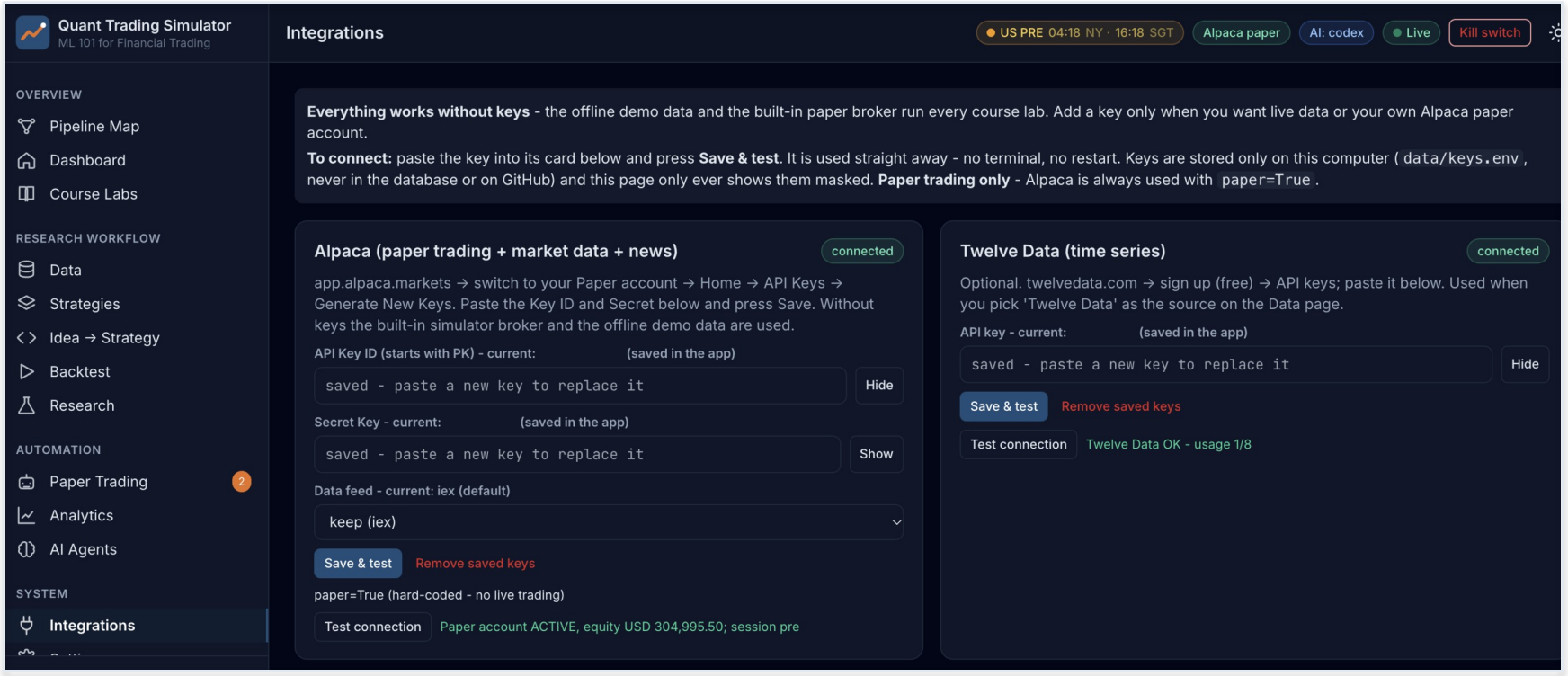


Check that the account selector shows 'Paper'. The banner confirms that no real money is being used. The API keys for the labs are generated from this account.

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GETTING STARTED

Connect Alpaca to the Simulator — Integrations



Paste the paper Key ID and Secret into the Alpaca card, then press Save & test. Keys stay on your computer and are only shown masked. Without keys, every lab still runs offline.

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GETTING STARTED

Sign Up for TradingView and Find the Tools You Will Use

Full step-by-step guide: Learner Guide, 'Platform Guides'.

1. Sign up at tradingview.com (email, Google or Apple)
2. Keep the free plan
3. Products > Supercharts; search NASDAQ:AAPL or SGX:D05
4. Set time zone to Singapore
5. Find Indicators, Pine Editor, Strategy Tester, Trade

The free plan covers Labs 2, 5 and 6. Webhook alerts in Lab 17 need a paid plan - the lab includes a local simulator.

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TradingView — Chart, Watchlist and Symbol Panel



Search for a symbol, pick the interval, and add Indicators. Pine Script and the Strategy Tester open from the panel below the chart, and Trade opens Paper Trading.

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GETTING STARTED · LOW CODE

No Coding Needed - and a Prompt Library if You Want the Code

Every lab, no code

Each lab runs in the Trading Simulator: choose settings, press Run, read the result. The lab manual gives every step and check.

Want to see the code?

labs/Prompt_Library.html has one prompt per lab. One click opens Claude or ChatGPT with the prompt filled in - the AI writes the code for you.

Run it without installing

Paste the code into Google Colab and press Run, or paste a strategy into the simulator's Strategy Builder. Never paste API keys into an AI chat.

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PLATFORM · THE IDEA THAT MATTERS MOST

Paper first. Always.

Every strategy in this course is tested on history, then forward-tested on a paper account with automatic orders, before anyone would consider real capital. That discipline is the course.

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TOPIC 1

01

Quant and Algo Trading Foundations

LEARNING OUTCOME

Develop hypotheses for implementing data analytics and computational modelling activities in algorithmic trading, and identify possible big data applications across the forms of data within financial services.

Day 1 · Labs 1–4

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Quant Success Stories

1

Ed Thorp - Princeton Newport · 1969

First market-neutral quant fund: about 20% a year after fees for two decades by hedging warrants and convertibles

2

Renaissance - Medallion · 1988

Purely model-driven; about 66% a year before fees, 39% after (1988-2018); closed to outsiders, employee-owned

3

D.E. Shaw · 1988

Founded by a computer-science professor from Morgan Stanley's quant group; models and software; about US$60-65bn

4

Man AHL · 1987

Systematic trend following from hand-typed price histories to momentum models across 800+ markets

5

Two Sigma · 2001

AI, machine learning and distributed computing applied to trading; about US$70bn under management

6

Jane Street · 2024

Quant market maker: about US$20.5bn net trading revenue; nearly a quarter of primary US ETF activity

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Spotlight: Renaissance Technologies and the Medallion Fund

1

Founded by a mathematician

Jim Simons founds Monemetrics in 1978; renamed Renaissance Technologies in 1982

2

Medallion fund, 1988

The flagship fund trades purely on mathematical models of market data

3

About 66% a year before fees

Average gross return 1988-2018, about 39% after its high fees; closed to outsiders, employee-owned

4

The lesson for this course

The edge came from data, rigorous testing, costs and risk control - not from one clever indicator

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What the Success Stories Have in Common

1. **Hypothesis first**
   Each started from a testable idea about prices, not from a tool

2. **Data and testing**
   Clean data and out-of-sample tests before capital

3. **Many small edges**
   Diversified, market-neutral bets rather than one big call

4. **Costs and risk**
   Execution cost, leverage and risk limits decide survival - see LTCM 1998

5. **Citadel Securities**
   Quant market maker estimated to handle about a quarter of all US equities trading

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TOPIC 1 · DEFINITION

Algorithmic trading

Using computer programs that follow predefined rules on price, timing, volume or models to generate and execute orders automatically, with limited or no human intervention.

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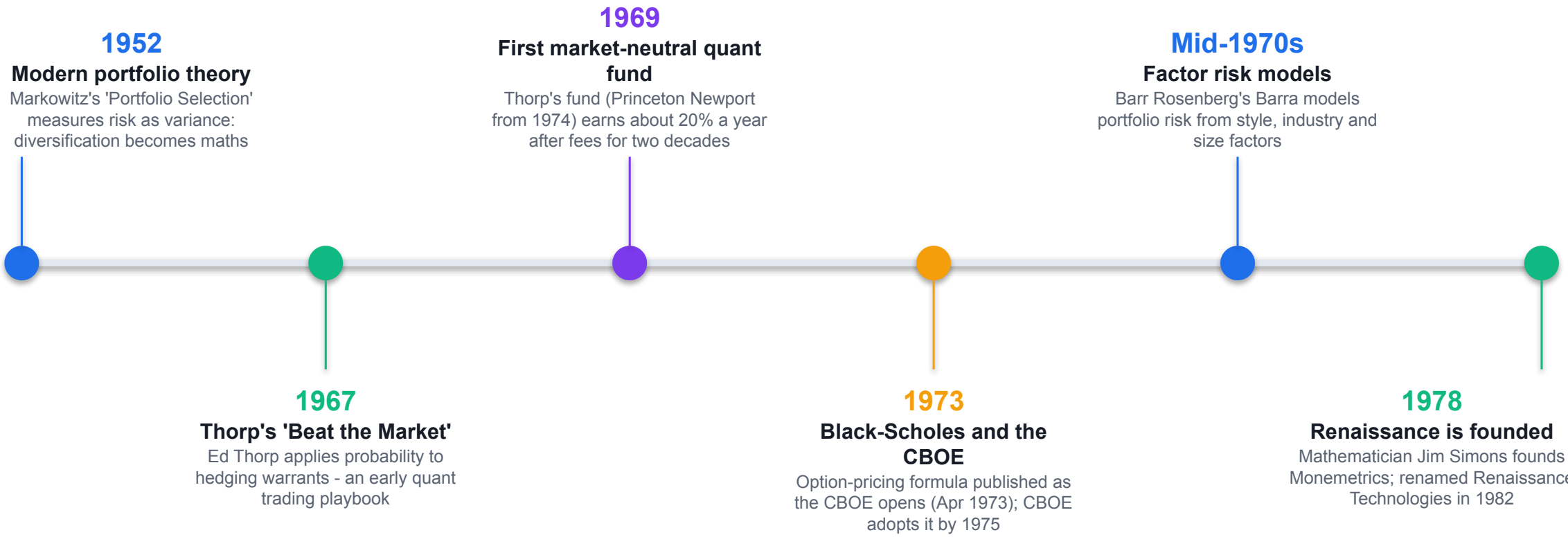
The Evolution of Quant: From Thorp to HFT to AI

1. 1950s-70s: portfolio theory, arbitrage and option pricing
2. 1970s-80s: factor risk models and program trading
3. 1980s-90s: statistical arbitrage and pairs trading
4. 1988 on: fully systematic signal mining (Renaissance, D.E. Shaw)
5. 2000s: electronic market making and HFT
6. 2010s: speed race - co-location, microwave, microseconds
7. 2020s: ML on alternative data, then LLM and agentic quant

Each era kept the last era's discipline - hypothesis, data, testing, costs, risk - and added speed or new data.

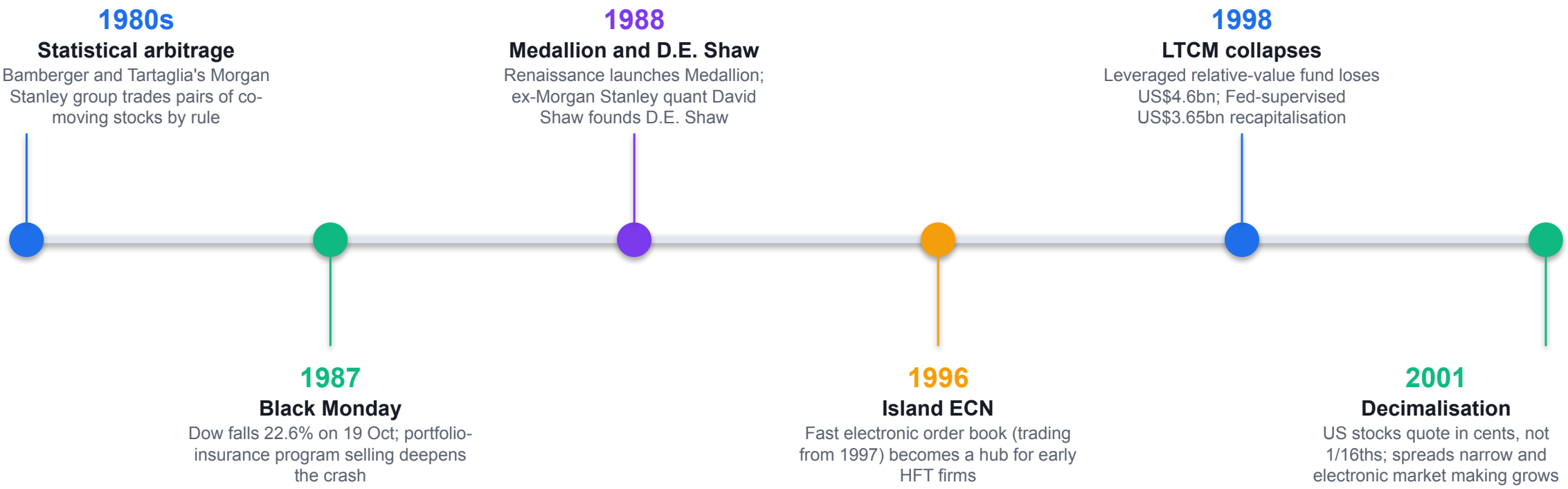
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The Evolution of Quant and Algorithmic Trading (1 of 4)



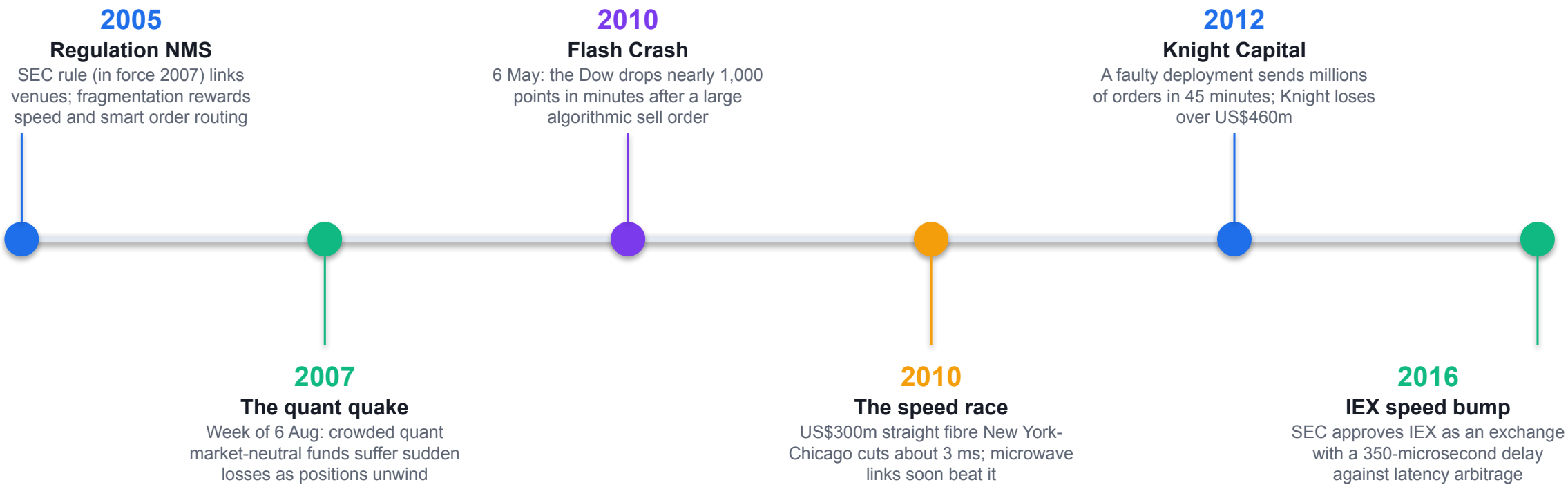
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The Evolution of Quant and Algorithmic Trading (2 of 4)



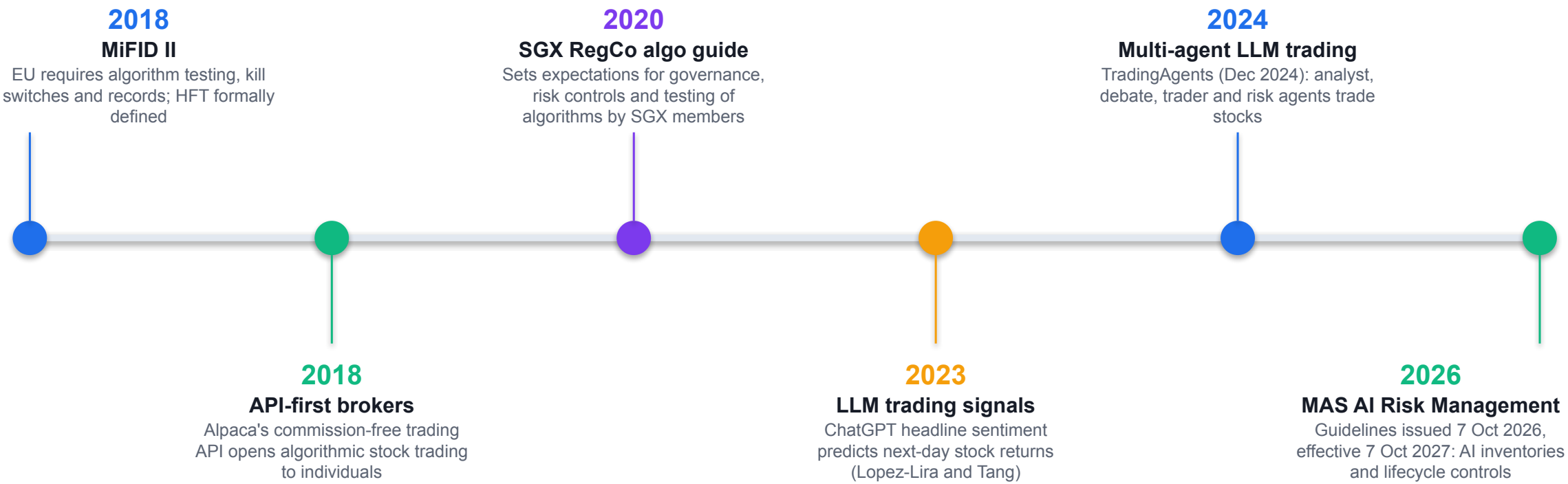
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The Evolution of Quant and Algorithmic Trading (3 of 4)



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The Evolution of Quant and Algorithmic Trading (4 of 4)



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Six Paradigm Shifts in Quant and Algorithmic Trading

1

From intuition to probability · 1950s-1970s

Markowitz, Thorp and Black-Scholes make risk, hedging and option value mathematical

2

From stocks to factors · 1970s-1980s

Barra factor models and program trading treat portfolios as baskets of risk exposures

3

From factors to statistical arbitrage · 1980s-2000s

Morgan Stanley's pairs group, D.E. Shaw and Renaissance harvest many small market-neutral edges

4

From floors to electronic speed · 1996-2010s

ECNs, decimalisation and Reg NMS enable HFT: co-location, microwave links, microseconds

5

From statistics to machine learning · 2000s-2010s

Models learn non-linear patterns from prices, order books and alternative data

6

From writing code to directing AI · 2023-present

AI writes and runs the code; quants own the big concepts - hypothesis, algorithm logic, testing, risk

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TOPIC 1

Why Algorithms - and What Can Go Wrong

Benefits

- Speed: reacts and executes in milliseconds
- Discipline: rules remove fear, greed and fatigue
- Backtesting: test ideas on historical data first
- Lower costs: split large orders to reduce market impact
- Scale: monitors many markets and assets around the clock

Risks

- Technical failure: bugs and outages (Knight Capital, 2012)
- Overfitting: great backtests that fail on live data
- Black swans and regime change the model has never seen
- Feedback loops between algorithms (2010 Flash Crash)
- Set-up, data and compliance costs; no guaranteed profit

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The Eras of AI in Trading

1. **Symbolic AI · 1956-1980s**
   Hand-written rules: expert systems and rule-based program trading; brittle when markets change

2. **Machine learning · 1990s-2000s**
   Models learn from data: regression, tree and boosting signals; statistical arbitrage at scale

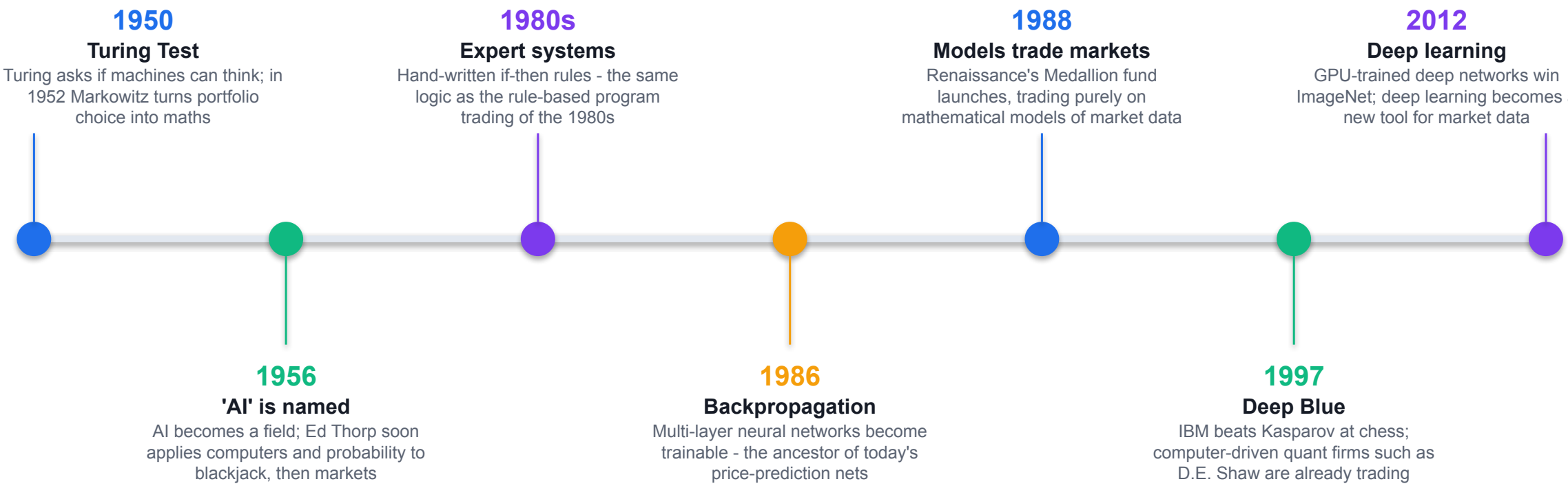
3. **Deep learning · 2012-2017**
   Many-layer networks on GPUs find non-linear patterns in prices, order books and alternative data

4. **Generative AI · 2017-2023**
   Transformers and LLMs turn news, filings and earnings calls into trading signals

5. **Agentic AI · 2024-present**
   LLM agents research, code, backtest and paper-trade strategies under hard risk limits

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AI Milestones and Their Trading Counterparts (1 of 2)



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AI Milestones and Their Trading Counterparts (2 of 2)



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Paradigm Shift: Machine-Learning Quant versus LLM-Era Quant

	Machine-learning quant	LLM-era quant
Main data	Structured: prices, volumes, fundamentals	Adds unstructured text: news, filings, earnings calls, research
Features	Hand-engineered by the quant (lags, RSI, factors)	Text read directly; LLM scores sentiment and events
Model building	Fit regressions, trees, boosting on labelled data	Prompt or fine-tune pretrained LLMs; agents propose alphas
Who writes the code	The quant, line by line	AI writes it; the quant directs and reviews
Research workflow	The quant writes every line of code	Coding agents draft code and backtests; the quant reviews
Alpha decay	Signals fade as more funds trade them	LLM signals fade too as LLM adoption rises
New risks	Overfitting; look-ahead in features	LLM training-data look-ahead and 'distraction'; homogeneous factors
Human role	Model builder	Supervisor: hypothesis, testing, risk sign-off

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Impact of LLMs on Quant

1

Text becomes a signal

ChatGPT headline scores predict next-day returns, strongest for small stocks and bad news (Lopez-Lira and Tang)

2

Finance-specific LLMs

BloombergGPT (2023): 50bn parameters trained with 363bn tokens of Bloomberg financial data

3

Automated alpha mining

Alpha-GPT turns ideas into formulaic alphas; AlphaAgent targets alpha decay and crowding

4

Backtests need new controls

LLMs may 'know' in-sample history: test after the knowledge cutoff and anonymise firms (Glasserman and Lin)

5

Faster crowding

Strategy returns decline as LLM adoption rises - prices become more efficient (Lopez-Lira and Tang)

6

Agents still need limits

Agents propose trades; hard-coded risk checks and humans approve them

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Agentic AI versus AI Agents in Trading

Agentic AI - the capability

The capability of AI to plan and act: it calls tools, uses skills and memory, and follows instructions to complete multi-step work when it is asked.

Trading example: On request, an LLM fetches a stock's bars and news through tools, checks its memory of past trades and proposes an order within its instructions

AI Agent - always on, 24/7

A special class of agentic AI that stands by 24/7 and works all the time: always on, woken by schedules, events and messages, with its own long-term memory.

Examples: OpenClaw, Hermes Agent, OpenAI Dots.

Trading example: A round-the-clock trading agent that watches prices and news overnight and proposes paper trades within risk limits

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AI Agents vs Agentic AI - Side by Side

	AI Agents	Agentic AI
What it is	A special class of agentic AI that is always on	Capability to plan and act with tools, skills, memory
Runtime	Persistent: stands by 24/7 on a server or PC	Runs when invoked: a chat, workflow or pipeline
Trigger	Schedules, events, messages - unprompted	A user prompt or a pipeline step
Memory	Long-term memory across days and tasks	Session or task memory; can use long-term memory
Building blocks	Agentic AI + always-on runtime and scheduler	LLM + tools + skills + memory + instructions
Examples	OpenClaw, Hermes Agent, OpenAI Dots	Claude Code, ChatGPT with tools, TradingAgents
Trading example	24/7 market and news monitor proposing trades	On-demand multi-agent analysis before a trade
Main risks	Acting unsupervised overnight; runaway loops	Hallucination; prompt injection; tool misuse

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From Chatbot to Always-On AI Agent

1

1 · Chatbot: Answers one prompt at a time; no tools, no actions

2

2 · Agentic AI: Plans and calls tools, skills, memory and instructions when asked

3

3 · Multi-agent agentic system: Specialist agents coordinate: analysts, debate, trader, risk manager

4

4 · AI agent (always on): Stands by 24/7, wakes on schedules and events, works continuously within limits

An AI agent is agentic AI that never switches off - so its limits, kill switch and monitoring matter most.

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Roles in a Multi-Agent Trading System

1. **Fundamental Analyst**
   Reads filings, earnings and balance sheets to judge intrinsic value

2. **Technical Analyst**
   Reads price, volume and indicators such as MACD and RSI

3. **Sentiment Analyst**
   Gauges mood from social media, analyst ratings and options flow

4. **News Analyst**
   Tracks news wires, press releases and macro events

5. **Bull / Bear Researchers**
   Argue for and against the trade so weak ideas are challenged

6. **Trader**
   Combines the reports into a buy, sell or hold call with timing and size

7. **Risk Manager**
   Checks volatility, liquidity, exposure and drawdown limits

8. **Portfolio Manager**
   Approves or rejects the proposal and fits it into the whole portfolio

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Stock-Trading Agent Frameworks (1 of 2)

1. **TradingAgents (Xiao et al., 2024)**
   Simulated stock-trading firm: fundamental, sentiment, news and technical analysts, bull/bear debate, trader, risk team

2. **FinMem (Yu et al., 2023)**
   LLM trading agent with profiling and layered memory making Buy / Sell / Hold decisions on individual stocks

3. **FinAgent (Zhang et al., 2024)**
   Multimodal agent reading prices, news and chart images, with dual-level reflection; tested on stocks and crypto

4. **StockAgent (2024)**
   LLM investor agents trade a simulated stock market under macro, policy and event shocks; avoids test-set leakage

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Stock-Trading Agent Frameworks (2 of 2)

1

QuantAgent (2025)

Price-driven multi-agent system for high-frequency trading: Indicator, Pattern, Trend and Risk agents

2

AI Hedge Fund (open source)

Investor-persona agents (e.g. Warren Buffett), a risk manager and a portfolio manager decide trades for chosen tickers

3

Agentic Trading Lab (Open-Finance-Lab)

Agent-run stock portfolios with backtests and paper trading through the Alpaca API

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Risks and Controls for AI Agents in Trading

	What goes wrong	Control
Hallucination	An LLM states a wrong earnings figure or invents a signal with confidence	Ground answers in tool or RAG data, log the reasoning, review unusual calls
Prompt injection	Malicious text in news, web pages or files hijacks the agent's instructions	Input and output guardrails, least-privilege API keys, no secrets in prompts
Overtrading and runaway loops	Agents trade too often or amplify each other, compounding losses	Order and position caps, daily-loss circuit breaker, kill switch
Model and backtest risk	Overfitting, look-ahead bias and regime change: strong backtests, weak live results	Walk-forward tests, paper trade first, monitor drift and costs
Accountability	Unclear who answers for an autonomous agent's trade or a vendor's AI	The firm stays accountable: human sign-off, audit trails, clear owners
Regulation (MAS)	FEAT principles (2018) and MAS AI Risk Management Guidelines (issued Oct 2026)	Board oversight, AI inventory, materiality assessment and lifecycle controls

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TOPIC 1

Forms of Data within Financial Services

1

Market data

OHLCV bars, quotes, trades, order-book depth — Alpaca serves bars, quotes, trades, news.

2

Reference & corporate actions

Symbols, calendars, splits, dividends — unadjusted splits fake a crash.

3

Fundamental & macro

Statements, ratios, earnings; rates, inflation, central-bank decisions.

4

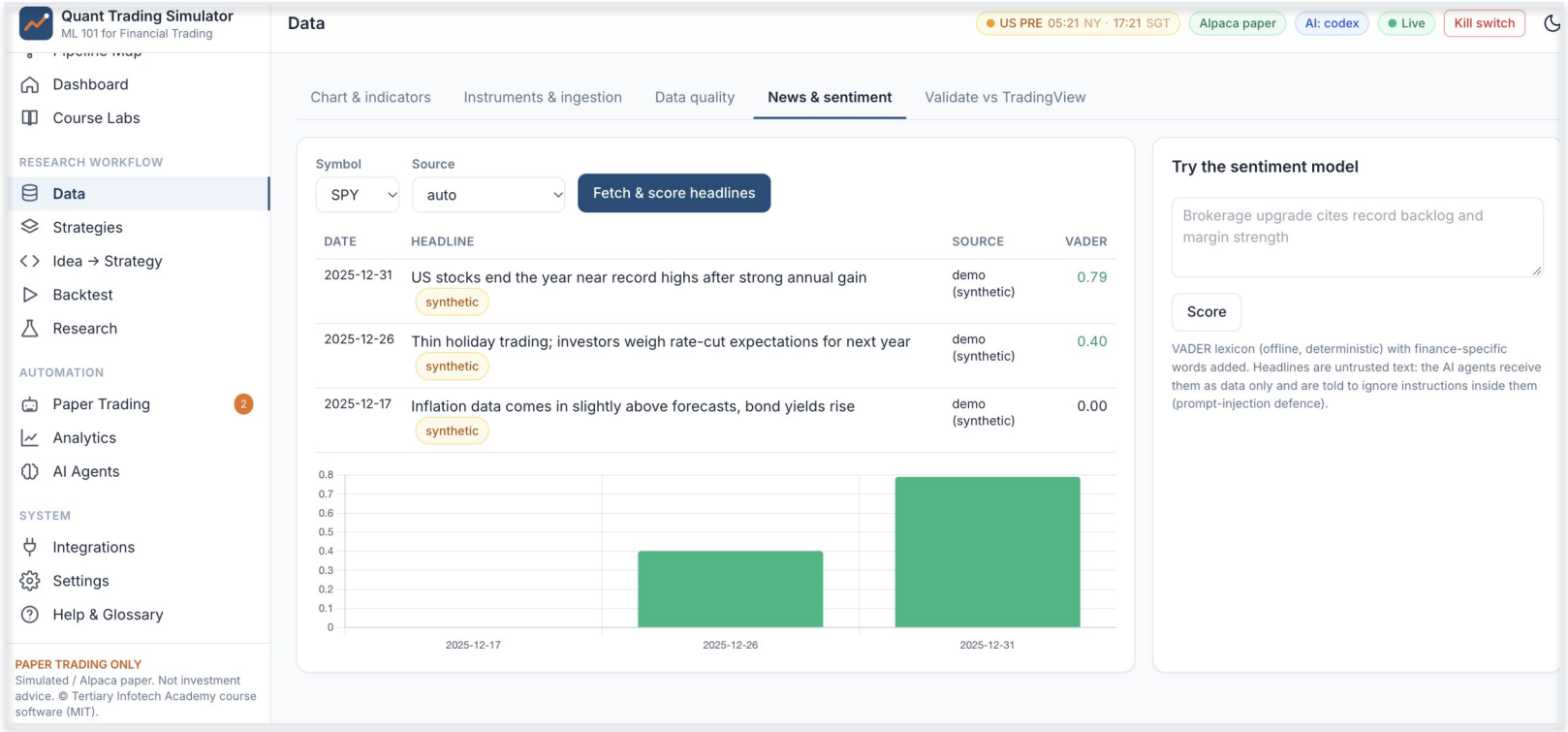
Alternative data

News and social sentiment, web traffic, satellite, card spend, ESG.

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TOPIC 1

Unstructured Data in Practice — Headlines and Sentiment



Data > News & sentiment: headlines (unstructured text) are scored by a lexicon model and stored as one sentiment number per symbol per day - structured data a strategy can use.

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TOPIC 1

Big Data in Trading — the Four Vs

1

Volume

Tick data runs to billions of rows per year for one exchange.

2

Velocity

Streaming quotes and news arrive in milliseconds.

3

Variety

Prices, text, images, transcripts — structured and unstructured.

4

Veracity

Bad ticks, survivorship, restatements — verify before you trust.

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TOPIC 1

Big Data Applications

1

Execution analytics

Tick and quote data to measure and cut slippage.

2

Sentiment and event trading

News and social text scored by NLP and LLMs.

3

Regime detection

Cross-asset returns clustered into market states.

4

Surveillance and risk

Order and trade data scanned for market abuse; intraday risk.

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TOPIC 1

Data Analytics and Modelling Use Cases

1

Signal generation

Find repeatable return — judged by out-of-sample, net Sharpe.

2

Execution

VWAP / TWAP / shortfall — judged by slippage in basis points.

3

Risk forecasting & sizing

Volatility and VaR models — judged by limit breaches.

4

Research at scale

LLMs read filings and news — judged by accuracy and hours saved.

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TOPIC 1

What Makes a Hypothesis Testable

1

Instrument and universe

e.g. SPY, daily bars, 2018 – 2025.

2

Signal and holding period

e.g. three down closes in a row; hold 5 days.

3

Expected effect and benchmark

Forward return exceeds the unconditional 5-day return.

4

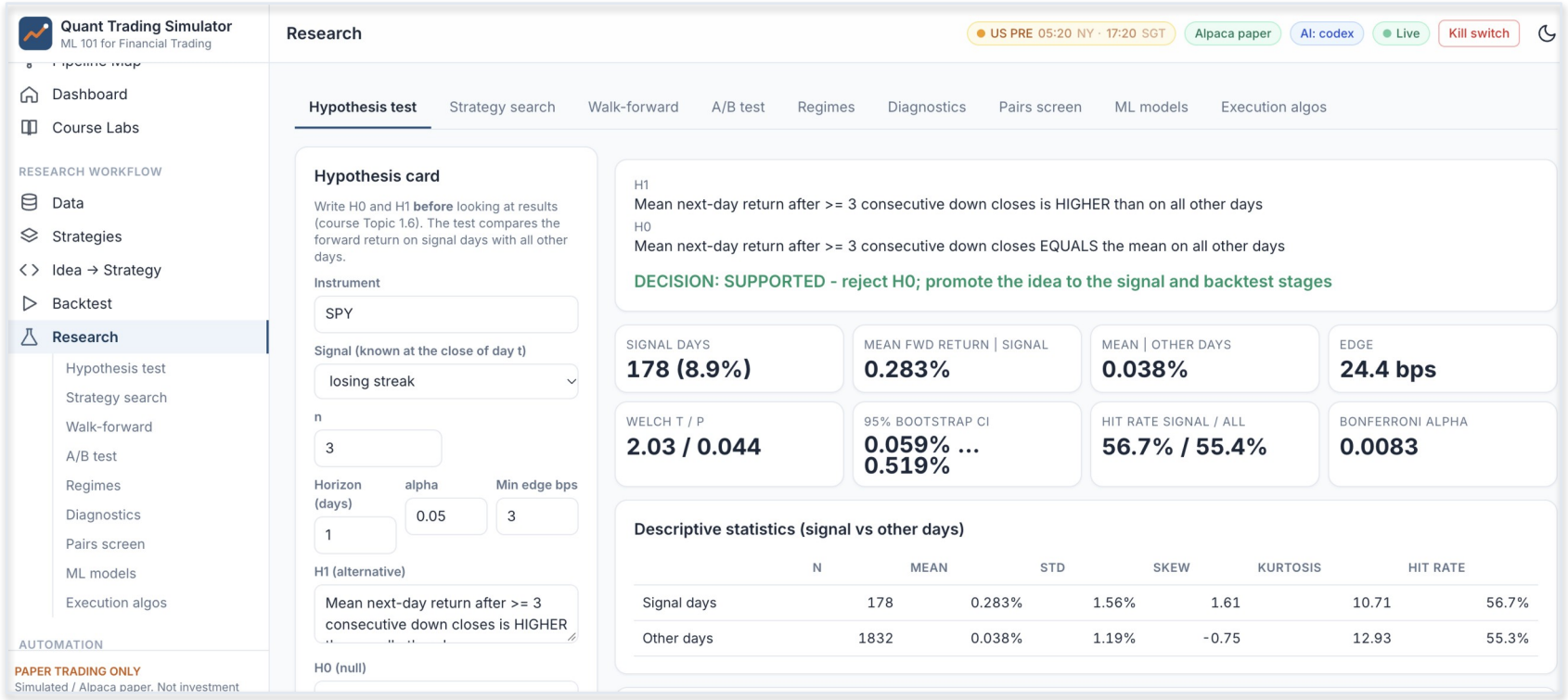
Written BEFORE the test

H0, H1 and the significance level chosen first.

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TOPIC 1

A Testable Hypothesis in Practice

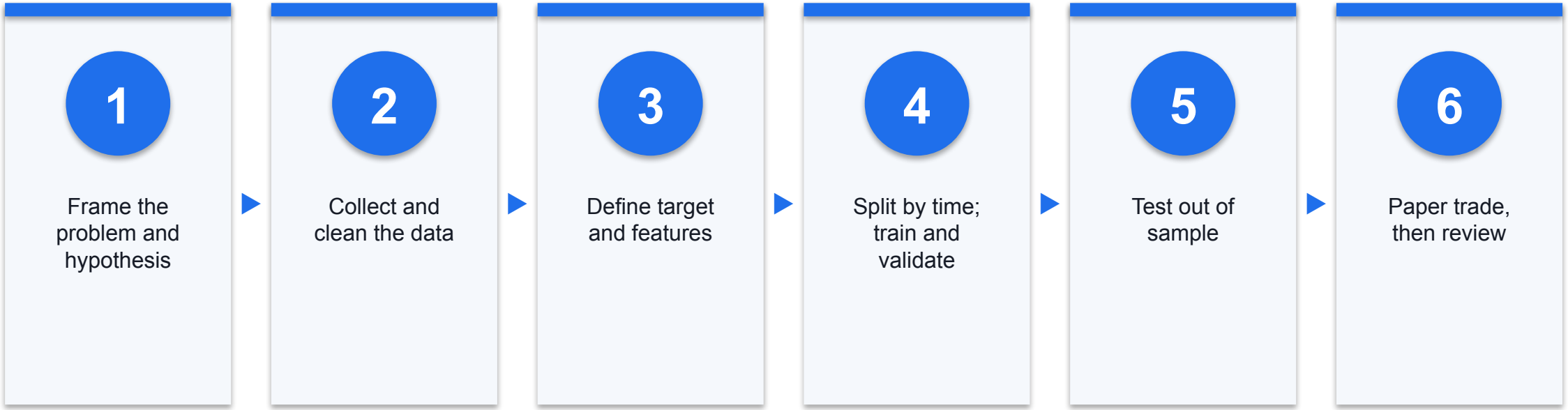


Research > Hypothesis test: H1 and H0 are written before the result, then tested with a Welch t-test, a bootstrap confidence interval and a minimum edge (Lab 4).

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TOPIC 1

Machine Learning Implementation Steps



A model that skips the out-of-sample test and the paper trade has not been tested.

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TOPIC 1 · HANDS-ON LAB

Pipeline Skeleton, Configuration and the Alpaca Paper Account

LAB 01

Get the simulator running, decide between the offline demo and an Alpaca paper account, and check every integration's status.

WORKFLOW

1. Simulator: Load the demo data
2. Simulator: Check the integrations
3. Simulator: Check the AI agent login
4. Simulator: Read the market clock
5. Simulator: Tour the Pipeline Map

YOU'LL PRODUCE

A working app with demo data loaded, a verified Alpaca paper account or the simulator, and a known AI-agent brain.

Duration
20 min

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LAB 01 · FOLLOW ALONG

Lab 01 — Step by Step

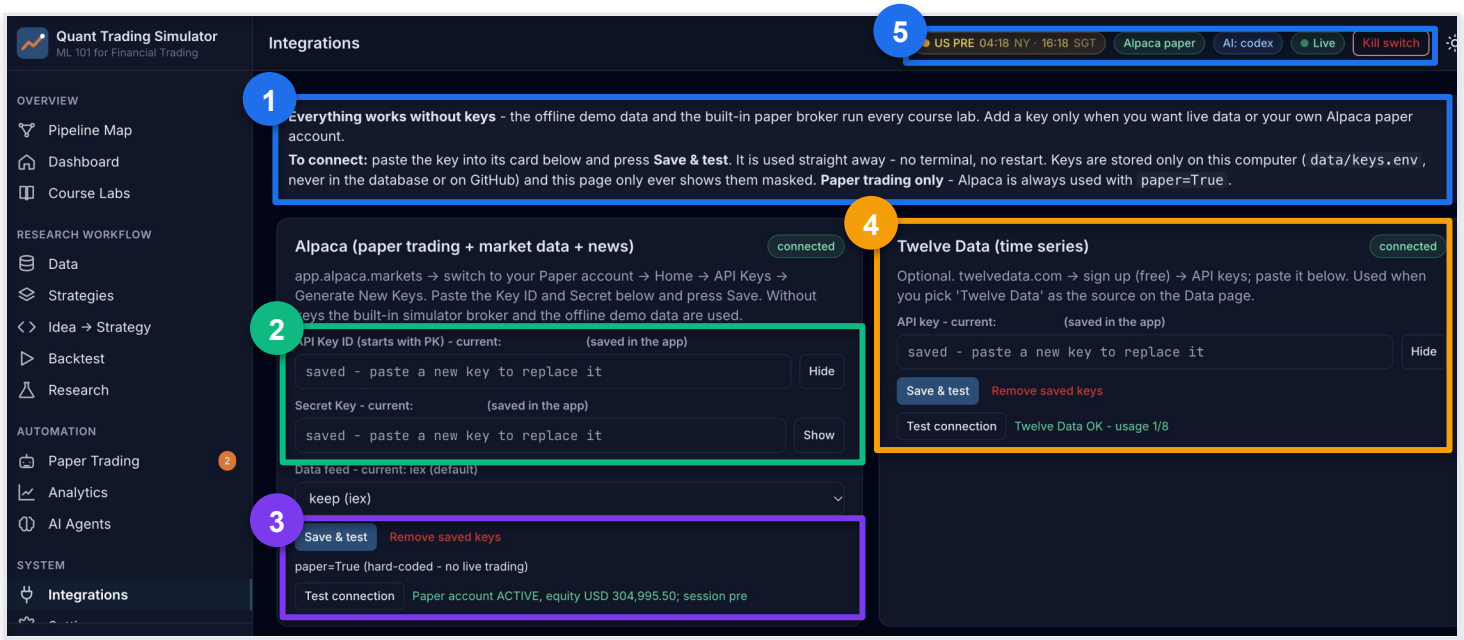
Course Labs > Lab 01 at http://localhost:8000 - each step's Go button opens its page pre-filled. Tick each 'You should see' before moving on.

	STEP · WHERE	DO THIS	YOU SHOULD SEE
1	**Load the demo data** ▸ Settings > Demo data	Open Settings and press 'Load / refresh demo data'.	✓ Settings shows about 29,000 bars and at least 8 backtests.
2	**Check the integrations** ▸ Integrations	Open Integrations. Without keys, Alpaca shows 'not configured' and the app uses the simulator broker and the offline data.	✓ Alpaca test says 'Paper account ACTIVE' (online) - or you understand the app runs fully offline.
3	**Check the AI agent login** ▸ Integrations > AI providers	On the Integrations page read the OpenAI Codex and Claude cards.	✓ You can say which brain the desk will use: codex, claude, openai or the rule-based fallback.
4	**Read the market clock** ▸ Paper Trading > clock	Look at the US session badge in the top bar.	✓ You can explain why orders placed during class become extended-hours LIMIT orders.
5	**Tour the Pipeline Map** ▸ Pipeline Map	Open the Pipeline Map and hover each of the 14 stages. Click 'Data ingestion' to see its recent activity.	✓ Data ingestion and Backtesting are green; you can name the stage that enforces the kill switch.

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LAB 01 · SIMULATOR

Lab 01 — How to Read the Screen



1 — No keys needed
Every lab runs on the offline demo data and the built-in paper broker; keys are optional.

2 — Paste your Alpaca keys
Key ID (starts with PK) and Secret from Alpaca > Paper account > API Keys.

3 — Save & test
Green 'Paper account ACTIVE' = connected. paper=True is hard-coded: no live trading.

4 — Optional data sources
Twelve Data and the other cards are extras; 'connected' shows the key works.

5 — Status bar
US session and clock, broker (Alpaca paper or Offline demo), AI brain and the kill switch.

Ready when the status bar shows your broker and every lab page opens - online with Alpaca paper or fully offline.

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TOPIC 1 · HANDS-ON LAB

Data-Validation Stage: Cross-Check Pipeline Bars against TradingView

LAB 02

Prove the app's prices are right: compare its bars with TradingView's Data Window and explain adjusted vs unadjusted prices.

WORKFLOW

1. Simulator: Read TradingView's Data Window
2. Simulator: Compare with RAW and ADJUSTED bars
3. Simulator: Repeat for SGX:D05 (DBS)
4. Simulator: Inspect the chart

YOU'LL PRODUCE

A PASS / FAIL validation for SGX:D05 and NASDAQ:AAPL on 2025-12-31, with the dividend-adjustment factor explained.

Duration
20 min

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LAB 02 · FOLLOW ALONG

Lab 02 — Step by Step

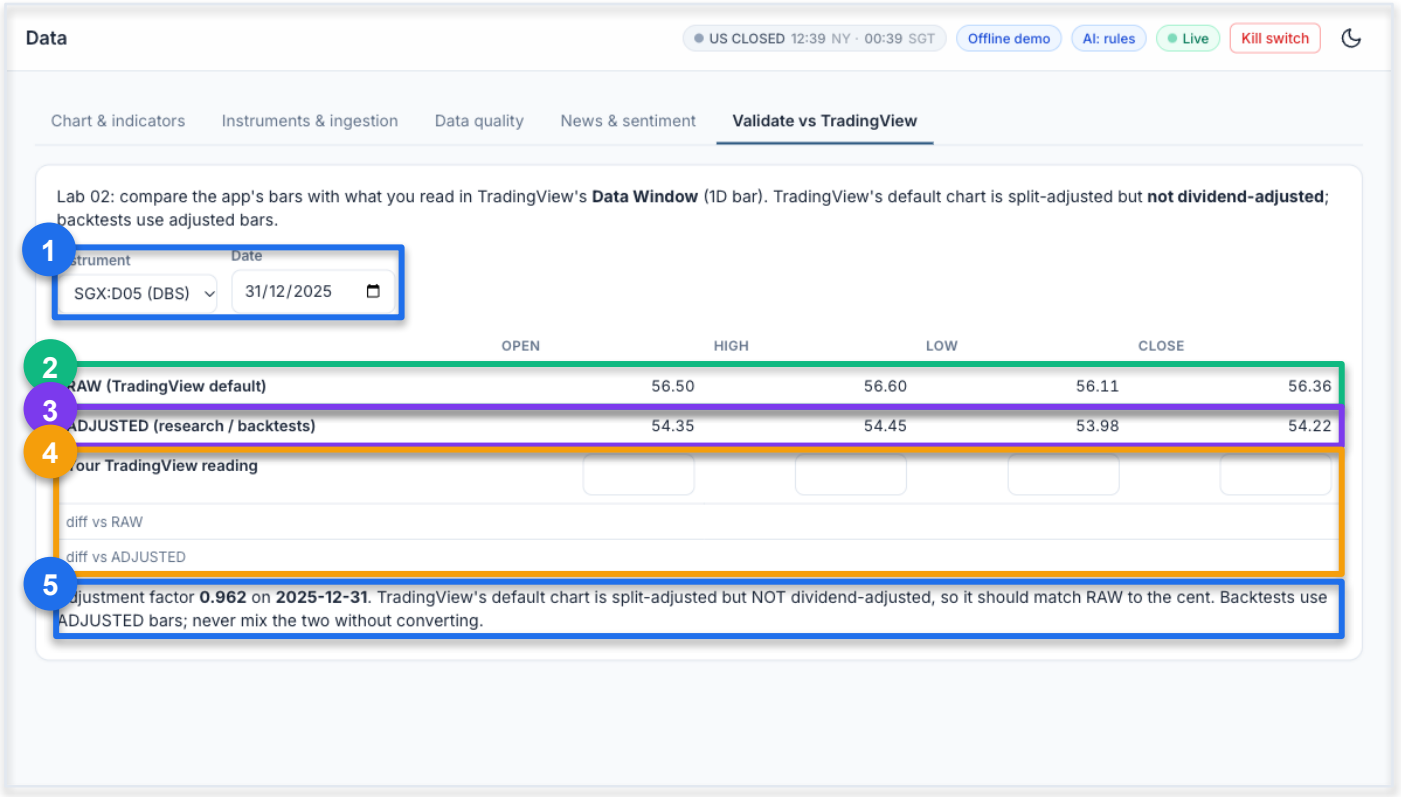
Course Labs > Lab 02 at http://localhost:8000 - each step's Go button opens its page pre-filled. Tick each 'You should see' before moving on.

	STEP · WHERE	DO THIS	YOU SHOULD SEE
1	**Read TradingView's Data Window** ▸ Data > Validate vs TradingView (AAPL)	In TradingView open NASDAQ:AAPL, daily timeframe, hover the bar of 2025-12-31 and read open/high/low/close in the Data Window (default settings: dividends NOT adjusted).	✓ You have four numbers from TradingView.
2	**Compare with RAW and ADJUSTED bars** ▸ Data > Validate vs TradingView	Type your TradingView values into the 'Your TradingView reading' row. The app shows the difference against RAW (split-adjusted only) and ADJUSTED (split + dividend) bars.	✓ Verdict 'PASS - TradingView matches RAW' (to within 0.1%).
3	**Repeat for SGX:D05 (DBS)** ▸ Data > Validate vs TradingView (D05)	Select SGX:D05. DBS pays large dividends, so the adjustment factor is far from 1.	✓ You can explain why the ADJUSTED close is lower than the RAW close and why a model trained on adjusted prices must never be fed raw live prices.
4	**Inspect the chart** ▸ Data > Chart AAPL	Open the AAPL chart and compare its shape with TradingView.	✓ The candles match TradingView's daily chart (allowing for the adjustment).

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LAB 02 · SIMULATOR

Lab 02 — How to Read the Screen



1 Instrument and date
SGX:D05 (DBS) or NASDAQ:AAPL on 31/12/2025 - the bar you read in TradingView.

2 RAW bars
Split-adjusted only, the TradingView default. Your reading should match this row.

3 ADJUSTED bars
Split + dividend adjusted, used by research and backtests - lower than RAW for DBS.

4 Your TradingView reading
Type the four prices; the diff rows show the gap. Within 0.1% of RAW = PASS.

5 Adjustment factor
0.962 for DBS: dividends shrink its history by about 4%.

PASS when TradingView matches RAW. Never feed raw live prices to a model trained on adjusted prices.

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TOPIC 1 · HANDS-ON LAB

Ingestion Stage: Multi-Timeframe Bars, Headlines and Data Quality

LAB 03

Ingest bars from Alpaca / Twelve Data / yfinance (or the demo files), run the data-quality checks and score news sentiment.

WORKFLOW

1. Simulator: Fetch data
2. Simulator: Read the data-quality report
3. Simulator: Switch timeframes
4. Simulator: Headlines and sentiment

YOU'LL PRODUCE

A cached universe, a data-quality report and scored headlines with daily sentiment.

Duration
20 min

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LAB 03 · FOLLOW ALONG

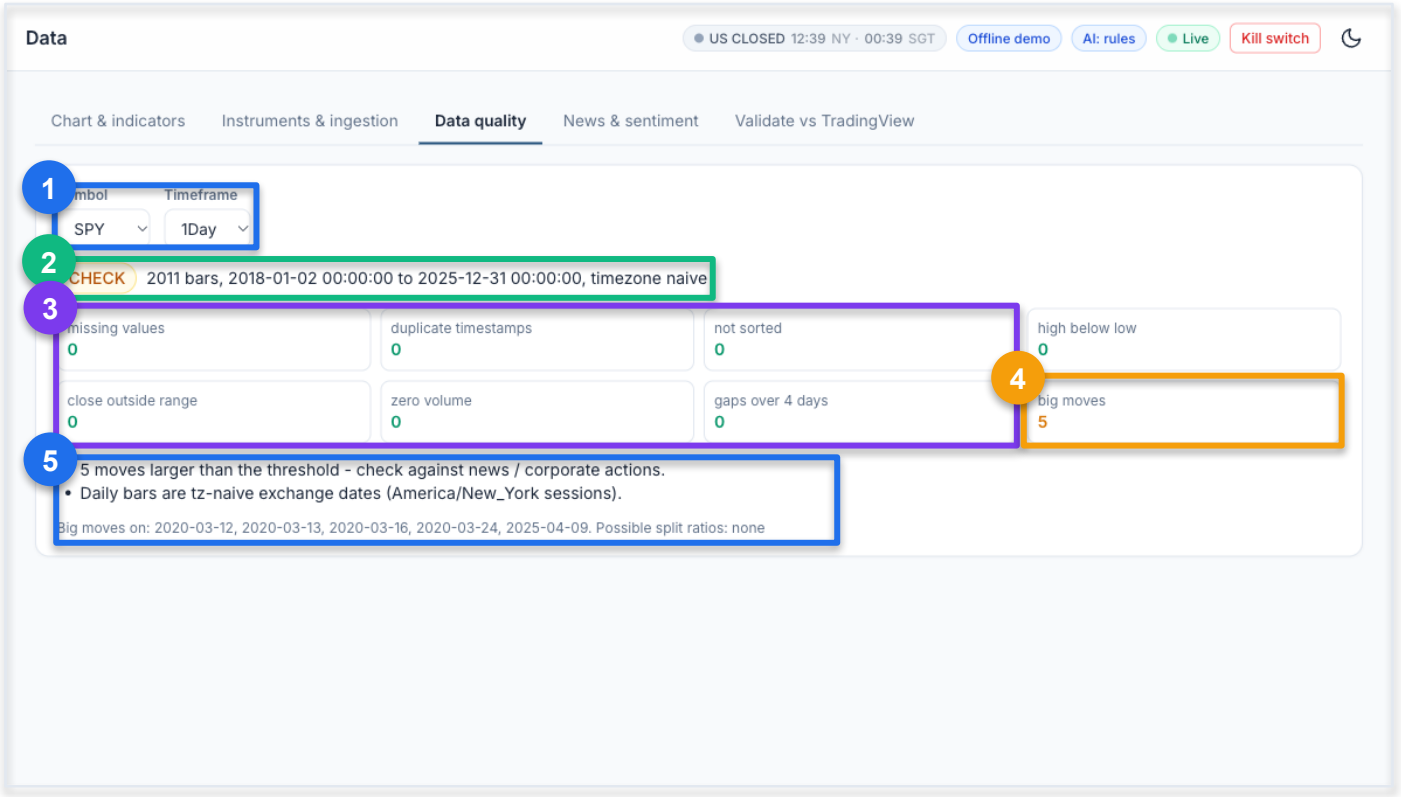
Lab 03 — Step by Step

Course Labs > Lab 03 at http://localhost:8000 - each step's Go button opens its page pre-filled. Tick each 'You should see' before moving on.

	STEP · WHERE	DO THIS	YOU SHOULD SEE
1	**Fetch data** ▸ Data > Fetch AAPL	Data > Instruments & ingestion: fetch AAPL with source 'auto' (Alpaca if keys, else the demo file). Online, try 'yfinance' or 'Twelve Data' too (Twelve Data needs TWELVEDATA_API_KEY).	✓ A toast reports the number of bars, the source and the quality status.
2	**Read the data-quality report** ▸ Data > Quality SPY	Open Data quality for SPY. Look at missing values, duplicate timestamps, gaps over 4 days and big moves (possible unadjusted splits).	✓ Status OK or CHECK; the big-move dates include March 2020.
3	**Switch timeframes** ▸ Data > Chart SPY weekly	On the chart, switch SPY between 1Day, 1Week, 1Month and 1Hour.	✓ Weekly and monthly bars are resampled from daily bars; hourly bars are US/Eastern.
4	**Headlines and sentiment** ▸ Data > News NVDA	Data > News & sentiment for NVDA: press 'Fetch & score headlines'. Online with NEWSAPI_KEY it calls NewsAPI.org. otherwise Alpaca News or the synthetic sample headlines.	✓ Each headline has a VADER score between -1 and +1; the daily sentiment chart appears.

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Lab 03 — How to Read the Screen



1 — Symbol and timeframe
SPY, 1Day. Weekly and monthly bars are resampled from the daily bars.

2 — Overall status
CHECK: 2,011 bars, 2018-01-02 to 2025-12-31 - usable, but something needs a look.

3 — Seven integrity checks
Missing, duplicate, unsorted, high < low, close outside range, zero volume, gaps: all 0.

4 — Big moves: 5
Days beyond the threshold - real news, or an unadjusted split?

5 — Explain each big move
March 2020 (COVID crash) and 2025-04-09 (tariff-pause rally); no split ratios found.

CHECK is not FAIL: every flagged date has a real-world cause, so the data are fit for research.

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Lunch Break

1 hour · see you at 13:30

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TOPIC 1 · HANDS-ON LAB

Hypothesis Stage: from a Written Card to an Automated Statistical Test

LAB 04

Write H0 and H1 before looking at results and test them with a Welch t-test and a bootstrap CI.

WORKFLOW

1. Simulator: Write the hypothesis card
2. Simulator: Run the test
3. Simulator: Robustness and multiple testing

YOU'LL PRODUCE

A decision (SUPPORTED / NOT SUPPORTED) with p-value, edge in bps and a robustness grid.

Duration
15 min

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LAB 04 · FOLLOW ALONG

Lab 04 — Step by Step

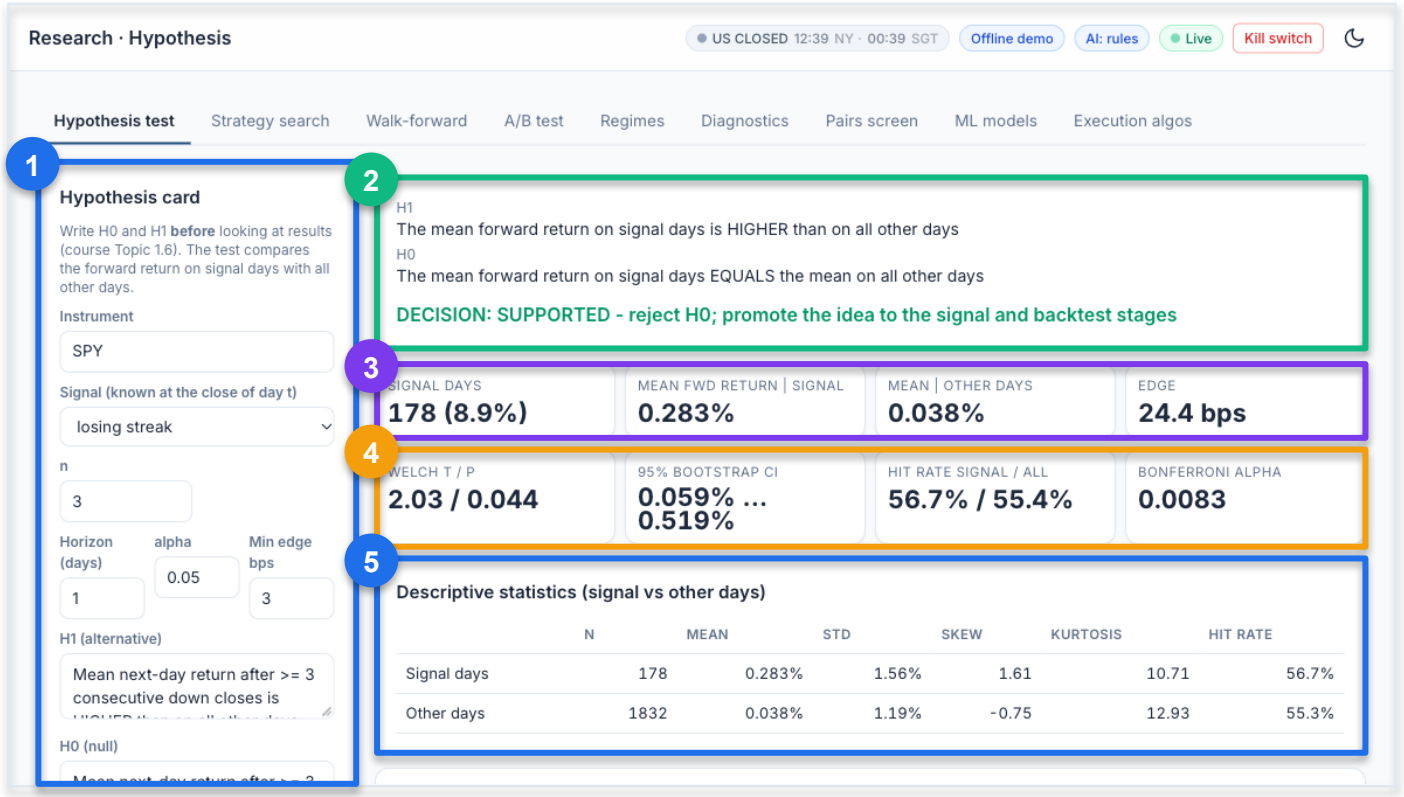
Course Labs > Lab 04 at http://localhost:8000 - each step's Go button opens its page pre-filled. Tick each 'You should see' before moving on.

	STEP · WHERE	DO THIS	YOU SHOULD SEE
1	**Write the hypothesis card** › Research > Hypothesis (SPY, losing streak 3)	Research > Hypothesis test. Keep signal 'losing streak' n = 3, horizon 1 day, alpha 0.05, min edge 3 bps. Read the H1 and H0 text.	✓ H0 and H1 are written before you press Run.
2	**Run the test** › Research > Hypothesis test	Press 'Run the test'.	✓ You can read the signal-day mean, the other-day mean, Welch t and p, and the 95% bootstrap CI.
3	**Robustness and multiple testing** › same page	Read the robustness grid (other streak lengths and QQQ) and the Bonferroni alpha.	✓ You can explain why trying many variants inflates false positives.

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LAB 04 · SIMULATOR

Lab 04 — How to Read the Screen



1 — Hypothesis card
SPY, losing streak n = 3, 1-day horizon, alpha 0.05, min edge 3 bps. Write H0 and H1 first.

2 — H1, H0 and the decision
SUPPORTED - reject H0: signal days beat the other days.

3 — The edge
178 signal days return 0.283% vs 0.038%: a 24.4 bps edge, above the 3 bps minimum.

4 — Is it real?
Welch p 0.044 < 0.05 and the bootstrap CI excludes 0. Stricter bar: Bonferroni alpha 0.0083.

5 — Descriptive statistics
Signal days are more volatile (1.56% std) and skewed - one reason to bootstrap.

Passes at alpha 0.05 but not at the Bonferroni 0.0083: promote it as a candidate and keep testing (Labs 10, 13, 18).

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Topic 1 Recap

1. Develop hypotheses for implementing data analytics and computational modelling activities
2. Identify possible big data applications
3. Data analytics and modelling business use cases
4. Forms of data within financial services

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TOPIC 2

02

Indicators, Strategies and Metrics

LEARNING OUTCOME

Identify appropriate statistical algorithms and data models to test trading hypotheses, and use appropriate analytics platforms, analytical tools and statistical methods given specific analytics and reporting requirements.

Day 1 · Labs 5–9

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TOPIC 2

Trend and Momentum Indicators

1

SMA / EMA

Average close over n bars; EMA weights recent bars by 2/(n+1).

2

MACD

EMA12 − EMA26 with a 9-EMA signal line; crosses flag momentum.

3

RSI (14)

100 − 100/(1+RS); above 70 overbought, below 30 oversold.

4

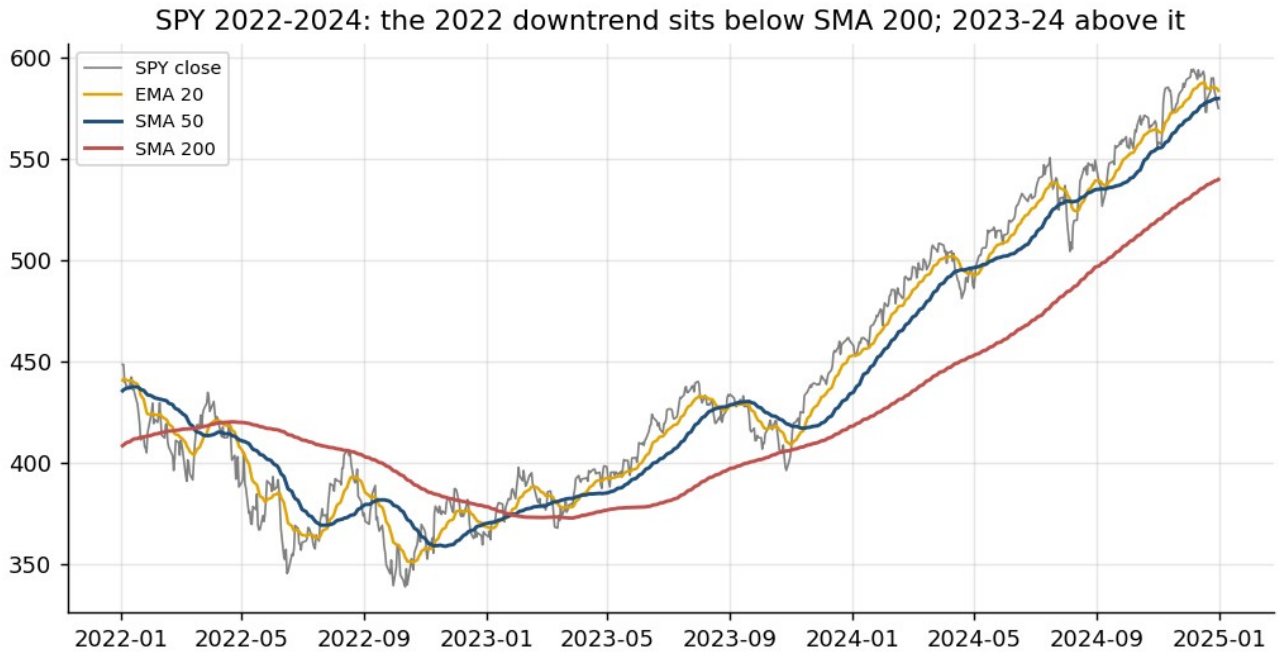
Donchian channel

Highest high / lowest low of n bars — the breakout level.

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TOPIC 2 · INDICATOR

Indicator: Moving Averages (SMA and EMA)



DEFINITION

The average closing price over the last n bars - a smoothed line that shows the trend. The EMA weights recent bars more, so it turns sooner.

FORMULA

SMA(n) = (C1 + C2 + ... + Cn) / n. EMA(t) = a x C(t) + (1 - a) x EMA(t-1), with a = 2 / (n + 1).

HOW TO READ IT

Price above a rising average = uptrend. The 50 / 200-day pair is the classic trend filter; EMA 20 tracks the short-term trend.

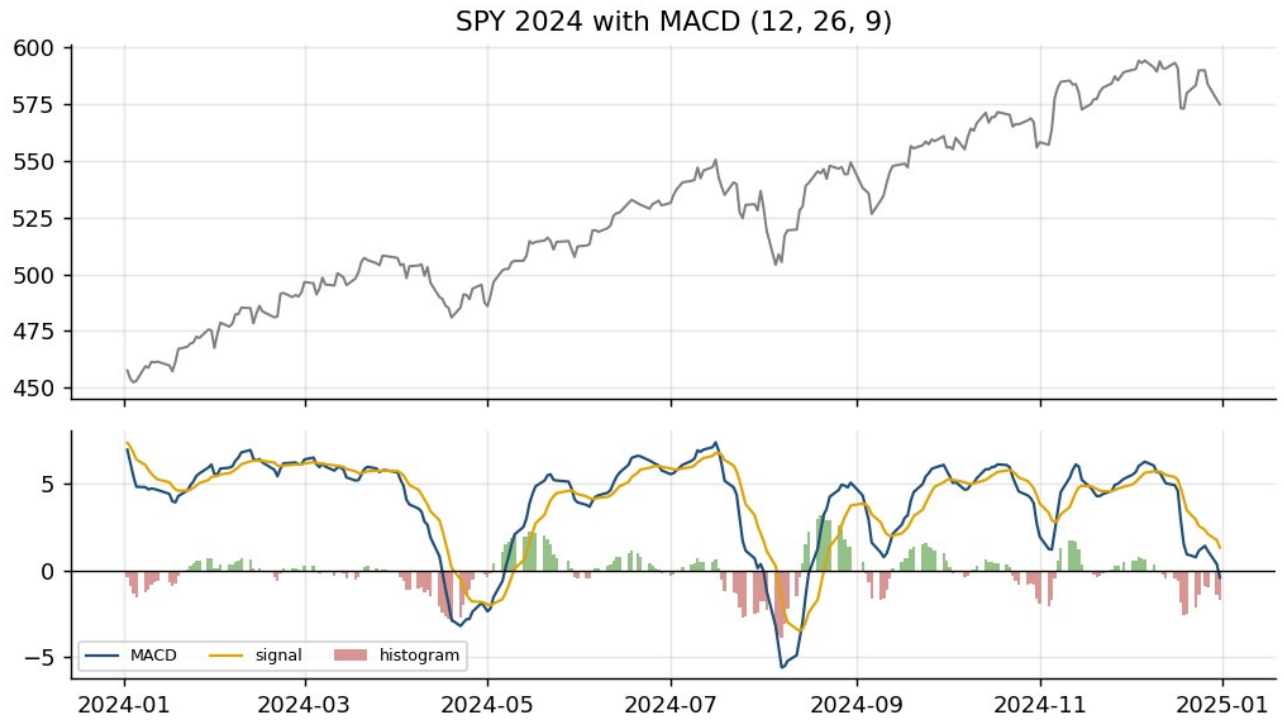
INSIGHT

Every average lags: the longer n, the smoother - and the later each signal. In sideways markets price whipsaws across it.

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TOPIC 2 · INDICATOR

Indicator: MACD (Moving Average Convergence Divergence)



DEFINITION

A momentum-of-trend gauge: the gap between a fast and a slow EMA, with a signal line and a histogram that show whether momentum is growing.

FORMULA

MACD = EMA12(C) - EMA26(C). Signal = EMA9(MACD). Histogram = MACD - Signal.

HOW TO READ IT

MACD crossing above its signal line = momentum turning up; MACD above zero = the 12-day trend is above the 26-day trend.

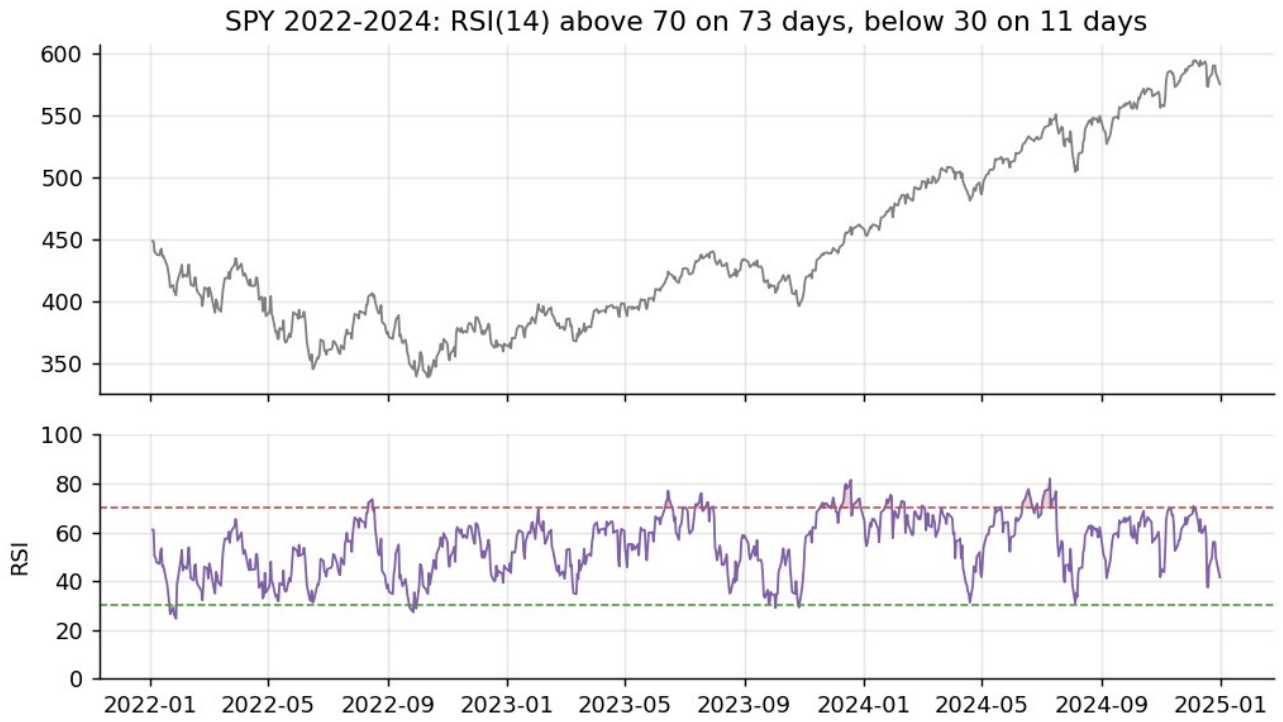
INSIGHT

A shrinking histogram warns that a move is losing strength before price turns - but it is built from EMAs, so it lags and whipsaws in ranges.

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TOPIC 2 · INDICATOR

Indicator: RSI (Relative Strength Index)



DEFINITION

An oscillator from 0 to 100 that compares the size of recent gains with recent losses - how stretched the latest move is.

FORMULA

RS = average gain / average loss over n bars (Wilder smoothing, n = 14). RSI = 100 - 100 / (1 + RS).

HOW TO READ IT

Above 70 = overbought, below 30 = oversold. Short look-backs such as RSI(2) are used for mean-reversion entries.

INSIGHT

In a strong trend RSI can stay above 70 for weeks: overbought is not a sell signal by itself. Test the thresholds - they are conventions.

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TOPIC 2 · INDICATOR

RSI(N): the Number Is the Look-back Period

	RSI(2) - short look-back	RSI(14) - Wilder's standard (1978)
Look-back	The last 2 bars (days on a daily chart)	The last 14 bars - about 3 trading weeks
Reacts to	1-2 day moves: fast and jumpy	Sustained moves: smooth and slower
How often extreme	Often: SPY RSI(2) < 10 on 9.6% of days	Rarely: SPY RSI(14) < 30 on 1.7% of days
Typical levels	Buy below 10, exit above 70 (simulator default)	Buy below 30, sell above 70
SPY, 10 Oct 2025 (-2.7% after a -0.3% day)	5.8 - deeply oversold: the RSI(2) rule buys	42.5 - neutral: no signal
Next trading day, 13 Oct	SPY +1.5% - the bounce is captured	No trade
Simulator strategy	RSI(2) mean reversion (Connors style)	RSI oversold / overbought

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TOPIC 2 · INDICATOR

The Number in Brackets Is Always the Look-back

1

RSI(N) formula

RS = average gain / average loss over the last N bars; RSI = 100 - 100 / (1 + RS). Mostly up days: near 100. Mostly down: near 0. Equal: 50.

2

SMA(50) and EMA(20)

SMA(50): average of the last 50 closes. EMA(20): exponential average over about 20 bars, weighting recent bars more.

3

Bollinger(20, 2) and ATR(14)

Bollinger: 20-day average with bands 2 standard deviations away. ATR(14): average true range over 14 bars.

4

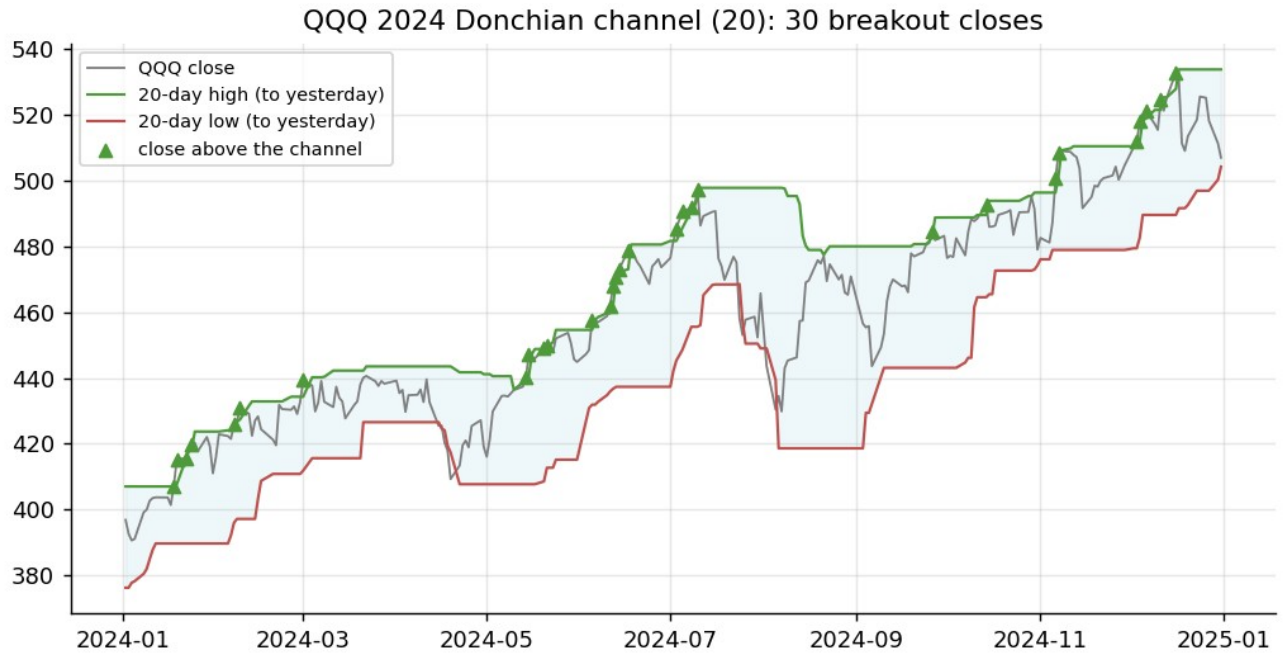
Where to change N

TradingView: the indicator's Settings > Inputs > Length. Simulator: the strategy's parameter (e.g. RSI length n) before you backtest.

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TOPIC 2 · INDICATOR

Indicator: Donchian Channel



DEFINITION

A channel drawn through the highest high and the lowest low of the last n bars - the levels a breakout must clear.

FORMULA

Upper(n) = max(High, last n bars). Lower(n) = min(Low, last n bars). Classic n = 20 (the Turtle traders used 20 and 55).

HOW TO READ IT

A close above the prior upper band = upside breakout; exit on a break of a shorter (e.g. 10-day) low.

INSIGHT

Breakout systems win rarely but big. Compare today's close with the channel up to YESTERDAY - including today's high is look-ahead.

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TOPIC 2

Volatility and Volume Indicators

1

Bollinger Bands

SMA20 ± 2 standard deviations; width tracks volatility.

2

ATR (14)

Average true range — position sizing and volatility stops.

3

VWAP

Volume-weighted average price — the execution benchmark.

4

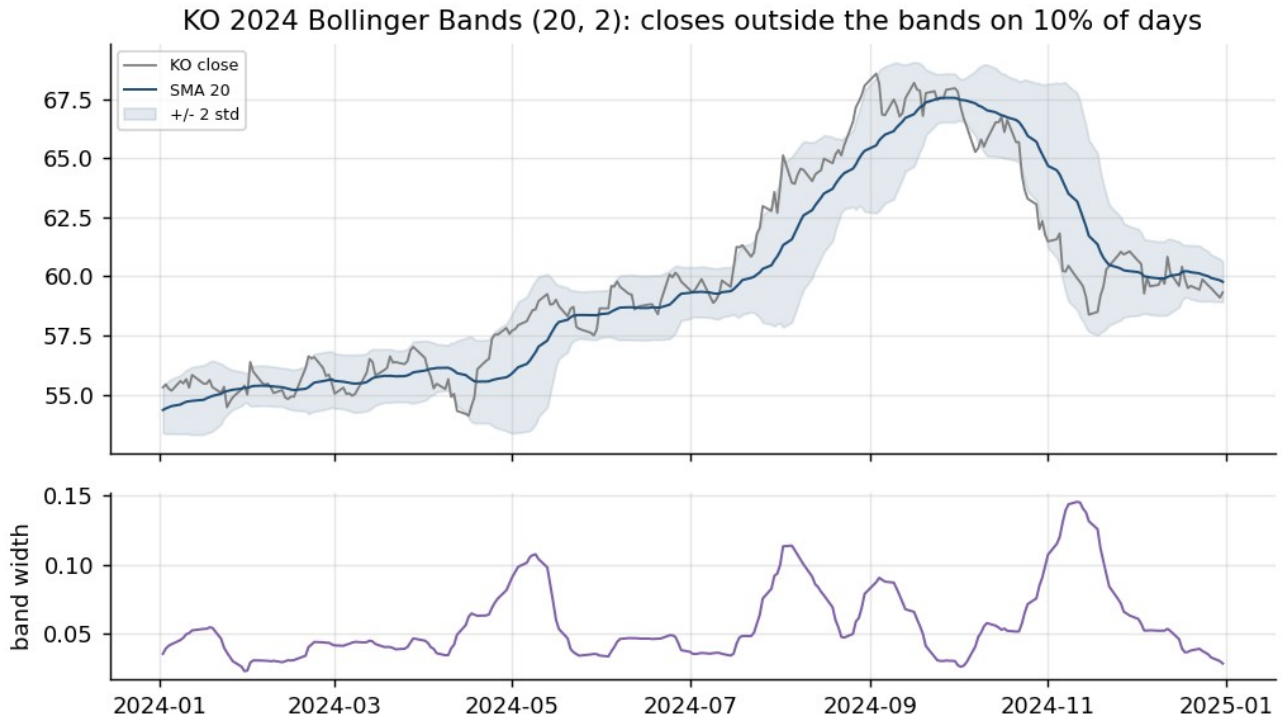
OBV

Adds volume on up days, subtracts on down days.

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TOPIC 2 · INDICATOR

Indicator: Bollinger Bands



DEFINITION

A moving average with bands set a number of standard deviations above and below it, so the bands widen and narrow with volatility.

FORMULA

Middle = SMA20(C). Upper / Lower = Middle +/- 2 x StDev20(C). %B = (C - Lower) / (Upper - Lower).

HOW TO READ IT

Mean reversion: buy a close below the lower band, exit at the middle. A 'squeeze' (narrow bands) often comes before a big move.

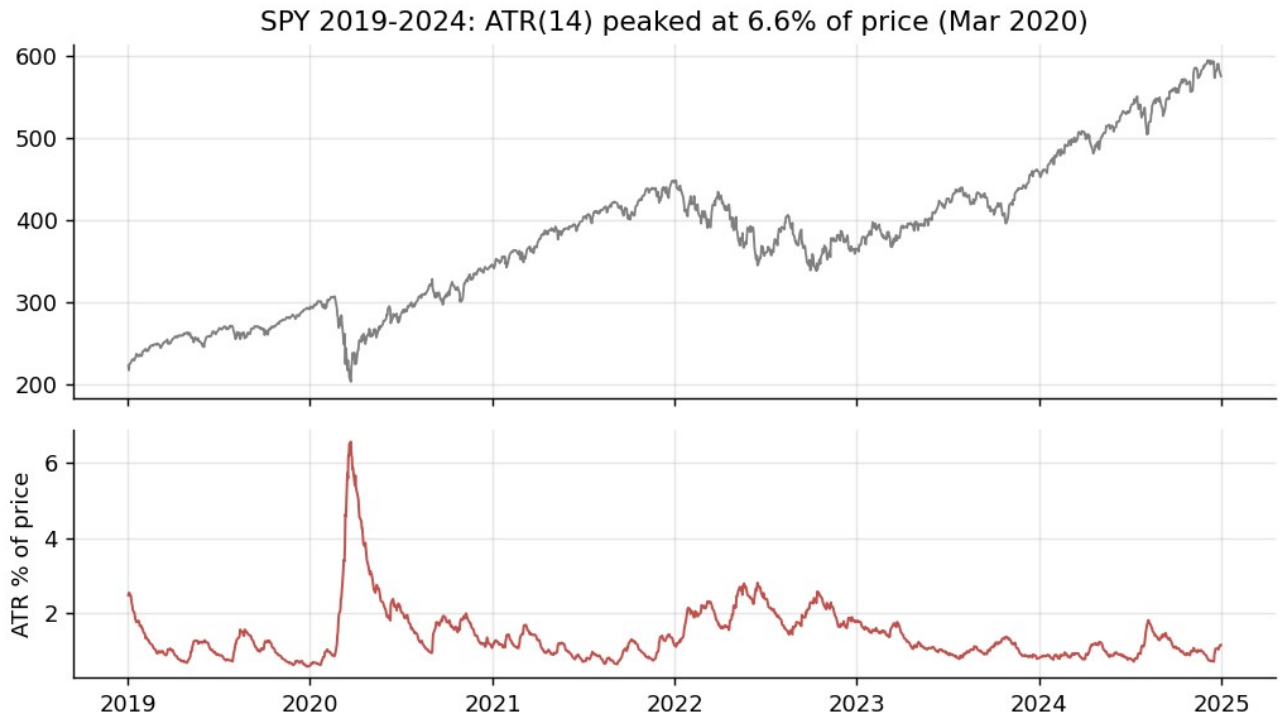
INSIGHT

Returns are not normal: closes fall outside the bands far more often than the 5% a normal curve implies. In a trend, 'oversold' keeps falling.

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TOPIC 2 · INDICATOR

Indicator: ATR (Average True Range)



DEFINITION

The average size of a day's price range, including overnight gaps - a volatility ruler in price units.

FORMULA

TR = max(H - L, |H - previous C|, |L - previous C|). ATR = average of TR over n bars (Wilder smoothing, n = 14).

HOW TO READ IT

Set stops at 2-3 x ATR and size positions as risk budget / (k x ATR), so every trade carries similar risk.

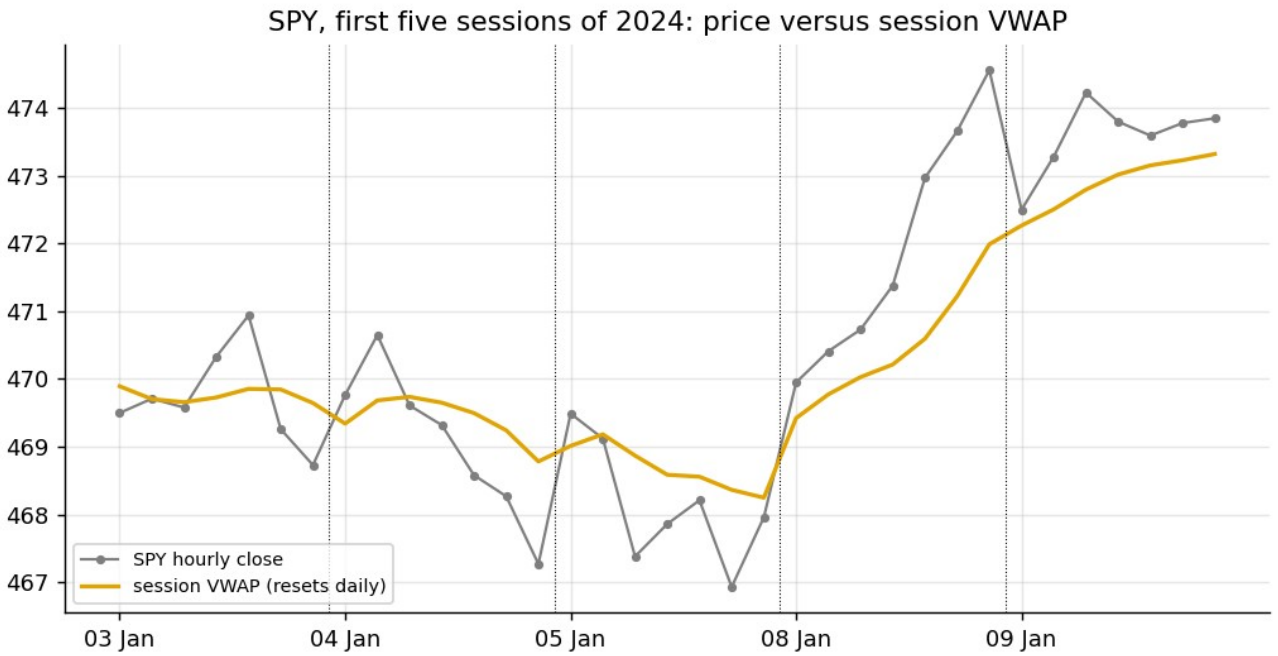
INSIGHT

ATR says how far, not which way. Compare instruments with ATR as a % of price, since ATR rises with the price level.

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TOPIC 2 · INDICATOR

Indicator: VWAP (Volume-Weighted Average Price)



DEFINITION

The average price paid during the session, weighted by the volume traded at each price. It resets at every session open.

FORMULA

Typical price P = (H + L + C) / 3. VWAP = sum(P x Volume) / sum(Volume), summed from the session open.

HOW TO READ IT

The execution benchmark: buying below VWAP is a good fill. Intraday traders read price above VWAP as buyers in control.

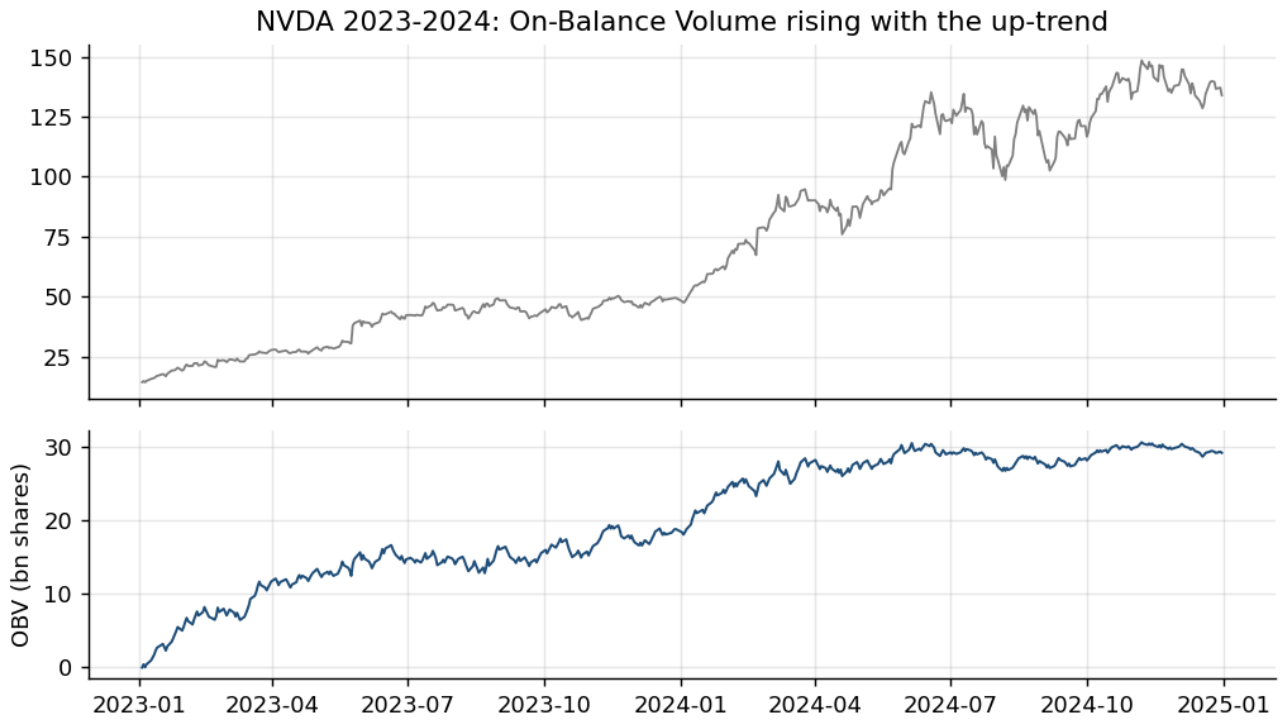
INSIGHT

Institutions judge execution algorithms against VWAP. It needs intraday data - on daily bars it is only an approximation.

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TOPIC 2 · INDICATOR

Indicator: OBV (On-Balance Volume)



DEFINITION

A running total of volume that adds a day's volume when the close rises and subtracts it when the close falls.

FORMULA

OBV(t) = OBV(t-1) + Volume(t) if C(t) > C(t-1); - Volume(t) if C(t) < C(t-1); unchanged if equal.

HOW TO READ IT

A rising OBV confirms an up-trend; OBV making new highs before price is read as accumulation. Read the slope, not the level.

INSIGHT

Volume shows the conviction behind a move. An OBV that falls while price rises (divergence) is a warning, not a signal on its own.

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TOPIC 2

Matching the Method to the Hypothesis

1

'Does the signal beat the benchmark?'

Backtest + t-test or bootstrap on returns.

2

'Will tomorrow be up?'

Classifier: logistic regression, random forest, XGBoost.

3

'What state is the market in?'

Clustering: K-means on return and volatility features.

4

'Which parameters?'

Grid search in-sample, confirm out-of-sample.

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TOPIC 2

Coding Languages for Algorithms and Signals

1

Pine Script v6

//@version=6 · ta.ema · ta.crossover · strategy.entry.

2

Python

Signals as pandas Series of +1 / 0 / −1 aligned to bars.

3

C++ / Java

Low-latency execution engines and exchange gateways.

4

R, SQL, MQL

Statistics, data stores, MetaTrader expert advisors.

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TOPIC 2

Order Types and Time in Force

1

Market and limit

Certainty of fill versus control of price.

2

Stop, stop-limit, trailing stop

Trigger on a price; protect or enter on breaks.

3

Bracket / OCO / OTO

Entry with attached take-profit and stop-loss.

4

day · gtc · opg · cls · ioc · fok

How long an order stays working.

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TOPIC 2

Performance and Risk Metrics

1

Return

Total return · CAGR = (end/start)^(1/years) − 1.

2

Risk

Volatility = σ(daily) × √252 · maximum drawdown.

3

Risk-adjusted

Sharpe · Sortino · Calmar = CAGR / |MaxDD|.

4

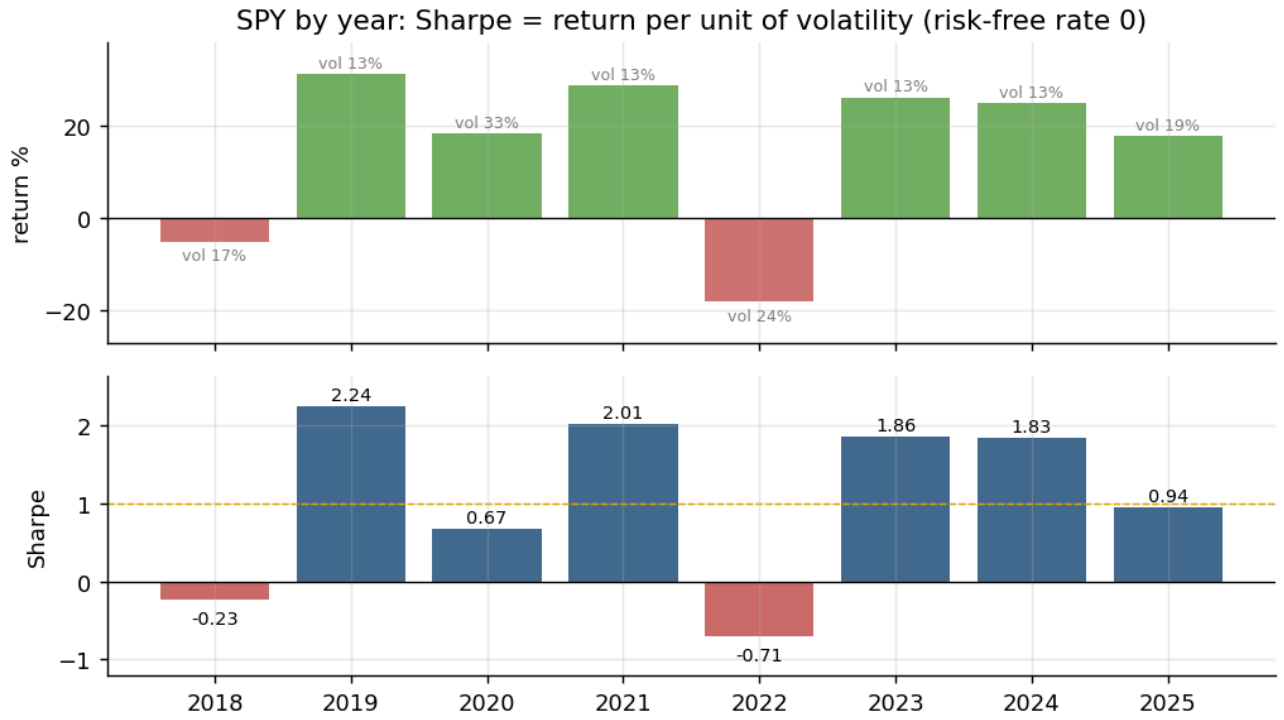
Trade quality

Win rate · profit factor · expectancy · turnover · exposure.

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TOPIC 2 · METRIC

Metric: Sharpe Ratio



DEFINITION

Return earned per unit of risk taken: the average excess return divided by its volatility, annualised.

FORMULA

Sharpe = (mean daily return - daily risk-free rate) / std of daily returns x sqrt(252).

HOW TO READ IT

Below 0.5 weak, about 1 good, above 2 rare for a daily strategy - check for errors. Always compare with buy-and-hold over the same period.

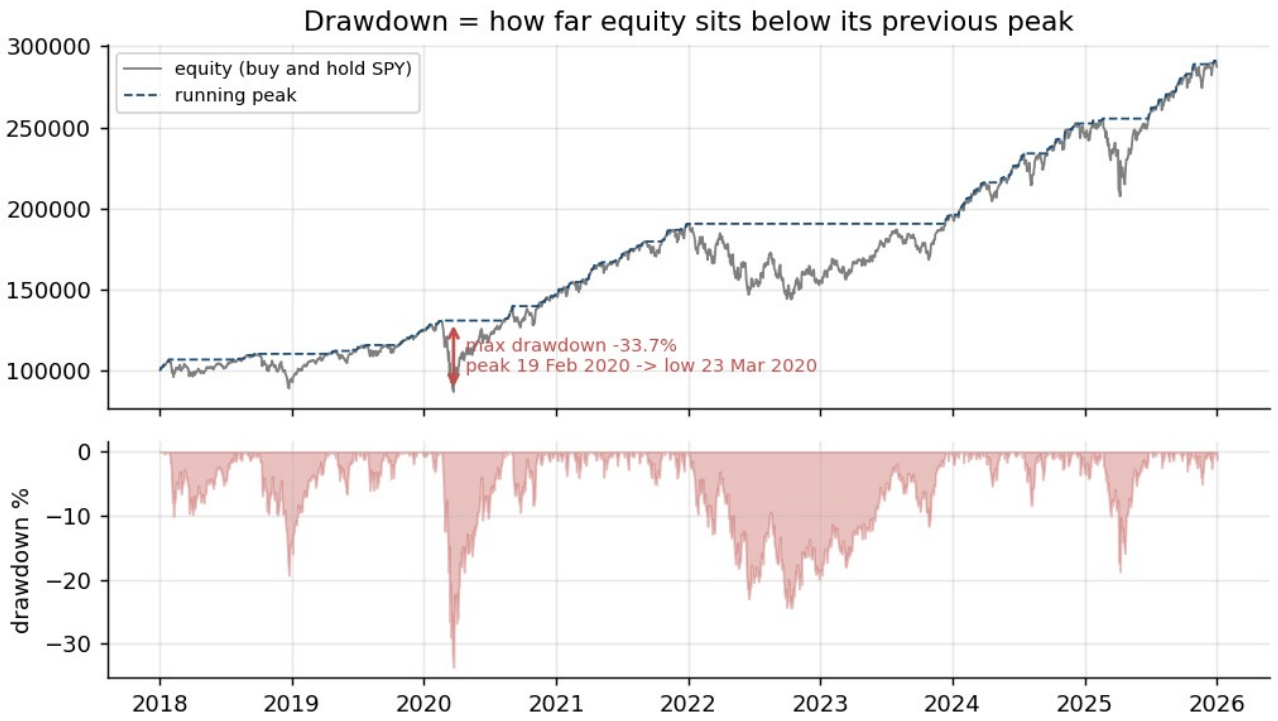
INSIGHT

Volatility counts as risk whether it is up or down, and a short or costless backtest inflates it - 2020 returned 18% but at 33% volatility, a Sharpe of only 0.67.

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TOPIC 2 · METRIC

Metric: Maximum Drawdown



DEFINITION

The largest fall of the equity curve from a previous peak to a later low - the worst loss an investor would have lived through before a new high.

FORMULA

Drawdown(t) = equity(t) / running peak(t) - 1. Max drawdown = the most negative drawdown over the period.

HOW TO READ IT

A -20% drawdown needs +25% to recover; -50% needs +100%. Compare with buy-and-hold and with the loss the investor can tolerate.

INSIGHT

One path-dependent number: it does not say how long recovery took, and a short backtest may simply miss a bad period. SPY 2018-2025: -33.7% in five weeks of 2020.

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Classification of Trading Algorithms by Purpose

	Purpose	Examples
Execution algorithms	Work a large order at low cost and market impact	VWAP, TWAP, POV, implementation shortfall, iceberg, smart order routing
Market making	Quote bids and offers, earn the spread, manage inventory	Passive market making with resting limit orders
Arbitrage	Capture mispricing between related products or markets	ETF versus its basket, index futures versus stocks
Directional / alpha	Take a view on the direction of price	Trend following, momentum, breakout, mean reversion, news
Statistical arbitrage and pairs	Hedged: long one asset, short a related one; trade the spread	Pairs trading, cointegration, market-neutral baskets
Factor and portfolio	Build and rebalance portfolios from return drivers and risk	Value, momentum, size factors; mean-variance optimisation
Hedging and risk	Offset unwanted exposure and enforce limits	Delta hedging, beta hedging, pairs hedge, kill switch

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SEC Classification of High-Frequency Strategies (2010)

1

Passive market making

Submits non-marketable resting bids and offers that provide liquidity at specified prices

2

Arbitrage

Captures pricing inefficiencies between related products or markets, e.g. an ETF and its basket

3

Structural

Exploits structural vulnerabilities in the market or its participants, e.g. stale quotes via faster data

4

Directional

Takes a short-term position anticipating an intraday price move in a particular direction

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Classification by Holding Period

	Horizon	Typical algorithms
High-frequency	Microseconds to seconds	Market making, ETF and latency arbitrage
Intraday	Minutes to hours	VWAP / TWAP execution, intraday momentum
Swing	Days to weeks	Mean reversion, pairs trading
Position	Weeks to months	Trend following, factor portfolios

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Classification by Model Type

1. **Rule-based**
   Fixed if-then rules on indicators, e.g. a moving-average crossover

2. **Statistical**
   Econometric tests and models: cointegration, regression, t-tests

3. **Machine learning**
   Models learn from labelled data: logistic regression, random forest, XGBoost

4. **Reinforcement learning**
   Learns a policy from rewards, e.g. order-execution research

5. **LLM and agentic**
   Language models read text and agents plan, code and act under limits

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TOPIC 2

Long and Short Positions

	Long	Short
What you hold	Shares you own	Shares you borrowed and sold - you owe them back
Opening trade	Buy	Sell (the borrowed shares)
Closing trade	Sell	Buy back ('buy to cover') and return the shares
You profit when	The price rises	The price falls
Example	Buy SPY at $500, sell at $550: +$50	Sell SPY at $500, buy back at $450: +$50
Maximum loss	What you paid - a price cannot go below $0	Unlimited - a price can keep rising
Extra requirements	Only the cash to pay for the shares	A margin account (Alpaca: $2,000+ equity); borrow fees on hard-to-borrow stocks (Alpaca: $0 on easy-to-borrow)

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TOPIC 2

Long and Short in Practice

1

'Long = buy, short = sell' is shorthand

Opening a long is a buy; opening a short is a sell. Selling shares you already own only closes the long - you go flat, not short.

2

Pine Script (TradingView)

strategy.entry("Long", strategy.long) opens a buy; strategy.close("Long") sells back to flat; strategy.short opens a short.

3

Trading Simulator

Most strategies are long-only: buy, then sell back to cash. 'RSI oversold / overbought' has an allow_short option that goes short above RSI 70.

4

Pairs trading (Lab 14)

Long the cheap stock and short the rich one at the same time: the trade profits from the gap closing, whichever way the market moves.

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TOPIC 2

Strategy Families at a Glance (1 of 2)

1

Trend following

Rides sustained moves in the direction of the trend; cuts losers quickly - e.g. Buy when the 50-day moving average crosses above the 200-day

2

Mean reversion

Bets that prices stretched far from their average will snap back - e.g. Buy when price closes below the lower Bollinger Band (20-day, 2 std dev)

3

Momentum and breakout

Buys recent winners or prices breaking out of a range, expecting them to keep going - e.g. Buy on a close above the 20-day high (Donchian channel breakout)

4

Statistical / pairs arbitrage

Trades the spread between two related assets when it drifts from its normal level - e.g. Short the rich leg, buy the cheap leg when the spread z-score exceeds 2

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TOPIC 2

Strategy Families at a Glance (2 of 2)

1

Execution algorithms

Split a large order to minimise cost and market impact: VWAP, TWAP, POV, implementation shortfall - e.g. POV: trade 10% of market volume until the order is filled

2

Market making

Continuously quotes buy and sell prices, earning the spread while managing inventory - e.g. Quote bid and ask around mid-price; skew quotes as inventory builds

3

Index rebalancing / event-driven

Trades predictable flows around index changes, earnings and corporate events - e.g. Position ahead of announced index inclusions on the rebalance date

4

Sentiment / NLP

Turns news, filings and social media text into trading signals using NLP or LLMs - e.g. Go long when the average FinBERT headline sentiment is strongly positive

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Strategy Family: Trend Following

The idea

Prices that have started to move tend to keep moving. Buy when a fast average is above a slow one (or price is above its 200-day average); exit when the trend turns.

When it works

Long, persistent moves - bull or bear markets, macro trends lasting months. A few big winners pay for many small losses.

When it will not work

Sideways, choppy markets: price crosses the averages again and again (whipsaws), every crossing costs money, and signals arrive late at turns.

Lab 7: EMA 20/50 on SPY

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Strategy Family: Mean Reversion

The idea

Like a rubber band, short-term over-reactions snap back. Buy when price is stretched below its normal range (lower Bollinger Band, RSI(2) below 10) and sell when it returns to the average.

When it works

Range-bound, liquid markets and stable large stocks, where extremes tend to snap back within days.

When it will not work

A real trend or a news shock: 'cheap' keeps getting cheaper. A high win rate hides rare large losses - as in March 2020.

Lab 7: Bollinger 20 / k = 2.0 and 1.5 on KO

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Mean Reversion Rules in the Simulator

	Buy when 'too low'	Exit
Bollinger Band mean reversion	Close below the lower band (20-day mean - 2 standard deviations)	Price returns to the middle band (the 20-day mean)
RSI oversold / overbought	RSI(14) below 30 - oversold	RSI(14) above 70 - overbought
RSI(2) mean reversion	RSI(2) below 10 - a sharp 1-2 day drop	RSI(2) above 70 - the bounce
Dip buy after a losing streak	3 or more down closes in a row (the Lab 4 hypothesis)	After the holding horizon, e.g. the next day

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Mean Reversion versus Trend Following

	Mean reversion	Trend following / momentum
Belief	Extremes snap back to the average	What is moving keeps moving
After a big drop	Buy	Sell or stay out
Works best in	Range-bound, sideways markets	Strongly trending markets
Typical horizon	Short: days	Longer: weeks to months
Win rate	High - many small wins	Low - few but large wins
Main danger	A rare big loss when the dip becomes a crash	Many small losses in choppy markets

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Mean Reversion: the Main Risk and Three Safeguards

The risk

The rule assumes the average holds. After a real change - a crash, bad news, a broken business - price does not come back: buying the dip becomes catching a falling knife.

1 Stop-loss or time limit

Cap the rare big loss: exit at a fixed loss, or after n days if the bounce has not come.

2 Trend filter

Only buy dips while price is above its 200-day SMA, so you are not buying into a bear market.

3 Stable markets

Liquid, defensive stocks such as KO, and pairs trading (Lab 14: the spread between two linked stocks) revert more reliably.

Lab 7: Bollinger on KO - high win rate, but losses larger than wins: many small wins, rare big losses.

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Strategy Family: Momentum

The idea

Winners keep winning for months. Buy what has risen over the last 6-12 months (time-series) or what has risen most among its peers (cross-sectional).

When it works

Persistent market or sector themes, spread across many assets and rebalanced regularly.

When it will not work

Sharp reversals - 'momentum crashes' such as the 2009 rebound, when last year's losers rally hardest - and choppy markets that churn the portfolio.

Lab 7: 126-day time-series momentum on QQQ

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Strategy Family: Breakout

The idea

A close above the recent range (for example the 20-day high, the Donchian channel) signals the start of a new move: buy the breakout, exit on a shorter low.

When it works

Volatility expanding after a quiet period and a new trend starting - indices and commodities in strong regimes.

When it will not work

False breakouts in a range: price pokes above the high and falls back. A low win rate, so exits must be disciplined and costs kept low.

Lab 7: Donchian 20 / 10 on QQQ

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TOPIC 2 · HOW THE ALGORITHMS WORK

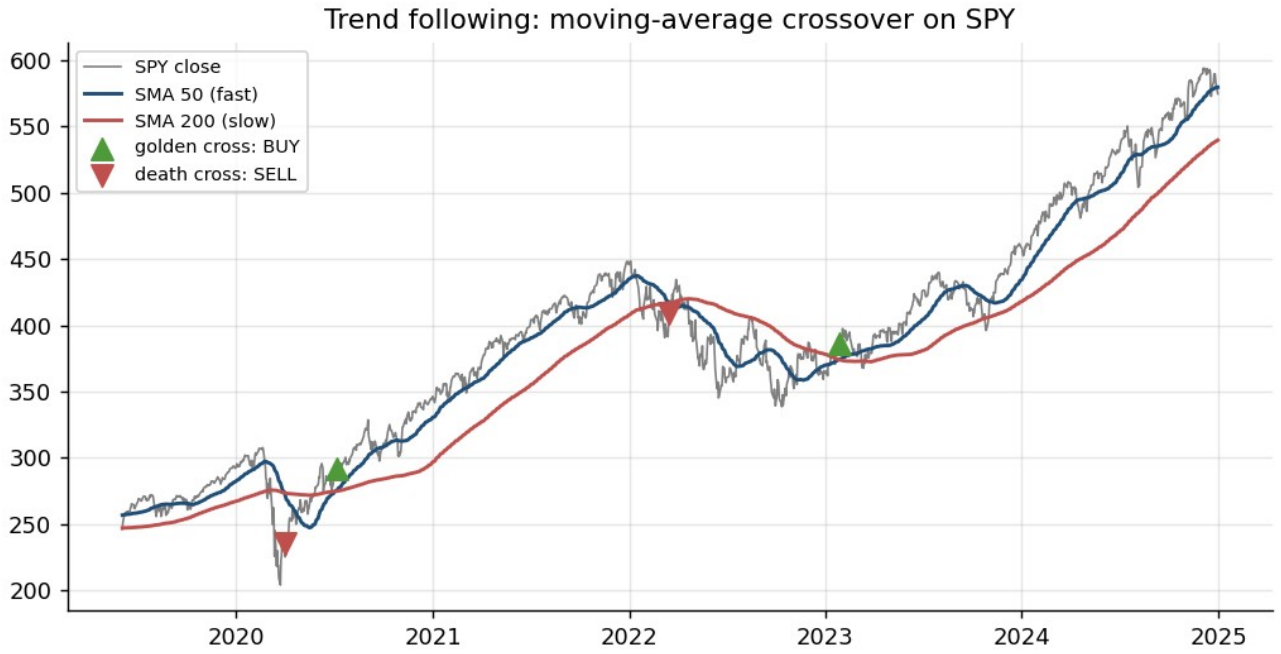
Concepts over code

Every algorithm below is coded for you - by the simulator or an AI assistant. Learn what it does, when it works and what can go wrong; let AI handle the syntax.

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TOPIC 2

Algorithm: Moving-Average Crossover (Trend Following)



HOW IT WORKS

Two moving averages of the close: a fast one reacts quickly, a slow one shows the trend.

RULE

Fast crosses ABOVE slow (golden cross): buy. Fast crosses BELOW (death cross): sell or go flat.

WORKS BEST WHEN

Markets trend for months - it rides the move and cuts losers when the trend turns.

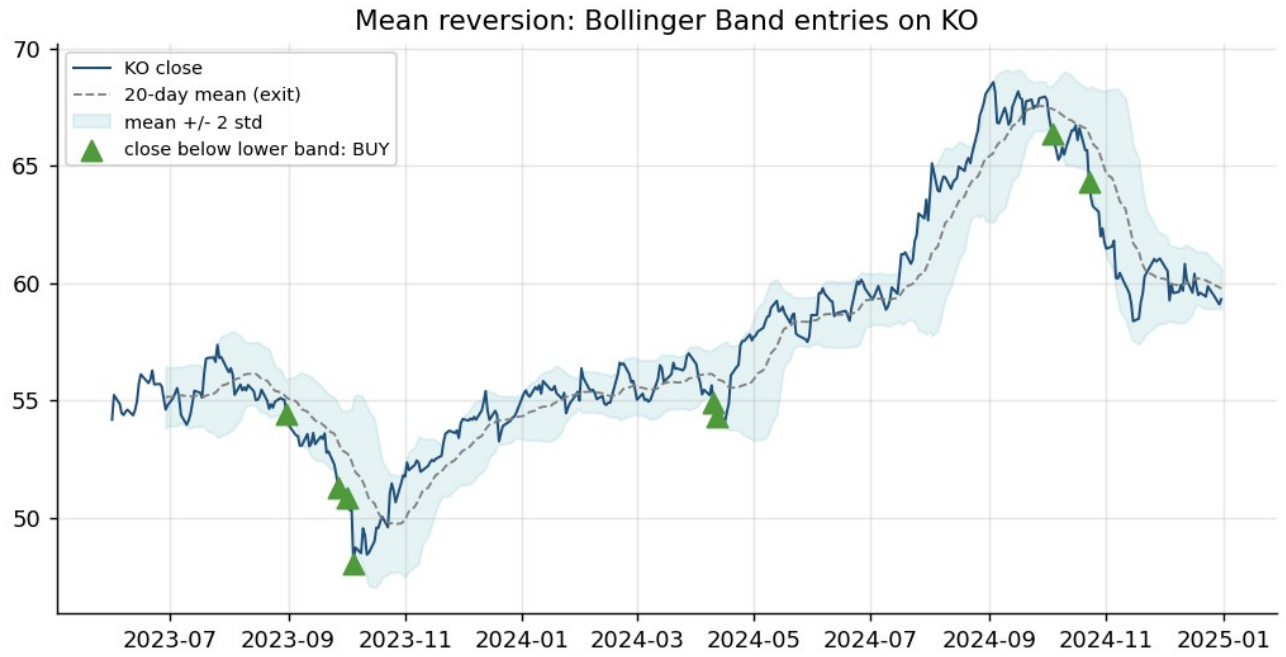
WATCH OUT FOR

Whipsaws in sideways markets; signals lag; trade on the next bar, not the crossing bar.

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TOPIC 2

Algorithm: Bollinger Band Mean Reversion



HOW IT WORKS

Bands = 20-day mean +/- 2 standard deviations. A close outside the band is 'stretched'.

RULE

Close below the lower band: buy. Exit when the price returns to the middle band (the mean).

WORKS BEST WHEN

Range-bound markets where over-reactions snap back, e.g. a defensive stock such as KO.

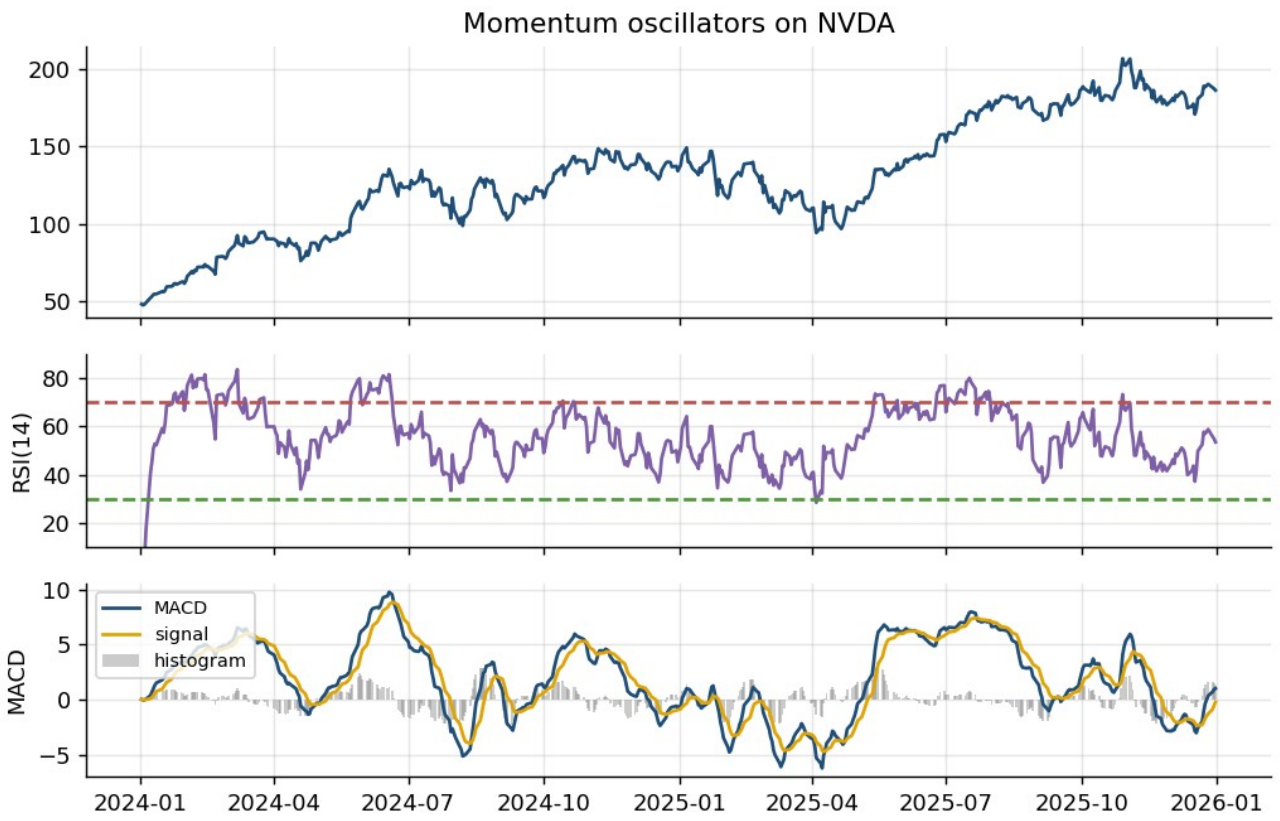
WATCH OUT FOR

Strong trends keep 'stretching' - add a stop-loss or a trend filter; many small wins, rare big losses.

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TOPIC 2

Algorithm: RSI and MACD Momentum Oscillators



HOW IT WORKS

RSI = 100 - 100/(1+RS), RS = average gain / average loss over 14 bars. MACD = EMA12 - EMA26.

RULE

RSI < 30 oversold / > 70 overbought. MACD crossing above its 9-EMA signal line = momentum buy.

WORKS BEST WHEN

As filters or confirmation for a trend or mean-reversion rule, not as stand-alone predictors.

WATCH OUT FOR

Oscillators can stay overbought in strong trends; RSI and MACD are highly correlated with price.

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TOPIC 2

Algorithm: Cross-Sectional Momentum



HOW IT WORKS

Rank a universe by past return (e.g. 12 months, skipping the last month) each month.

RULE

Hold the top-ranked assets, avoid or short the bottom-ranked; rebalance monthly.

WORKS BEST WHEN

Information spreads slowly and winners keep winning for several months.

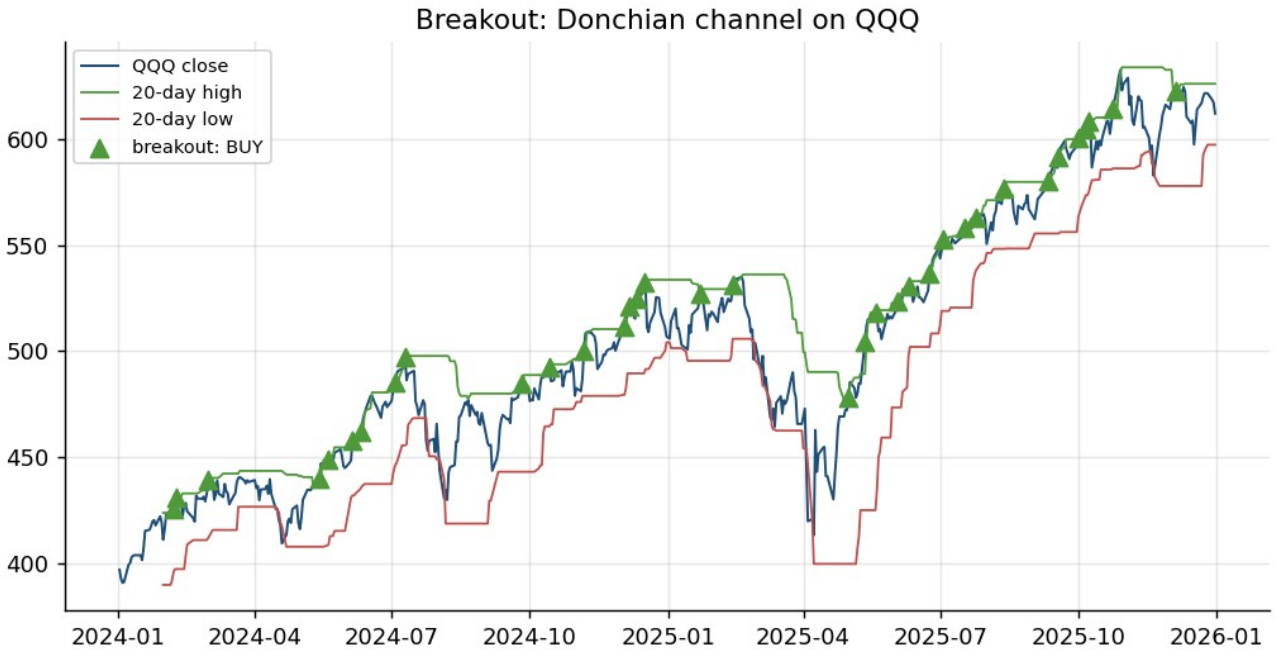
WATCH OUT FOR

Sharp 'momentum crashes' when markets reverse; turnover costs at every rebalance.

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TOPIC 2

Algorithm: Donchian Channel Breakout



HOW IT WORKS

Track the highest high and lowest low of the last n bars (n = 20 here).

RULE

Close above the 20-day high: buy the breakout. Exit below the 20-day low (or a trailing stop).

WORKS BEST WHEN

New trends start with a break of a range - the rule used by the 1980s 'Turtle Traders'.

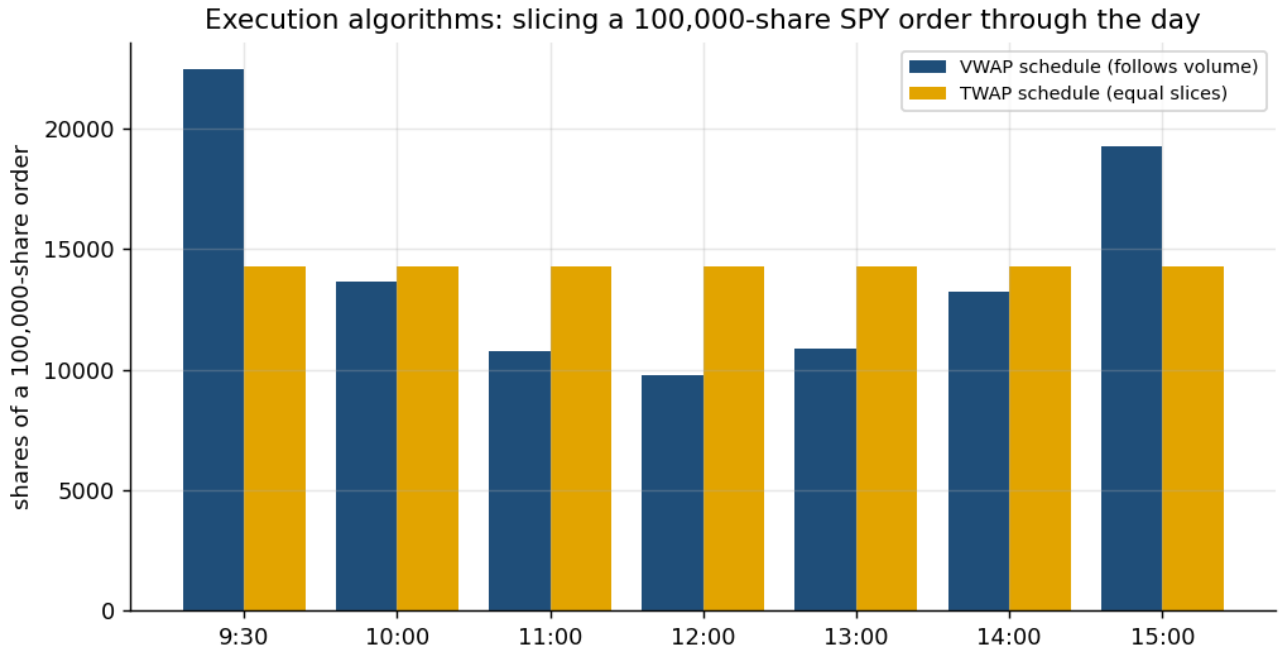
WATCH OUT FOR

False breakouts in choppy markets; the channel must use PREVIOUS bars only (no look-ahead).

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TOPIC 2

Algorithm: VWAP and TWAP Execution



HOW IT WORKS

Split a large parent order into child orders through the day to reduce market impact.

RULE

TWAP: equal slices per time interval. VWAP: slices proportional to the typical volume curve.

WORKS BEST WHEN

Working client or fund orders too large to trade at once; benchmarked to the day's VWAP.

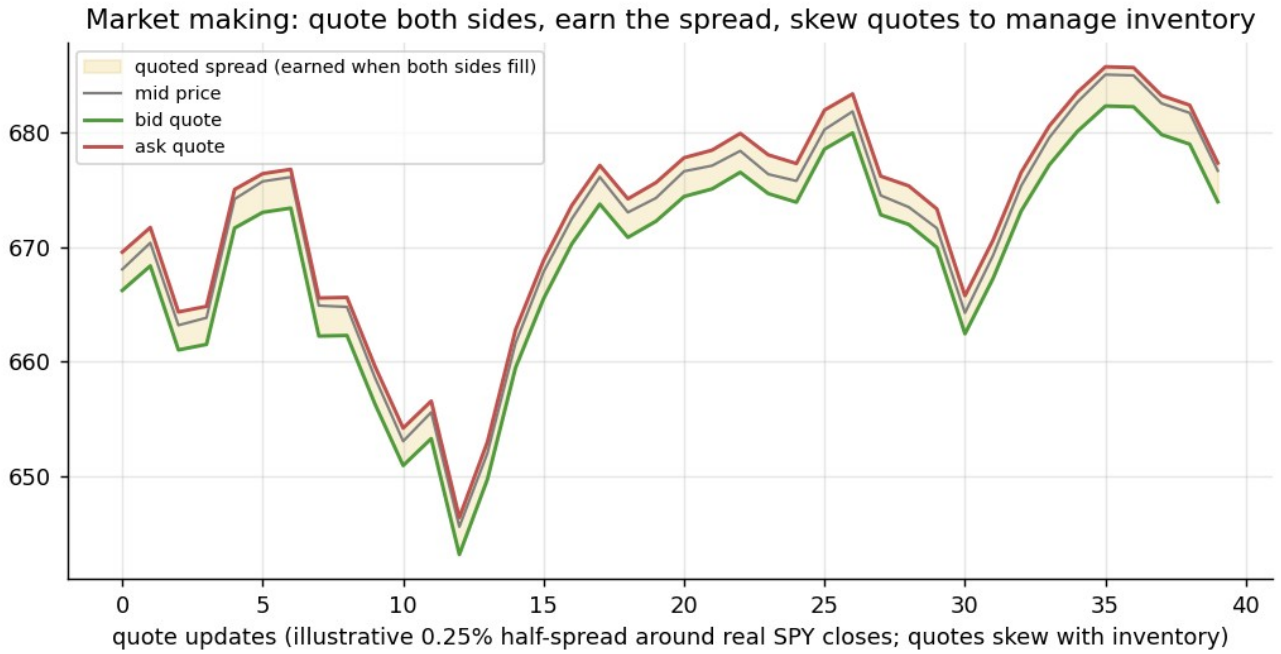
WATCH OUT FOR

VWAP relies on a volume forecast; both are predictable to other traders - add randomisation.

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TOPIC 2

Algorithm: Market Making



HOW IT WORKS

Quote a bid below and an ask above the mid price at the same time.

RULE

Earn the spread when both sides fill; skew quotes down when long, up when short, to cut inventory.

WORKS BEST WHEN

Liquid markets with steady two-way flow - the business of quant market makers.

WATCH OUT FOR

Adverse selection by better-informed traders; inventory risk in fast moves; needs low latency.

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TradingView versus Alpaca

	TradingView	Alpaca
What it is	Charting and strategy-design platform	API-first US broker for automated trading
Markets	Global charts incl. SGX, US, FX, crypto	US stocks and ETFs, options, crypto
Strategy design	Pine Script v6 indicators and strategies	Through the API - in this course the Trading Simulator does it for you
Backtesting	Strategy Tester (simplified costs and fills)	None built in - the Trading Simulator runs the cost-aware backtest
Paper trading	Built-in Paper Trading account from the chart	Paper account mirroring the live API
Automation	Alerts; webhooks on paid plans	Trading API: orders, positions, clock; 24/5 trading
Data	Charts and Data Window	Market Data API: bars, quotes, trades, news (IEX free)
Machine learning	Not supported	Through code - in this course the Trading Simulator has built-in logistic regression, random forest and XGBoost
Best used for	Designing, visualising and first-testing a rule	Executing and forward-testing it automatically

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Paper Trading on TradingView, Step by Step

Full step-by-step guide: Learner Guide, 'Platform Guides'.

1. Supercharts > Trade > Paper Trading > Connect
2. Set starting capital and commission
3. Order ticket: side, quantity, Market / Limit / Stop
4. Add stop loss and take profit; place order
5. Track Positions, Orders, Account History
6. Close the position; reset when done

Paper fills use TradingView's prices but cannot show your own market impact - compare them with the backtest's assumed costs.

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Paper Trading on Alpaca, Step by Step - Manual and Automated

Full step-by-step guide: Learner Guide, 'Platform Guides'.

1. Dashboard: Paper account > search SPY > Buy
2. Watch Orders: new > accepted > filled
3. Outside US hours: extended-hours LIMIT orders
4. Simulator: Paper Trading > Order ticket
5. Simulator: + New bot > Start
6. Check dashboard; kill switch to stop

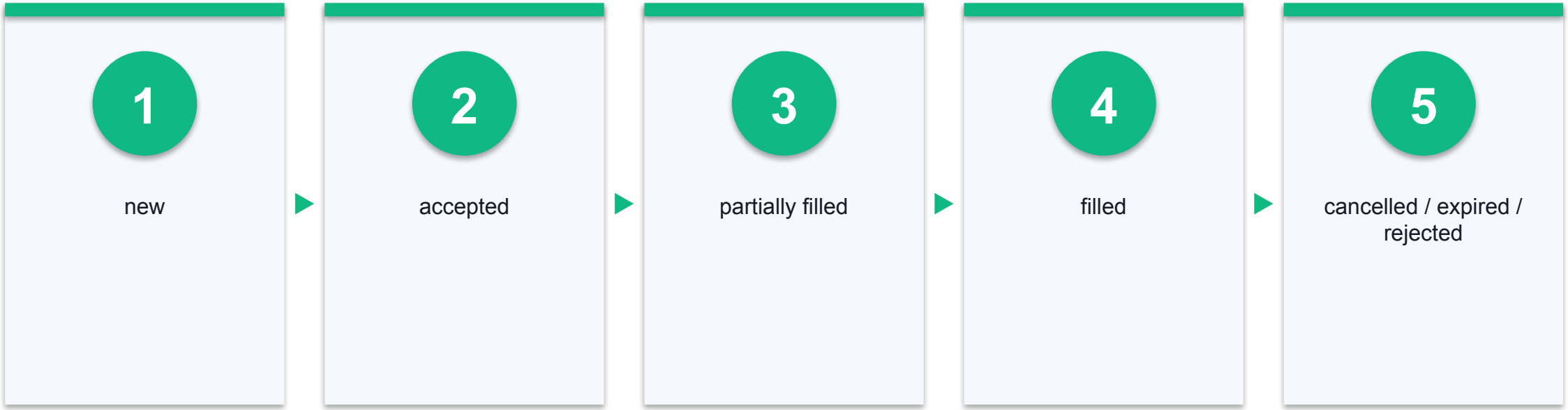
From Singapore the US regular session is 21:30-04:00 (22:30-05:00 in northern winter); class-time orders use Alpaca's 24/5 extended-hours limit orders.

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TOPIC 2

The Order Lifecycle

A program must read the status — never assume a fill.

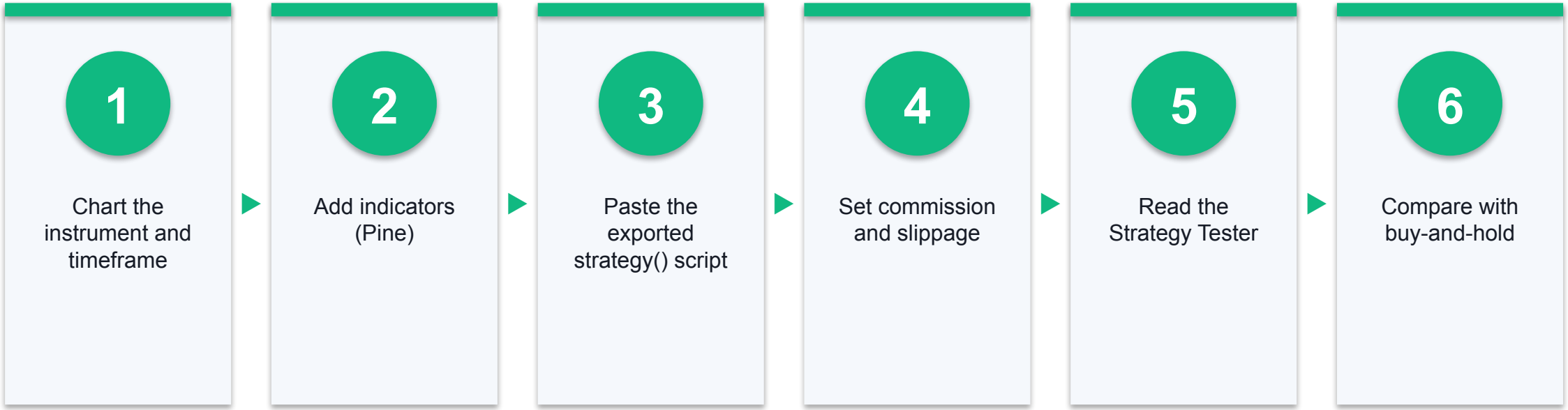


Bracket orders protect a position even if your program stops running.

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TOPIC 2

From Idea to Tested Rule on TradingView



The Strategy Tester is a first filter; the full test with costs runs in the Trading Simulator.

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TOPIC 2 · HANDS-ON LAB

Indicator Stage: Pine Script v6 Indicators and their Python Twins

LAB 05

Plot SMA/EMA, Bollinger Bands, RSI and MACD and export a Pine Script v6 strategy for TradingView.

WORKFLOW

1. Simulator: Overlay trend indicators
2. Simulator: Oscillators
3. Simulator: Export Pine Script v6

YOU'LL PRODUCE

A chart with overlays and an oscillator pane, and a Pine Script pasted into TradingView whose values match.

Duration
25 min

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LAB 05 · FOLLOW ALONG

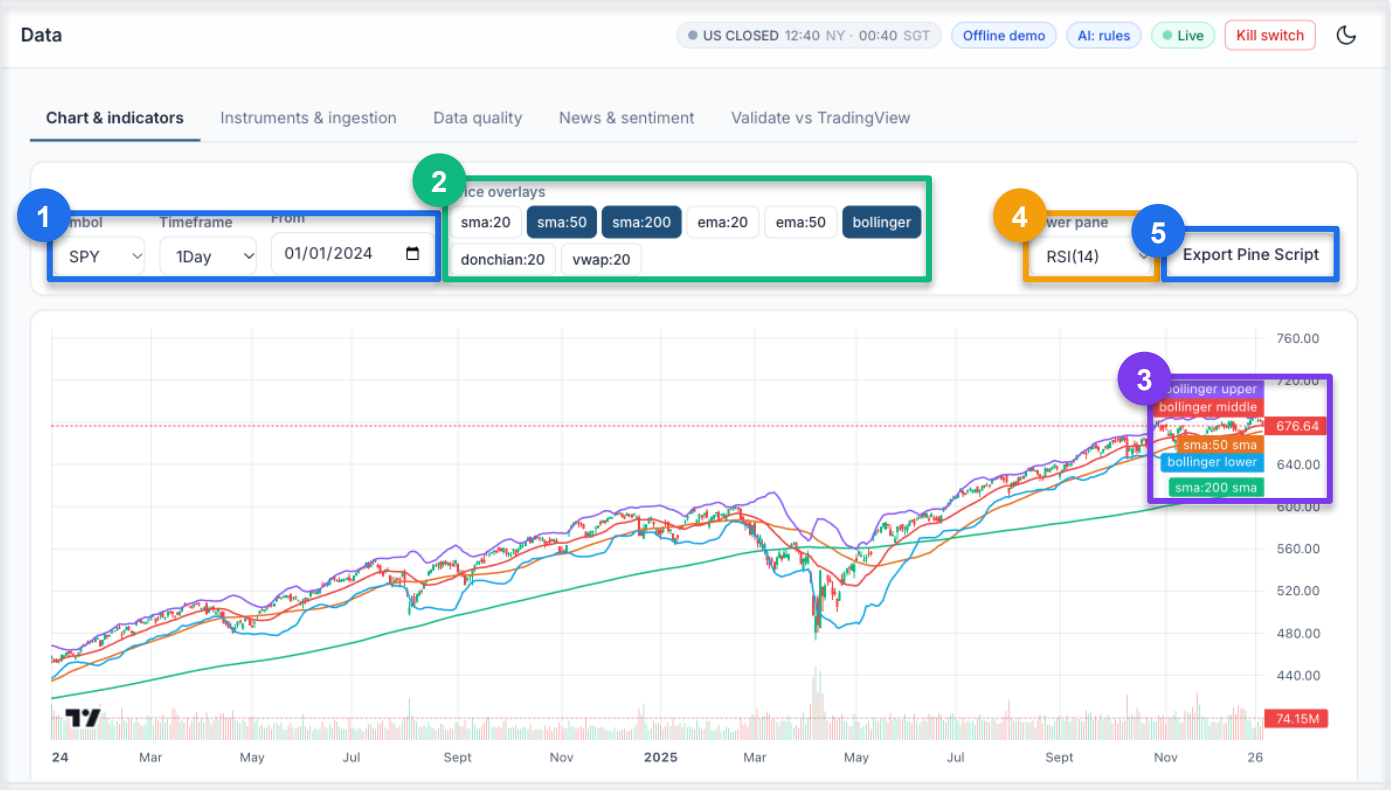
Lab 05 — Step by Step

Course Labs > Lab 05 at http://localhost:8000 - each step's Go button opens its page pre-filled. Tick each 'You should see' before moving on.

	STEP · WHERE	DO THIS	YOU SHOULD SEE
1	**Overlay trend indicators** ▸ Data > Chart SPY with SMA 50/200 and Bollinger	Data > Chart for SPY: toggle sma:50, sma:200, ema:20 and bollinger.	✓ Lines draw over the candles (TradingView Lightweight Charts).
2	**Oscillators** ▸ Data > Chart SPY MACD	Switch the lower pane between RSI(14) and MACD.	✓ RSI has 30/70 guide lines; the MACD histogram changes colour above and below 0.
3	**Export Pine Script v6** ▸ Strategies > EMA crossover	Strategy Library > EMA crossover > Pine Script v6 > Copy. In TradingView open the Pine Editor, paste, 'Add to chart'.	✓ TradingView shows the two EMAs; their last values match the app's chart.

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Lab 05 — How to Read the Screen



1 — Symbol, timeframe, from
SPY, 1Day, from 01/01/2024.

2 — Price overlays
Toggle sma:50, sma:200, ema:20 and bollinger; dark buttons are switched on.

3 — Line labels
Each overlay's latest value; the Bollinger upper, middle and lower bands wrap the price.

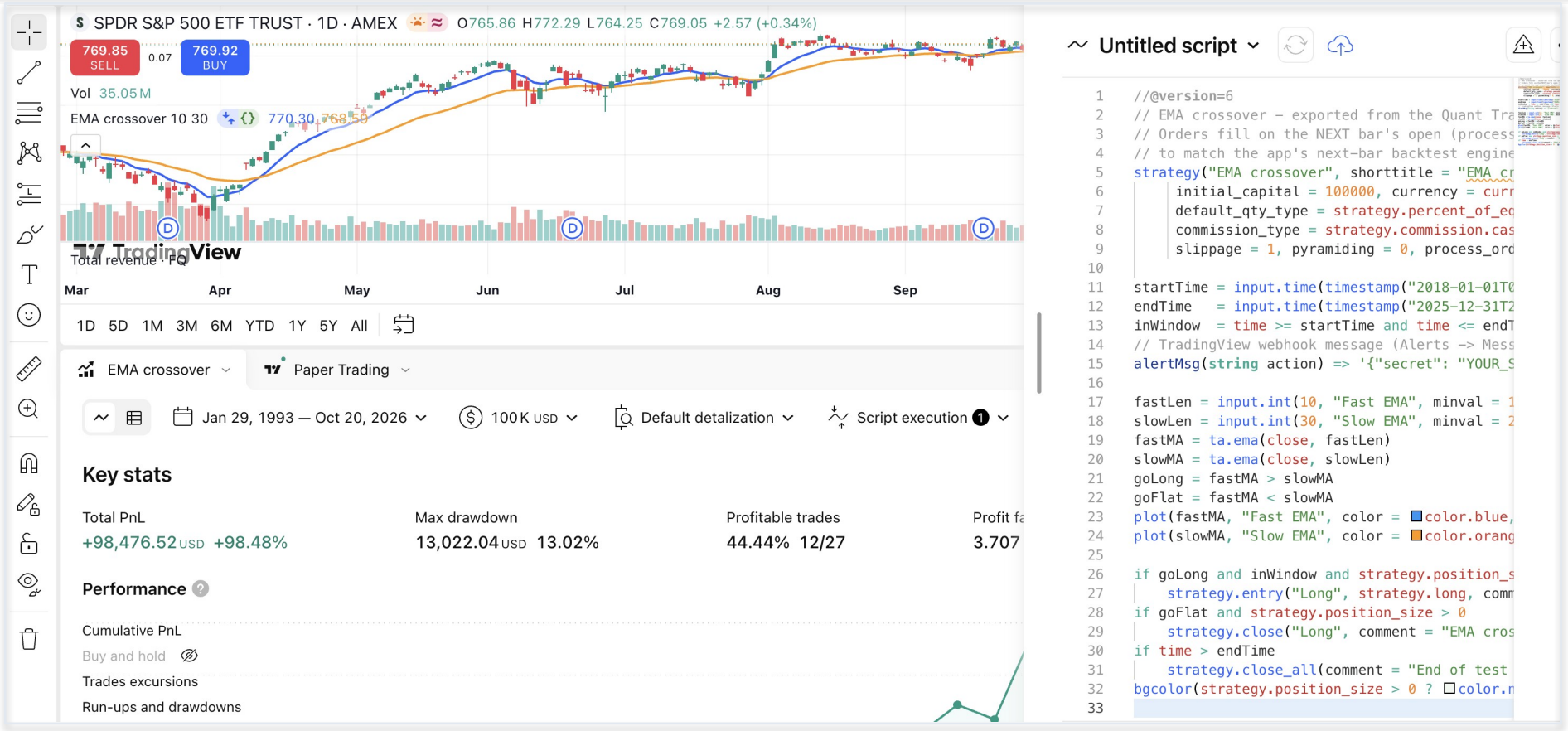
4 — Lower pane
RSI(14) with 30/70 guide lines, or MACD with its histogram above and below 0.

5 — Export Pine Script
Copy the v6 script, paste it into TradingView's Pine Editor and press 'Add to chart'.

The indicators agree when TradingView's last EMA and SMA values equal the app's line labels.

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Lab 05 — The Exported Strategy in TradingView



The EMA crossover exported from the simulator as Pine Script v6, pasted into TradingView's Pine Editor and added to the SPY chart: the Strategy Tester reports PnL, drawdown and profitable trades to compare with the simulator's backtest.

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TOPIC 2 · HANDS-ON LAB

Strategy Prototype: TradingView Strategy Tester, Paper Trading from the Chart and Pipeline Reconciliation

LAB 06

Run the same MA-crossover in TradingView's Strategy Tester and in the app, reconcile the results, and try TradingView Paper Trading.

WORKFLOW

1. Simulator: Backtest the SMA 50/200 crossover in the app
2. Simulator: Run it in TradingView
3. Simulator: Reconcile and paper trade on TradingView

YOU'LL PRODUCE

Trade counts and net profit that agree within the expected differences (fills, costs, data).

Duration
25 min

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LAB 06 · FOLLOW ALONG

Lab 06 — Step by Step

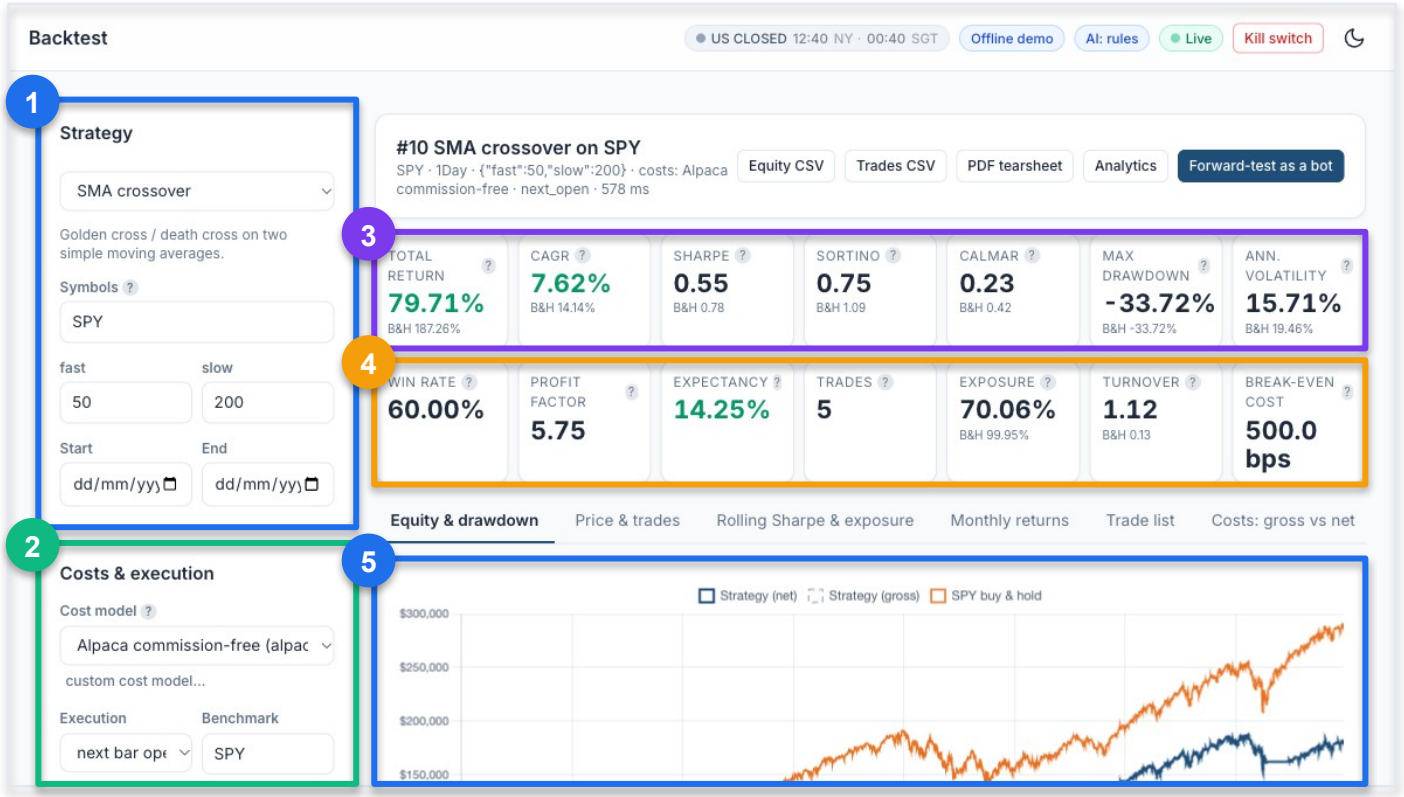
Course Labs > Lab 06 at http://localhost:8000 - each step's Go button opens its page pre-filled. Tick each 'You should see' before moving on.

	STEP · WHERE	DO THIS	YOU SHOULD SEE
1	**Backtest the SMA 50/200 crossover in the app** ▸ Backtest SPY SMA 50/200, Alpaca costs	Run the preset backtest: SMA crossover 50/200 on SPY, Alpaca-style costs, next-bar-open fills.	✓ About 5 round trips 2018-2025; note the net CAGR and Sharpe.
2	**Run it in TradingView** ▸ Strategies > SMA crossover > Pine	Export the SMA crossover Pine Script and add it to an AMEX:SPY daily chart. Open the Strategy Tester.	✓ The Strategy Tester list of trades has the same entry dates (TradingView also fills on the next bar's open).
3	**Reconcile and paper trade on TradingView** ▸ Backtest	Explain any differences (adjusted vs raw data, commission per order, slippage ticks). Optionally connect TradingView Paper Trading and place one order from the chart.	✓ You can list at least two reasons the two backtests differ.

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LAB 06 · SIMULATOR

Lab 06 — How to Read the Screen



1 — Strategy and parameters
SMA crossover on SPY, fast 50 / slow 200 - the golden / death cross.

2 — Costs and execution
Alpaca commission-free costs; fills at the next bar's open, as in TradingView.

3 — Strategy vs buy-and-hold
B&H sits under each tile: CAGR 7.62% vs 14.14%, Sharpe 0.55 vs 0.78.

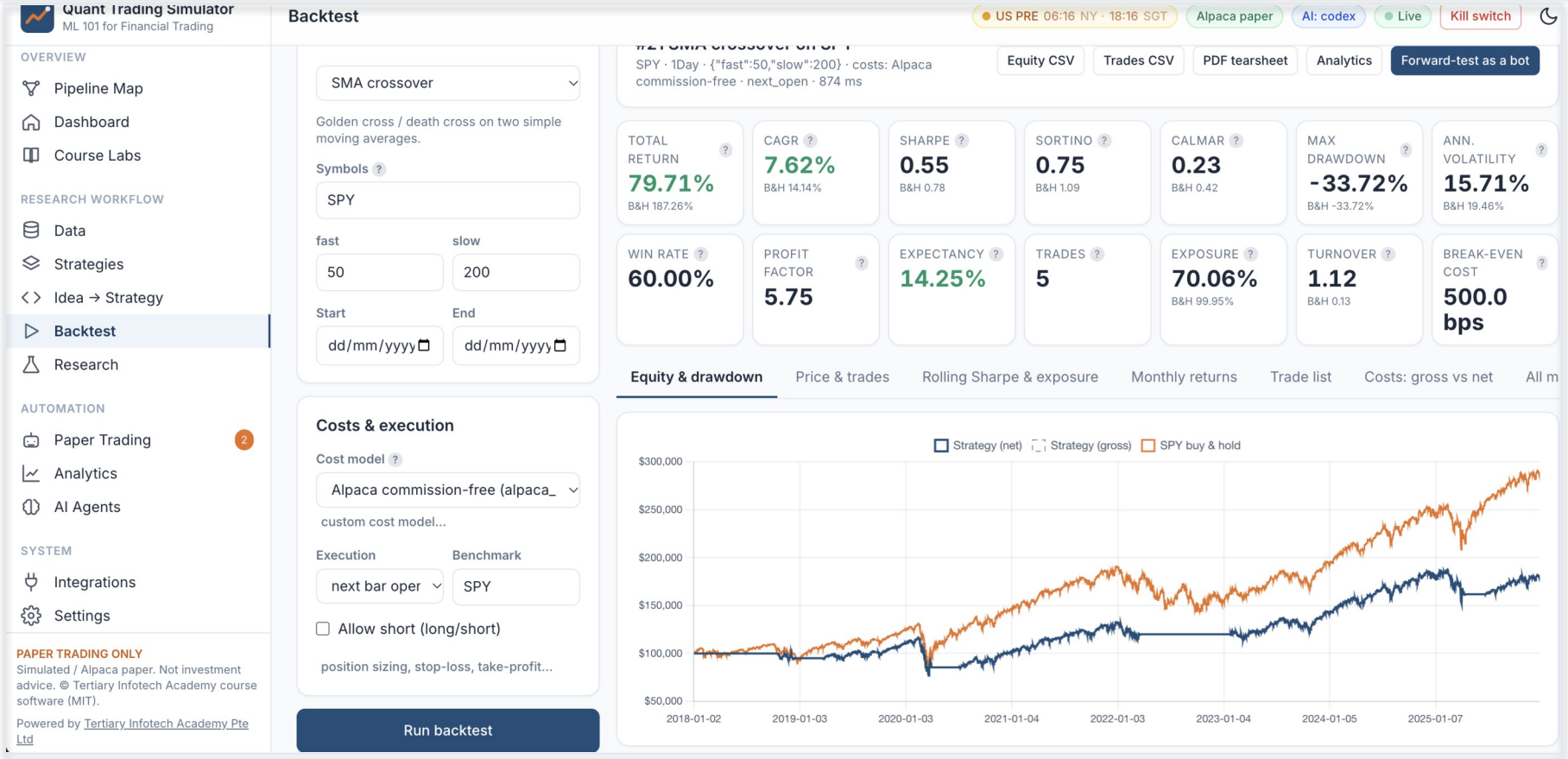
4 — Trades and break-even
Only 5 trades; break-even cost 500 bps - costs barely matter for a slow strategy.

5 — Equity curve
Net, gross and SPY. Compare the entry dates with TradingView's List of Trades.

Reconciled when the trade dates agree; explain any gap by adjusted vs raw data, commission and slippage.

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Lab 06 — The SMA 50/200 Backtest in the Simulator



Backtest: SMA crossover 50/200 on SPY with Alpaca commission-free costs and next-bar-open fills - return, CAGR, Sharpe, drawdown and trades beside buy-and-hold, ready to reconcile with TradingView's Strategy Tester.

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Tea Break

15 minutes · see you at 15:40

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TOPIC 2 · HANDS-ON LAB

Signal Stage: a Library of Trend, Mean-Reversion, Momentum and Breakout Signals

LAB 07

Pick and customise one strategy from each family and compare them after costs.

WORKFLOW

1. Simulator: Trend: EMA 20/50 on SPY
2. Simulator: Mean reversion: Bollinger on KO
3. Simulator: Momentum and breakout
4. Simulator: Read each strategy's explanation

YOU'LL PRODUCE

Four backtests with tuned parameters and an explanation of when each family works.

Duration
25 min

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LAB 07 · FOLLOW ALONG

Lab 07 — Step by Step

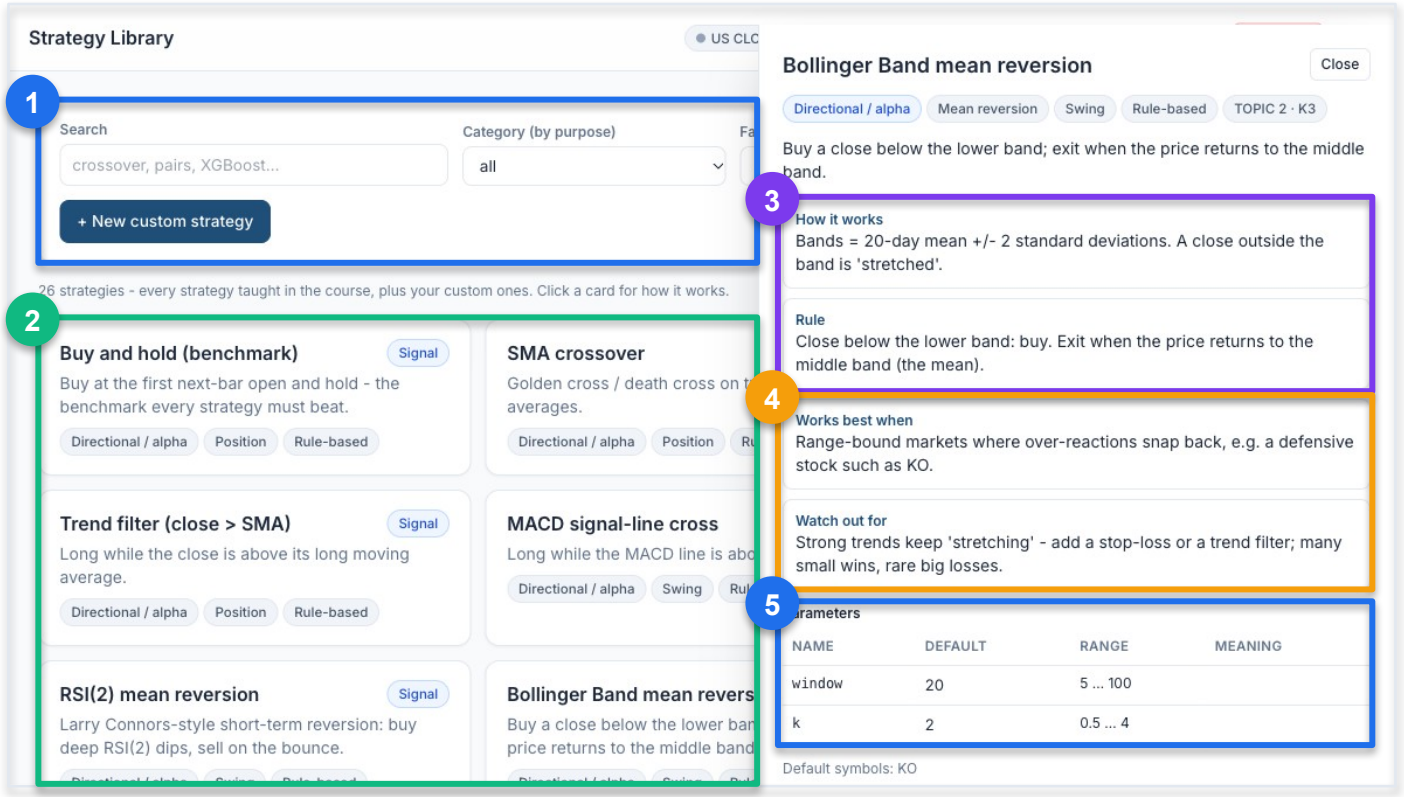
Course Labs > Lab 07 at http://localhost:8000 - each step's Go button opens its page pre-filled. Tick each 'You should see' before moving on.

	STEP · WHERE	DO THIS	YOU SHOULD SEE
1	**Trend: EMA 20/50 on SPY** ▸ Backtest SPY EMA 20/50, Alpaca costs	Run the preset.	✓ Note Sharpe and max drawdown vs buy-and-hold.
2	**Mean reversion: Bollinger on KO** ▸ Backtest KO Bollinger 20/2	Run Bollinger Band mean reversion on KO (window 20, k 2.0), then change k to 1.5 and run it again. Set Benchmark to KO so the buy-and-hold figures under each metric are KO's, not SPY's.	✓ Win rate is high, but losses are larger than wins - 'many small wins, rare big losses'.
3	**Momentum and breakout** ▸ Backtest QQQ Donchian 20/10	On the Backtest page choose Time-series momentum, look-back 126, symbol QQQ, Benchmark QQQ, and run it. Then run the Donchian breakout on QQQ (entry 20, exit 10).	✓ Open the 'Price & trades' tab: buy arrows follow 20-day highs.
4	**Read each strategy's explanation** ▸ Strategy Library	Strategy Library: open each card and read How it works / Rule / Works best when / Watch out for.	✓ You can say which family suits a trending market and which a range-bound one.

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LAB 07 · SIMULATOR

Lab 07 — How to Read the Screen



1. **Search and filter**
   Find a strategy by name or by purpose; 26 strategies in the library.

2. **Strategy cards**
   Trend, mean reversion, momentum, breakout - click a card to open it.

3. **How it works and the rule**
   Bollinger: buy a close below the lower band, exit back at the middle band.

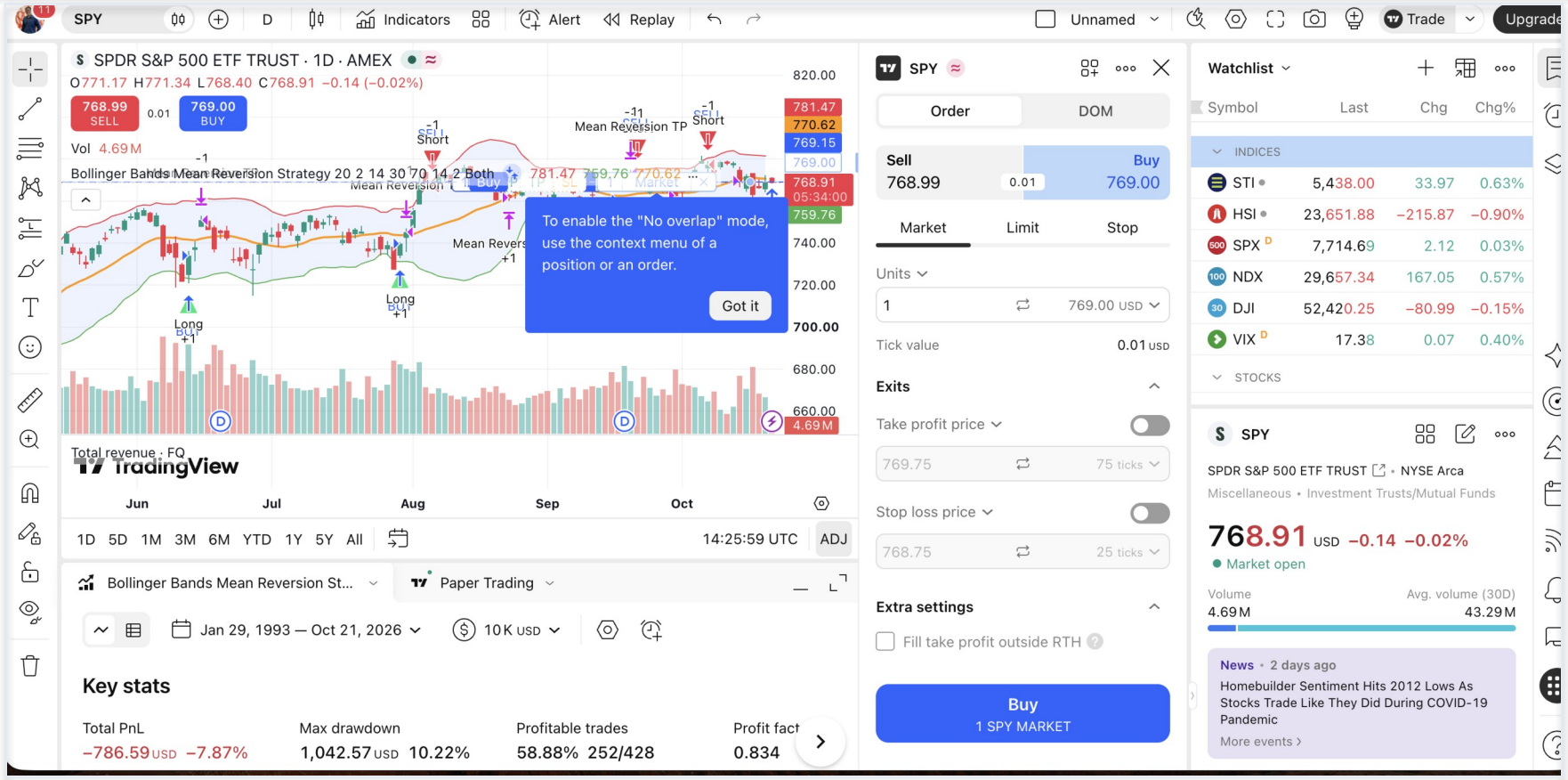
4. **When it works / watch out**
   Range-bound stocks such as KO; strong trends keep 'stretching' - add a stop.

5. **Parameters**
   window 20 and k 2 with their ranges - the values you change in the backtest.

Trend and breakout suit trending markets, mean reversion suits range-bound ones - compare all four after costs.

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Lab 07 — Mean Reversion in TradingView, with Paper Trading



Bollinger Bands mean reversion (20, 2) on SPY with TradingView Paper Trading's order panel open. It lost 7.87% since 1993 though 59% of trades won: buying dips in a trending index hurts, so the lab runs it on KO.

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TOPIC 2 · HANDS-ON LAB

Execution Stage: an Order Manager that Trades the Paper Account in Any Session

LAB 08

Place market, limit, stop and bracket paper orders, including an extended-hours limit order, and follow each order's lifecycle.

WORKFLOW

1. Simulator: Market order in the regular session
2. Simulator: Extended-hours limit order (Singapore class hours)
3. Simulator: Bracket order
4. Simulator: Queue while closed and cancel

YOU'LL PRODUCE

A blotter with filled, working, queued and cancelled orders and the risk checks for each.

Duration
25 min

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LAB 08 · FOLLOW ALONG

Lab 08 — Step by Step

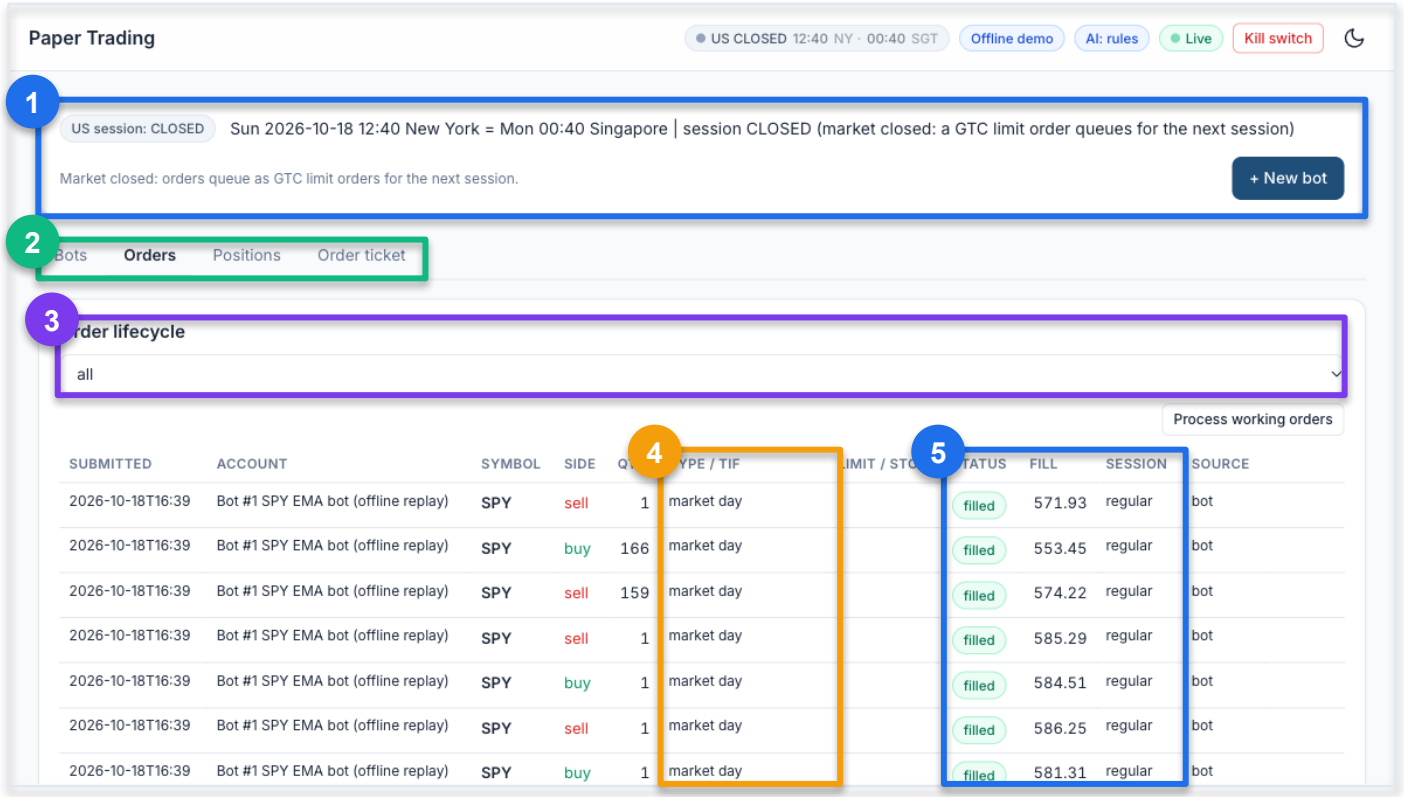
Course Labs > Lab 08 at http://localhost:8000 - each step's Go button opens its page pre-filled. Tick each 'You should see' before moving on.

	STEP · WHERE	DO THIS	YOU SHOULD SEE
1	**Market order in the regular session** ▸ Paper Trading > Order ticket	Open Paper Trading > Order ticket, fill in the fields below and press Send paper order. A market order buys straight away at the current price; it is accepted in the regular session.	✓ The Result panel shows ACCEPTED - all checks passed, 'BUY 10 SPY MARKET DAY -> filled', and each risk check (kill_switch, duplicate, session, allow_list, long_only, daily_loss.
2	**Extended-hours limit order (Singapore class hours)** ▸ same page	Keep the Step 1 ticket and change only the fields below, then press Send paper order.	✓ First send: ACCEPTED, 'BUY 10 SPY LIMIT DAY extended_hours limit <price> -> filled'. the Orders tab shows 'limit day ext-hrs', session overnight.
3	**Bracket order** ▸ same page	A bracket order is an entry plus two automatic exits: a take-profit (a sell limit above the price, to lock in a gain) and a stop-loss (a sell stop below it, to cap a loss).	✓ Orders tab: the parent 'buy 5 SPY market day bracket' is filled, and two sell legs appear with status new - 'limit gtc bracket (leg)' at the take-profit and 'stop gtc bracket (leg)' at the.
4	**Queue while closed and cancel** ▸ same page	Session (sim) = closed pretends the US market is shut (weekends and US holidays).	✓ Result: ACCEPTED, 'BUY 10 SPY LIMIT GTC limit <price> -> accepted', and the session check reads 'market closed - GTC limit order will queue for the next session'.

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LAB 08 · SIMULATOR

Lab 08 — How to Read the Screen



1 Session banner
New York vs Singapore time and the US session - it decides which orders are accepted.

2 Tabs
Order ticket to send, Orders for the blotter, then Positions and Bots.

3 Order lifecycle filter
Show all orders, or only the filled, working, queued or cancelled ones.

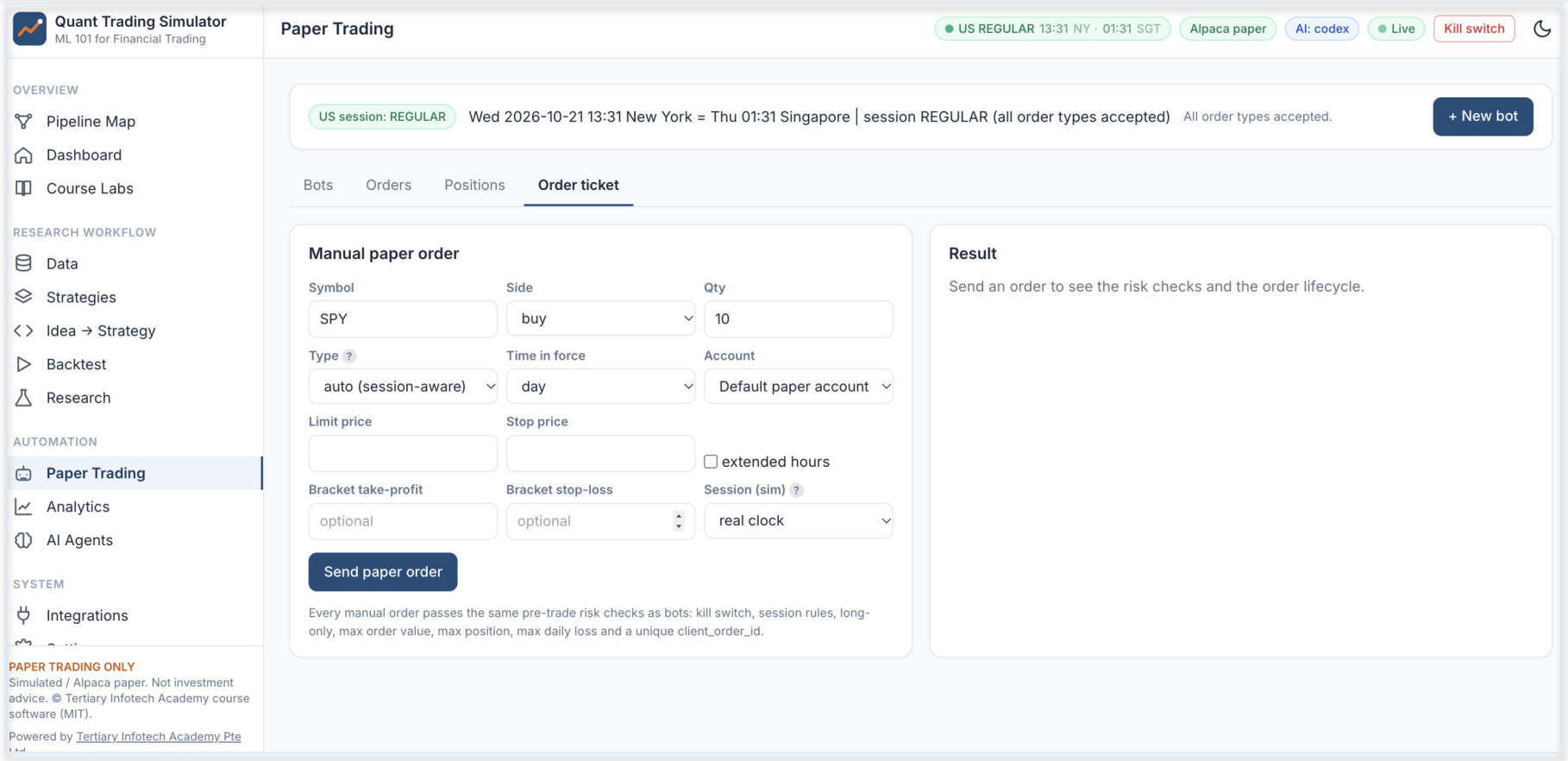
4 Type / TIF
market day, limit day ext-hrs, limit gtc, bracket legs - what you sent.

5 Status, fill and session
filled, accepted or canceled; click a status badge to see the order's lifecycle.

Every order lists the risk checks it passed. Overnight, only LIMIT orders with extended hours are accepted.

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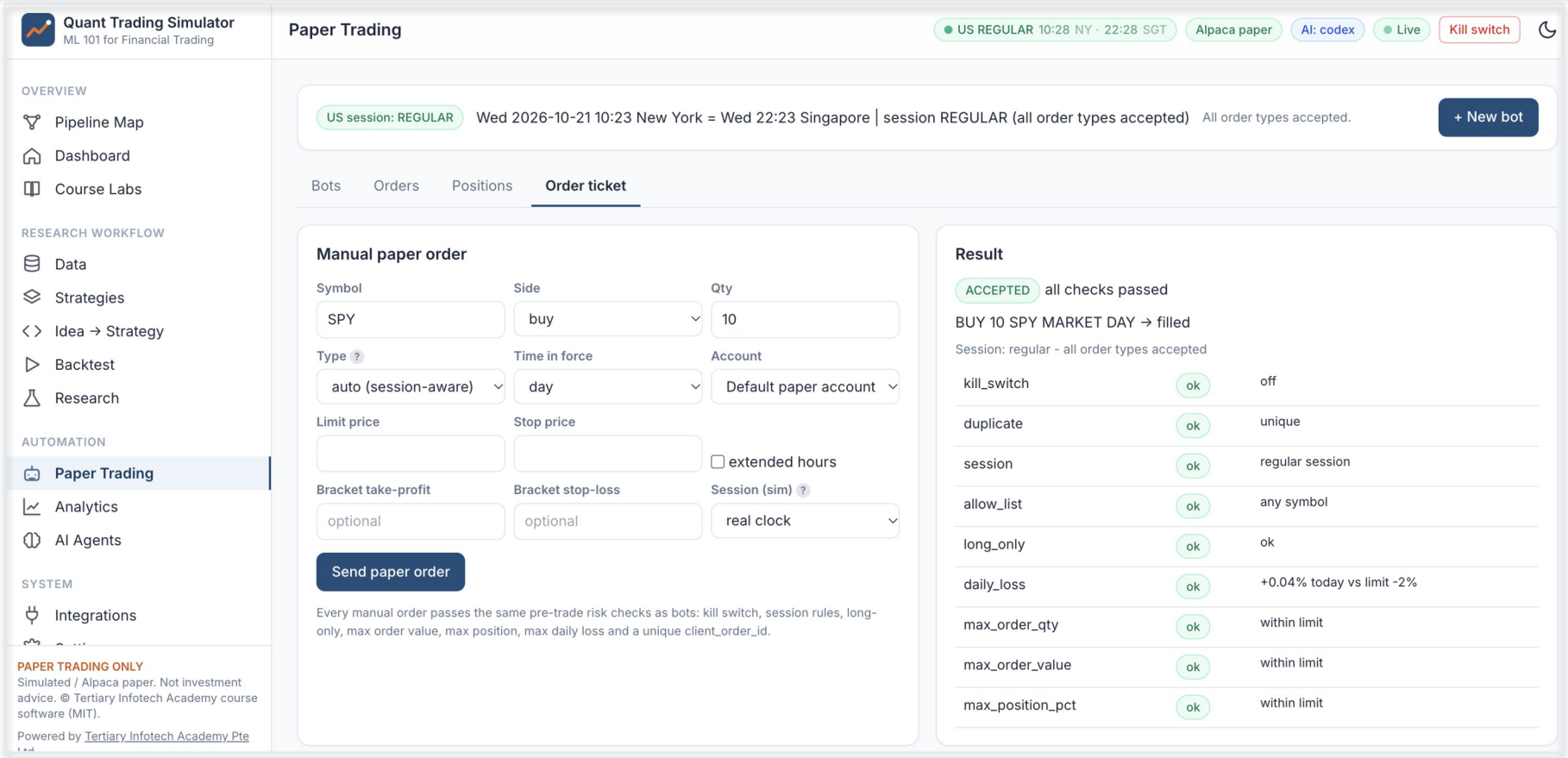
Lab 08 — The Paper Order Ticket



Paper Trading > Order ticket: buy 10 SPY, time in force day, default paper account. For Step 1 set Type = market and Session (sim) = regular; the banner shows the live session (here REGULAR, so all order types are accepted).

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Lab 08 — Risk Checks Passed, Order Filled



After Send paper order: ACCEPTED, all checks passed - BUY 10 SPY MARKET DAY -> filled. Every manual order passes the same pre-trade checks as the bots: kill switch, duplicate, session, allow list, long only, daily loss, max order quantity and value, max position.

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TOPIC 2 · HANDS-ON LAB

Evaluation Stage: Performance and Risk Metrics with a Benchmark Table

LAB 09

Read the full metrics table and compare a strategy with buy-and-hold.

WORKFLOW

1. Simulator: Open the analytics
2. Simulator: Compare with buy-and-hold
3. Simulator: Glossary

YOU'LL PRODUCE

A metrics table you can explain line by line.

Duration
25 min

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LAB 09 · FOLLOW ALONG

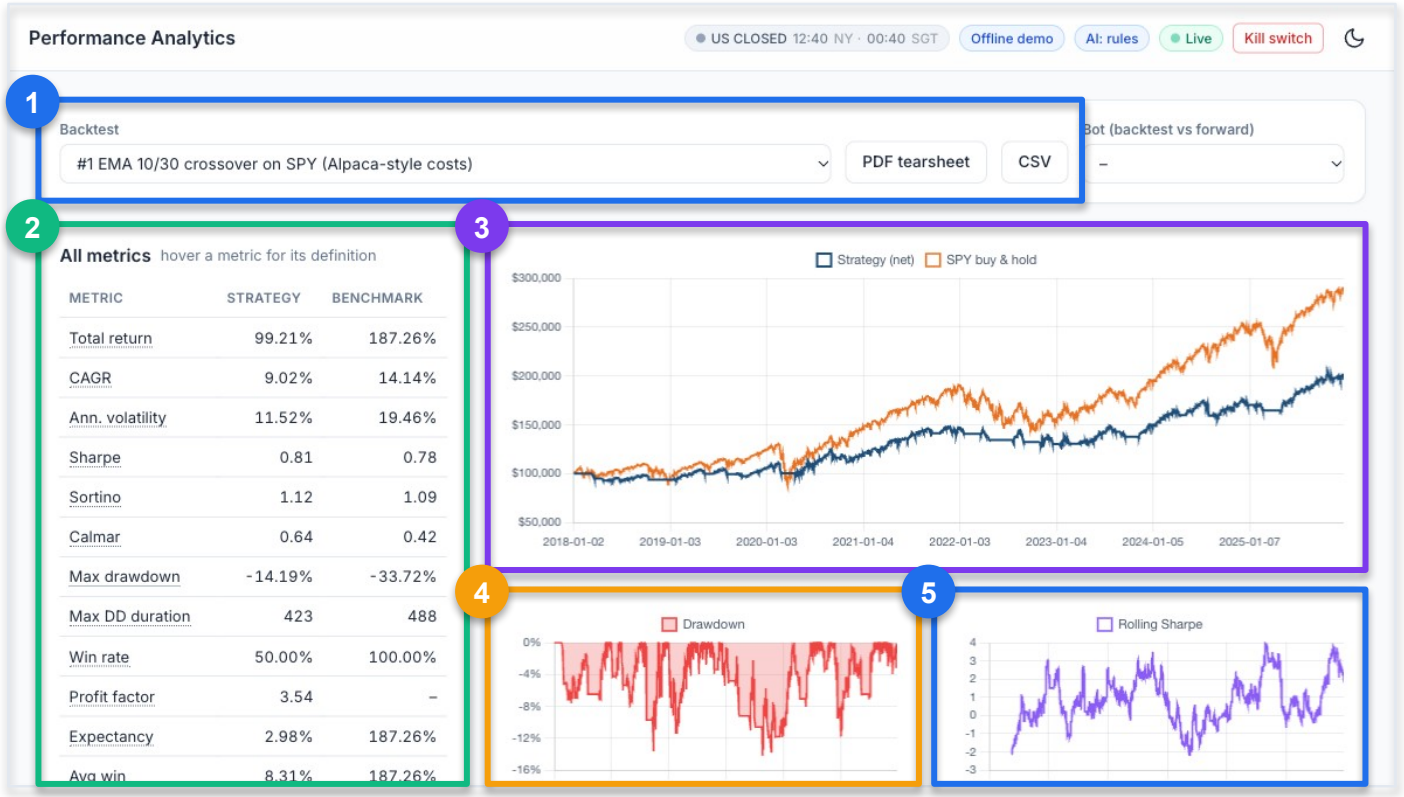
Lab 09 — Step by Step

Course Labs > Lab 09 at http://localhost:8000 - each step's Go button opens its page pre-filled. Tick each 'You should see' before moving on.

	STEP · WHERE	DO THIS	YOU SHOULD SEE
1	**Open the analytics** ▸ Analytics	Analytics: choose the EMA 10/30 seeded backtest. Hover every metric for its formula.	✓ You can define Sharpe, Sortino, Calmar, max drawdown, profit factor, expectancy, exposure and turnover.
2	**Compare with buy-and-hold** ▸ same page	Tick the EMA backtest and 'Buy and hold SPY' and press Compare.	✓ Buy-and-hold has higher CAGR but a deeper drawdown; the strategy has lower exposure.
3	**Glossary** ▸ Help & Glossary	Open Help & Glossary > Metric glossary.	✓ Each formula matches the course slides.

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Lab 09 — How to Read the Screen



1 Pick a backtest
EMA 10/30 on SPY; export a PDF tearsheet or CSV from here.

2 All metrics
Strategy vs benchmark; hover a metric for its formula. Sharpe 0.81 vs 0.78.

3 Equity vs buy-and-hold
SPY ends higher (CAGR 14.14% vs 9.02%) but swings much more.

4 Drawdown
Max drawdown -14.19% vs SPY's -33.72%: the strategy steps aside in sell-offs.

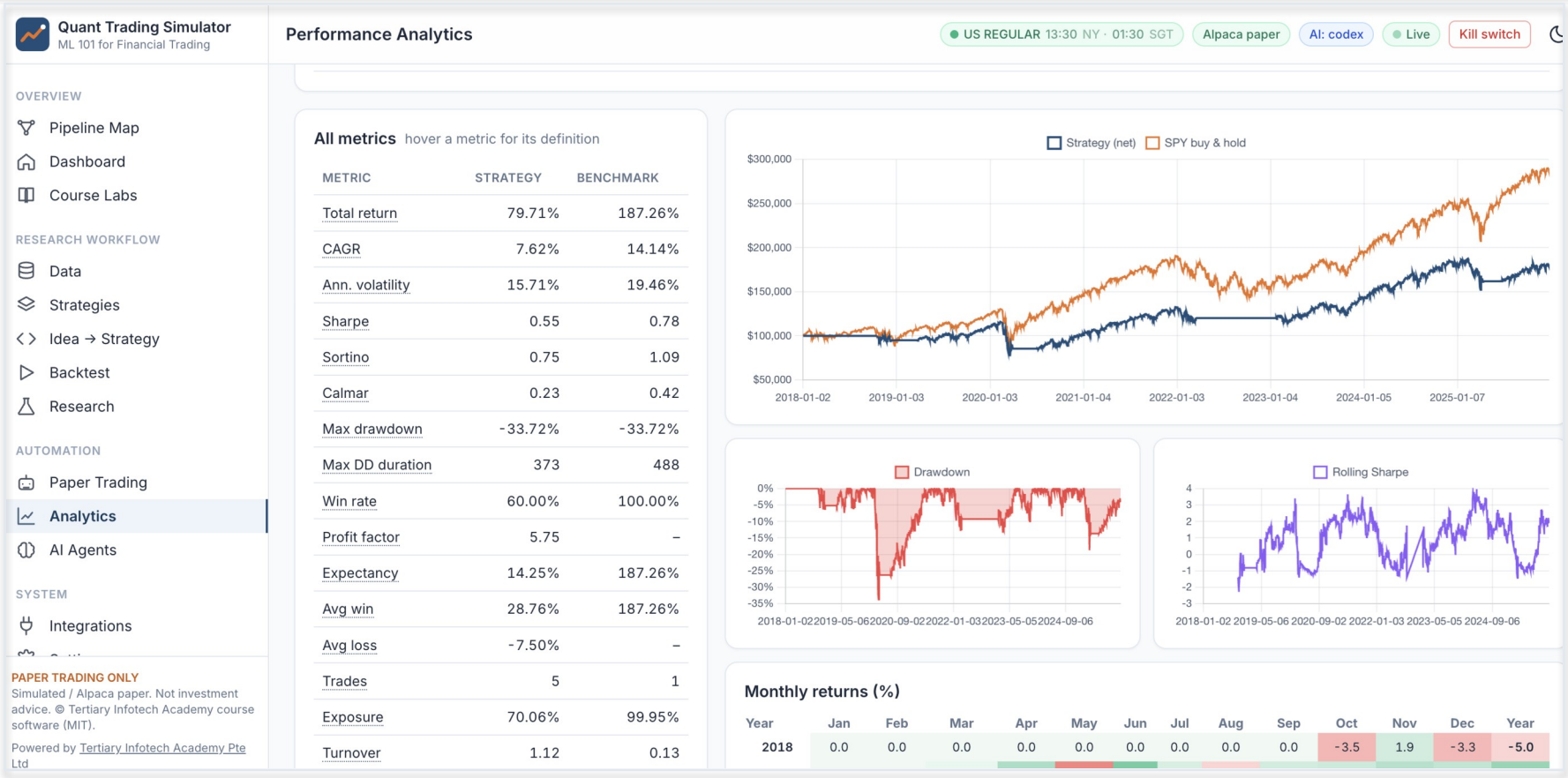
5 Rolling Sharpe
Risk-adjusted return over time; long negative spells flag a fading edge.

Lower return but far lower risk: judge a strategy on Sharpe, drawdown and exposure, not CAGR alone.

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LAB 09 · SIMULATOR

Lab 09 — Strategy versus Benchmark in Analytics



Analytics for Lab 6's SMA 50/200 backtest against SPY buy-and-hold: 79.71% versus 187.26% total return, Sharpe 0.55 versus 0.78. In Lab 9 choose the EMA 10/30 backtest; hover any metric for its definition.

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Topic 2 Recap

1. Identify appropriate statistical algorithms and data models to test hypotheses and/or theories
2. Use appropriate analytics platforms, analytical tools, statistical methods and analytical approaches given specific analytics and reporting requirements
3. Range and application of various statistical methods and algorithms
4. Usage of analytics platforms and tools
5. Coding languages for programming of algorithms and signals

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End of Day 1

See you at 09:30 tomorrow

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TOPIC 3

03

ML Models and A/B Testing

LEARNING OUTCOME

Conduct statistical modelling of market data with machine learning models to derive patterns and solutions, and perform A/B testing of trading strategies based on different hypotheses.

Day 2 · Labs 10–14

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TOPIC 3

Types of Statistical Analyses

1

Descriptive

What happened — mean return, volatility, skew, correlation.

2

Inferential

Is it real? — t-test, bootstrap, confidence intervals.

3

Predictive

What next? — regression, classification.

4

Prescriptive

What to do? — optimisation, position sizing.

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TOPIC 3

Families of Learning Algorithms

1

Supervised

Features → known target: up/down, next return.

2

Unsupervised

Structure without a target: regimes, PCA factors.

3

Reinforcement

Learn a policy from rewards — execution research.

4

Search & simulation

Grid search, Monte Carlo, bootstrap, walk-forward.

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TOPIC 3

Supervised Learning: Learning from Labelled Examples

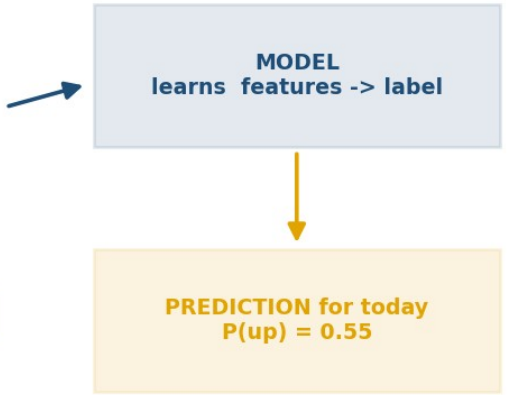
Supervised learning on SPY: learn from labelled history, predict the unknown label

FEATURES (known today)

date	5-day ret	RSI(14)	daily vol (21-day)	LABEL: next day
19 Dec 2025	+0.1%	54	0.76%	UP
22 Dec 2025	+0.9%	58	0.66%	UP
23 Dec 2025	+1.6%	60	0.64%	UP
24 Dec 2025	+3.1%	62	0.58%	DOWN
26 Dec 2025	+2.3%	62	0.55%	DOWN
29 Dec 2025	+1.1%	59	0.54%	DOWN
30 Dec 2025	+0.3%	58	0.53%	?

past rows: answer known -> TRAIN

last row = 'today': answer unknown -> PREDICT



DEFINITION

The model learns a mapping from inputs (features) to a known answer (the label), using history in which the answer is already known.

IN TRADING

Features known at today's close - 5-day return, RSI, volatility. Label: did the price rise the NEXT day (classification) or by how much (regression)?

WHAT IT GIVES YOU

A prediction for a new day - P(up) = 0.55 or an expected return. A rule such as 'long if P(up) > 0.55' turns it into a position.

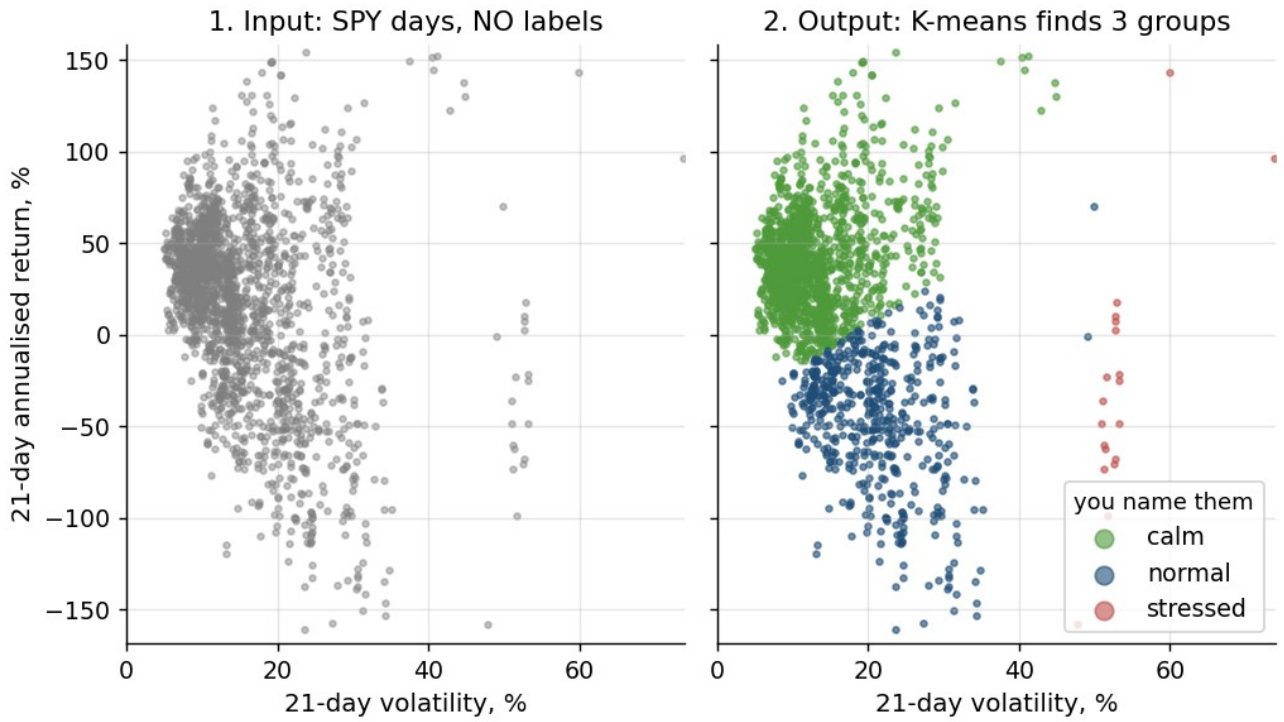
WATCH OUT FOR

Features must use only data known at decision time. Accuracy near 50% is normal for daily direction - judge the strategy net of costs.

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TOPIC 3

Unsupervised Learning: Finding Structure Without Labels



DEFINITION

There is no answer key. The algorithm groups or compresses the data by similarity, and you decide what the groups mean.

IN TRADING

Cluster days by return and volatility into regimes; PCA reduces many correlated assets to a few factors; cluster stocks for diversification.

WHAT IT GIVES YOU

A group per day or asset (calm / normal / stressed) or a few factors - inputs to a decision, such as cutting position size when stressed, not trades by themselves.

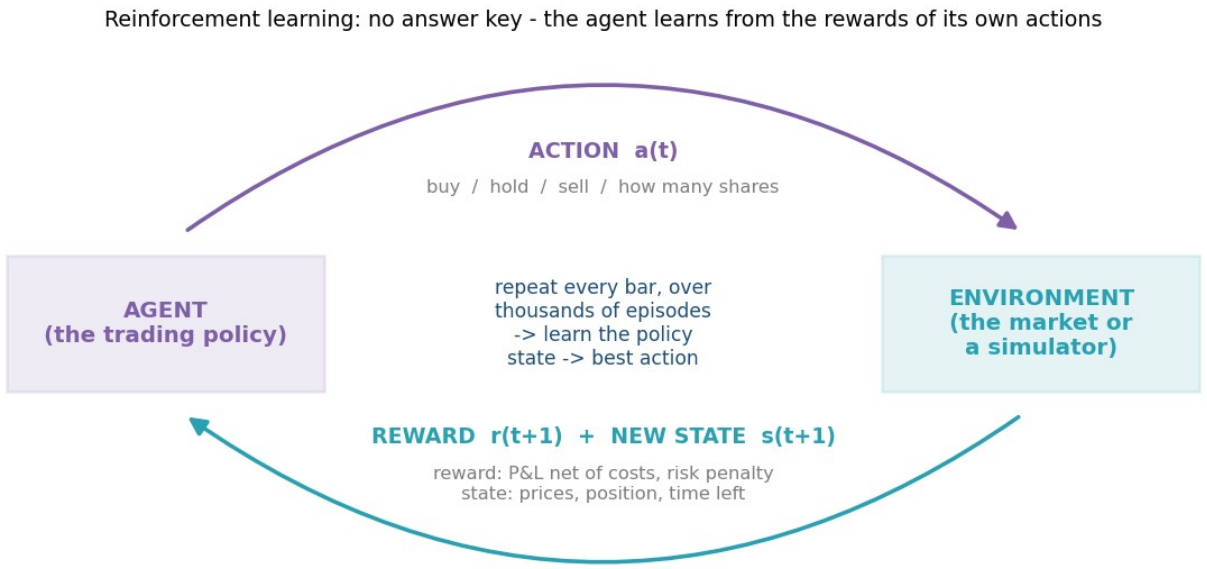
WATCH OUT FOR

Clusters have no built-in meaning and k is your choice. Fit on past data only and re-check as the market changes.

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TOPIC 3

Reinforcement Learning: Learning by Action and Reward



DEFINITION

An agent observes a state, takes an action and receives a reward; by trial and error it learns a policy that maximises total reward over time.

IN TRADING

State: prices, position, time left. Action: buy, hold, sell or how many shares. Reward: P&L net of costs, minus a penalty for risk.

WHAT IT GIVES YOU

A policy - the best action for each state. Used for optimal execution of large orders, market making and dynamic hedging, mostly at banks and research labs.

WATCH OUT FOR

Needs a realistic simulator and huge amounts of experience; it learns to exploit any flaw in the simulator. Unstable and hard to explain - rarely the first tool.

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TOPIC 3

Supervised vs Unsupervised vs Reinforcement Learning

	Supervised	Unsupervised	Reinforcement
Learns from	Labelled history: features + the known answer	Unlabelled data: features only	Its own experience: states, actions, rewards
Question it answers	What will happen? - predict a target	What patterns exist? - group or compress	What should I do? - choose the next action
Feedback	Error against the correct answer	None - judged by how useful the groups are	A reward, often delayed (profit, cost)
Typical algorithms	Logistic regression, random forest, XGBoost, LSTM	K-means, hierarchical clustering, PCA	Q-learning, policy gradients (PPO)
Trading example	P(SPY up tomorrow) from RSI and recent returns	Calm / normal / stressed market regimes	An agent slicing a 100,000-share order to cut cost
Main risk	Overfitting noise; look-ahead leakage	Meaningless or unstable clusters	Exploiting simulator flaws; unstable training

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TOPIC 3

Supervised Models for Trading Signals

1

Linear / logistic regression

Interpretable baselines; P(up) from a weighted sum.

2

Decision tree & random forest

Threshold splits; averaged trees reduce variance.

3

XGBoost

Sequential boosted trees — strongest on tabular features.

4

Neural networks

MLP, LSTM, Transformer — data-hungry, easy to overfit.

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TOPIC 3

Unsupervised Models for Regimes

1

K-means

Minimise within-cluster variance; choose k by elbow or silhouette.

2

Hierarchical clustering

Dendrogram of similar assets for diversification.

3

PCA

First component ≈ the market; yield curve = level, slope, curvature.

4

Fit on the past only

Assign new days to regimes as they arrive.

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TOPIC 3

Statistical Modelling Techniques

1. **Returns, not prices**
   Closer to stationary; avoids spurious regression.

2. **Time-ordered validation**
   TimeSeriesSplit and walk-forward — never shuffle.

3. **Leak-free features**
   Lags and rolling windows known at decision time.

4. **Regularise**
   L1/L2, depth limits, early stopping, fewer features.

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TOPIC 3

A/B Testing Two Strategy Variants

1

State H0 and H1 first

H0: equal mean returns; H1: B is higher. α = 5%.

2

Welch's t-test

Compare mean daily returns with unequal variances.

3

Bootstrap the Sharpe difference

Resample returns; 95% CI excluding 0?

4

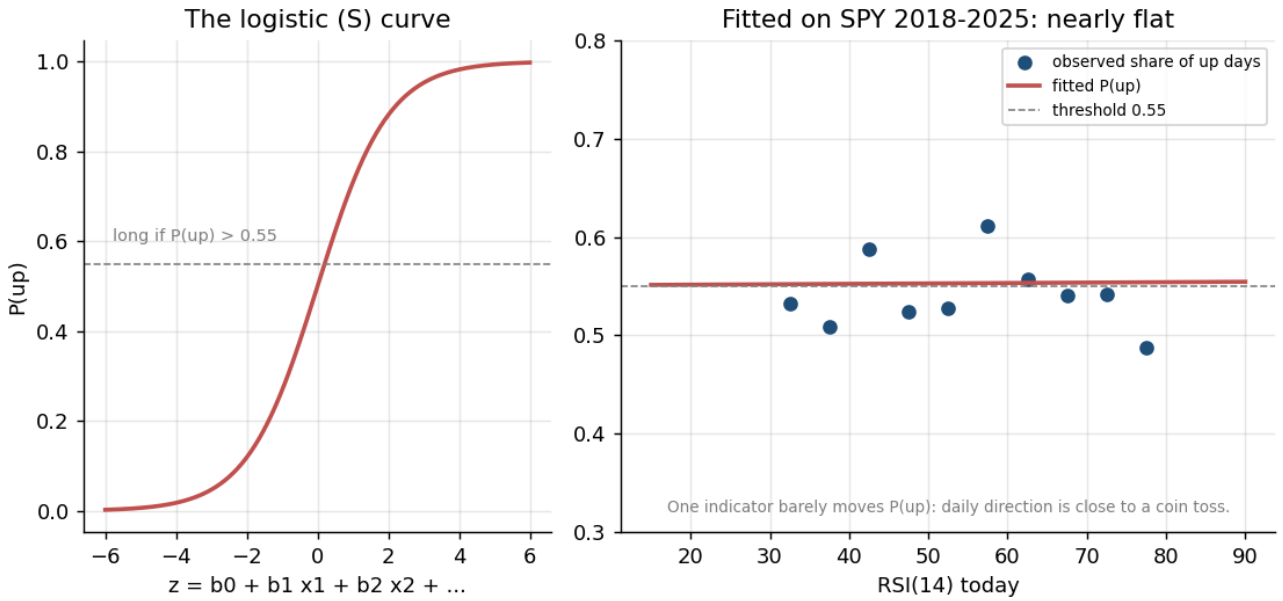
Mind multiple testing

100 variants → ~5 false 'winners' by chance.

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TOPIC 3

Algorithm: Logistic Regression for Direction



HOW IT WORKS

Fits P(up tomorrow) = 1 / (1 + e^-(b0 + b1 x1 + ...)) from features known today.

RULE

Go long when P(up) exceeds a threshold such as 0.55; otherwise stay flat.

WORKS BEST WHEN

A transparent baseline: each coefficient shows how a feature moves the probability.

WATCH OUT FOR

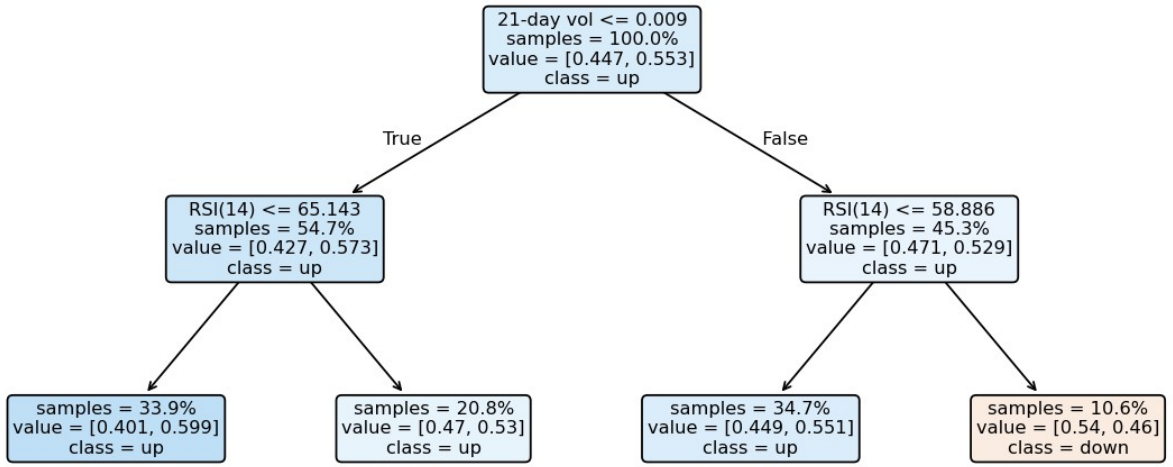
Only linear effects unless you engineer features; daily direction is close to a coin toss.

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TOPIC 3

Algorithm: Decision Tree and Random Forest

Decision tree (depth 2) on SPY: split on feature thresholds; a random forest averages many such trees



HOW IT WORKS

A tree splits the data on feature thresholds (e.g. RSI < 30) into leaves with an up/down share.

RULE

A random forest averages hundreds of trees, each grown on a random sample of rows and features.

WORKS BEST WHEN

Non-linear, interacting signals - e.g. oversold AND low volatility.

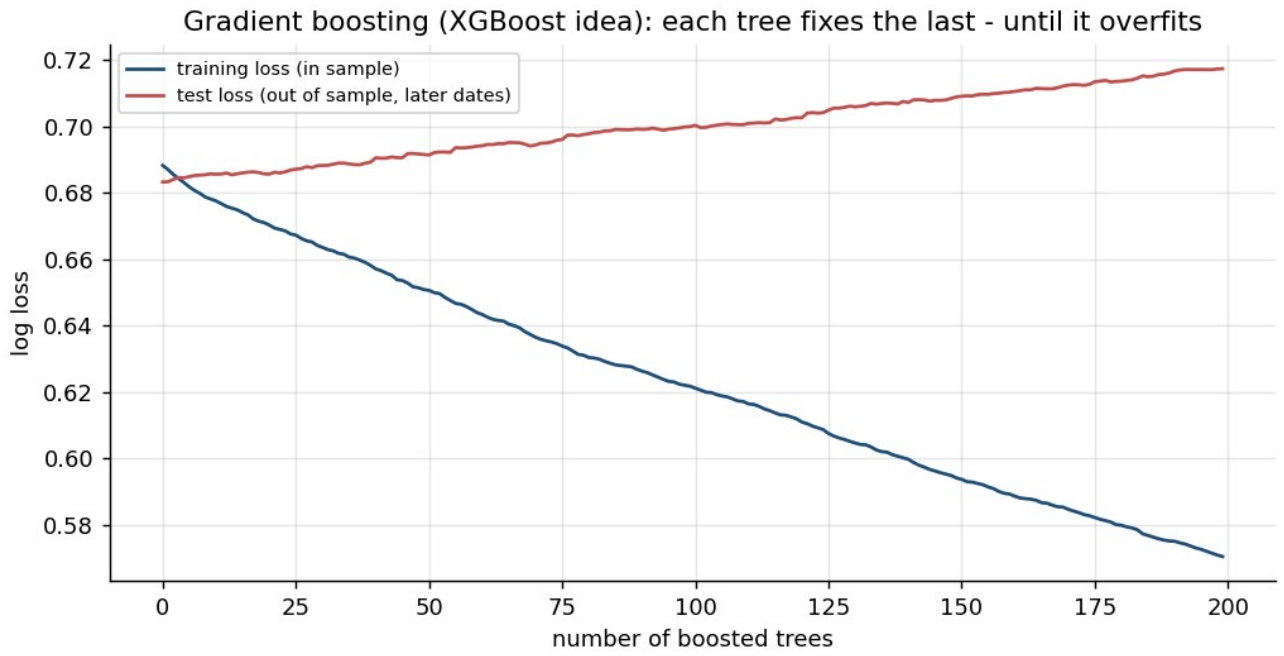
WATCH OUT FOR

Deep trees memorise noise; limit depth and leaf size and validate walk-forward.

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TOPIC 3

Algorithm: Gradient Boosting (XGBoost)



HOW IT WORKS

Builds trees one after another; each new tree corrects the errors of the ensemble so far.

RULE

Prediction = sum of many small trees x learning rate; stop early when test loss stops falling.

WORKS BEST WHEN

Tabular features such as returns, indicators and volatility - usually the strongest model.

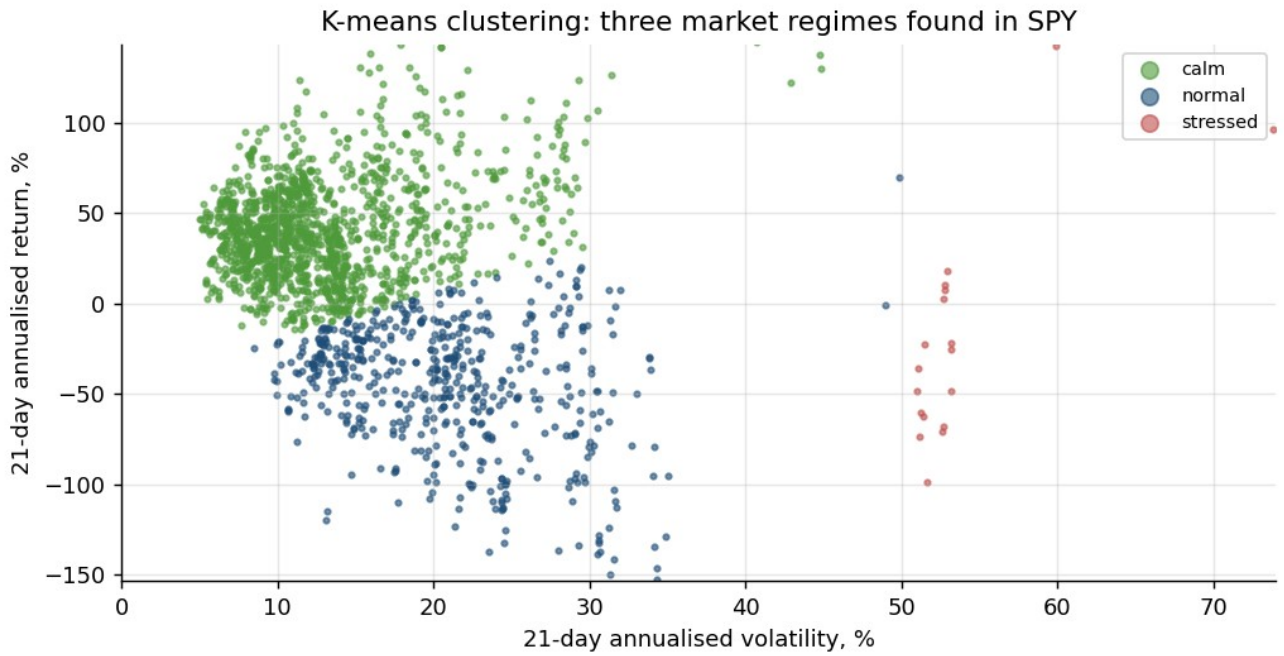
WATCH OUT FOR

Training loss keeps falling while test loss rises: classic overfitting (see the chart).

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TOPIC 3

Algorithm: K-Means Clustering for Market Regimes



HOW IT WORKS

Groups days into k clusters by minimising the distance of each day to its cluster centre.

RULE

Features: rolling return and volatility (standardised). Name clusters by their averages.

WORKS BEST WHEN

Switching or sizing strategies by regime: trend in calm markets, cut risk when stressed.

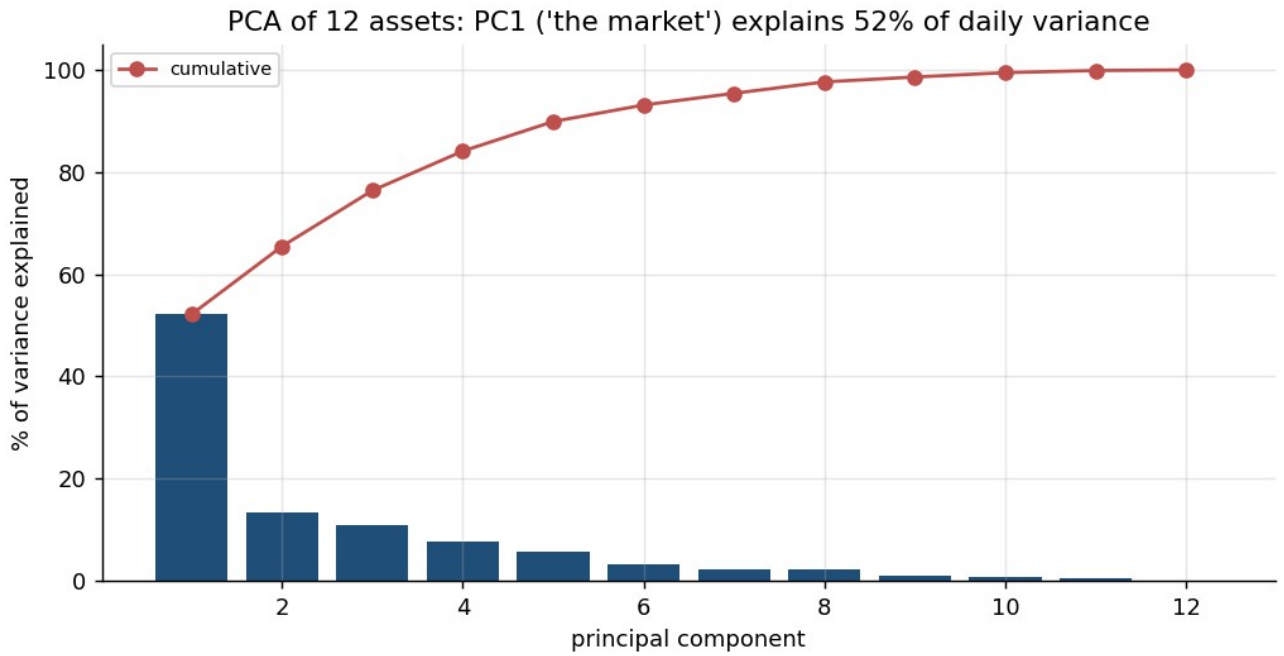
WATCH OUT FOR

Fit on past data only; choose k with the elbow or silhouette method; regimes shift.

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TOPIC 3

Algorithm: Principal Component Analysis (PCA)



HOW IT WORKS

Finds uncorrelated combinations of assets that explain the most variance of returns.

RULE

PC1 of a stock universe is 'the market'; later PCs are sector or style factors.

WORKS BEST WHEN

Reducing many correlated features; building factor-neutral and pairs portfolios.

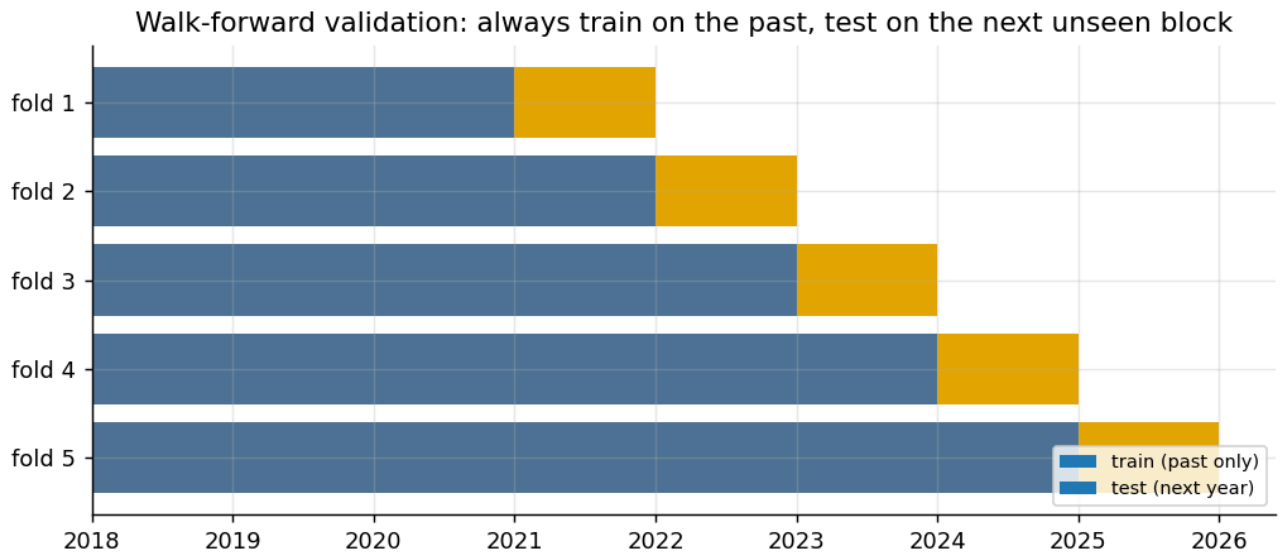
WATCH OUT FOR

Standardise first; components change over time; PCs are statistical, not economic, factors.

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TOPIC 3

Method: Walk-Forward Validation



HOW IT WORKS

Train on a past window, test on the next unseen block, then roll forward and repeat.

RULE

Never shuffle time series; each test block is genuinely out of sample.

WORKS BEST WHEN

Estimating how a model or parameter set would really have performed live.

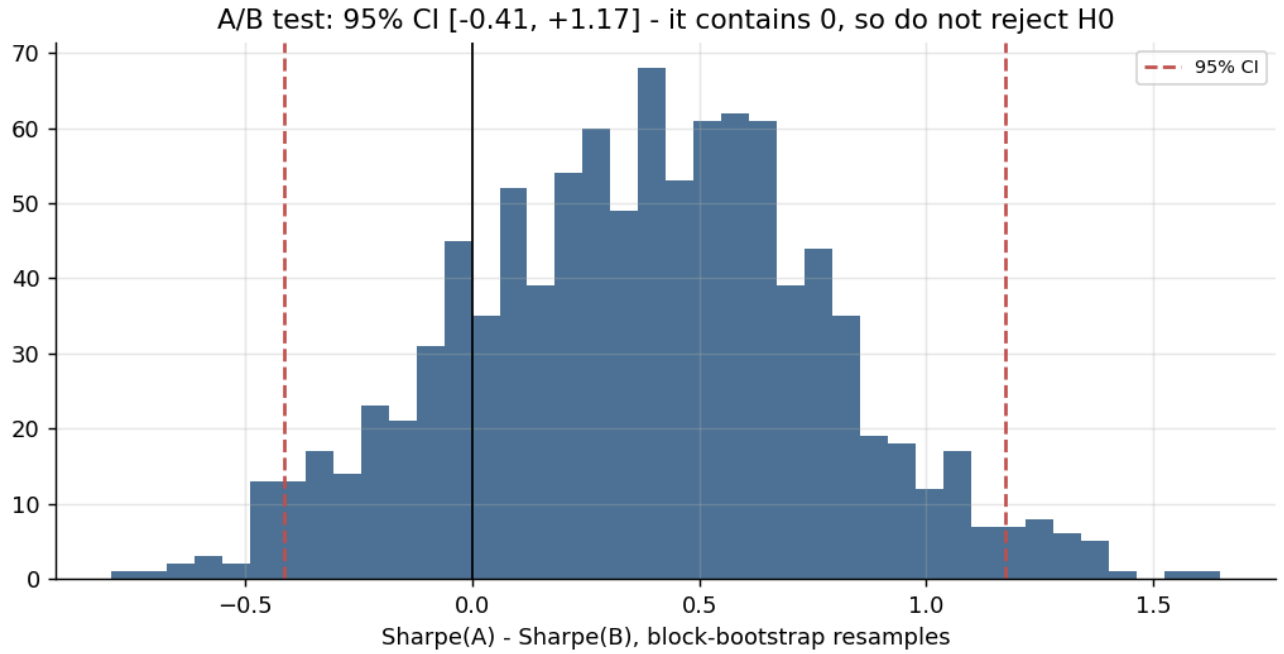
WATCH OUT FOR

Overlapping targets leak information - leave a gap between train and test.

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TOPIC 3

Method: A/B Testing Two Strategy Variants



HOW IT WORKS

Compare variant A and B on the same days: H0 'no difference' vs H1 'B is better'.

RULE

Welch / paired t-test on daily returns; bootstrap a 95% CI of the Sharpe difference.

WORKS BEST WHEN

Deciding which parameter set or filter to promote to paper trading.

WATCH OUT FOR

If the CI contains 0, do not reject H0; testing 100 variants gives ~5 false winners.

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TOPIC 3

From Probability to Position

1. Build leak-free features
2. Train on past window
3. Predict P(up) for tomorrow
4. Long if P(up) > 0.55, else flat
5. Backtest net of costs

Judge the model by the net-of-cost strategy, not by accuracy alone.

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Pairs Trading: a Hedging Algorithm

1

Why it is a hedge

Long the cheap stock, short the rich one in hedge-ratio proportion: market moves largely cancel

2

Choose the pair

Economically linked stocks (e.g. KO and PEP) that are cointegrated, not merely correlated

3

Hedge ratio and spread

Regress log price A on log price B: slope = hedge ratio, residual = the spread (Engle-Granger)

4

Test the spread

ADF test on the residual: a stationary spread tends to revert to its mean

5

Trade the z-score

Enter when $|z| > 2$, exit as z returns towards 0; trade both legs on the next bar

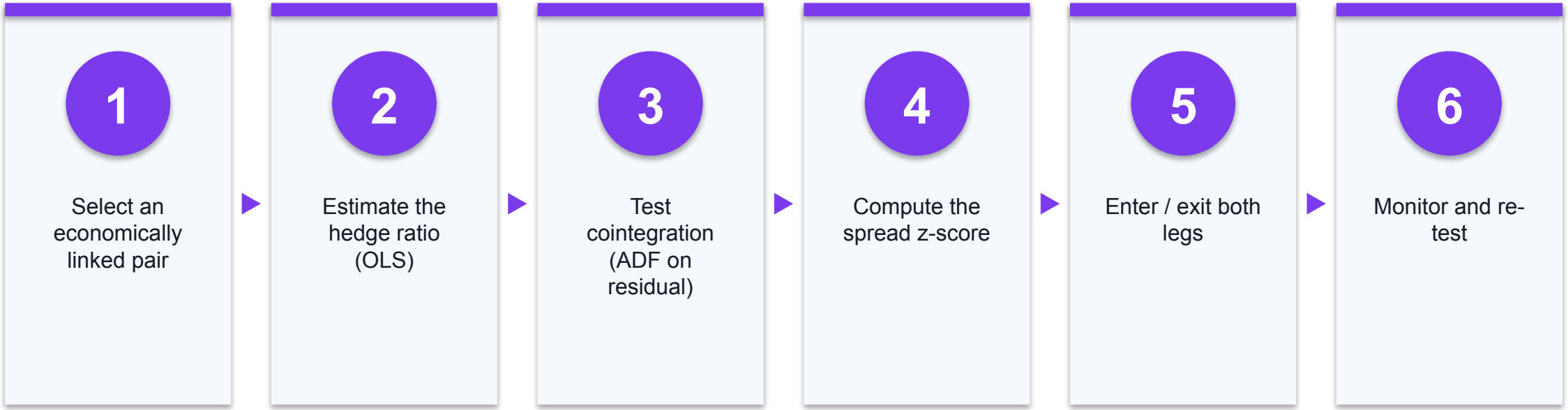
6

Risks

Cointegration breaks, hedge ratio drifts, double costs, short-borrow, multiple testing

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The Pairs-Trading Algorithm Step by Step

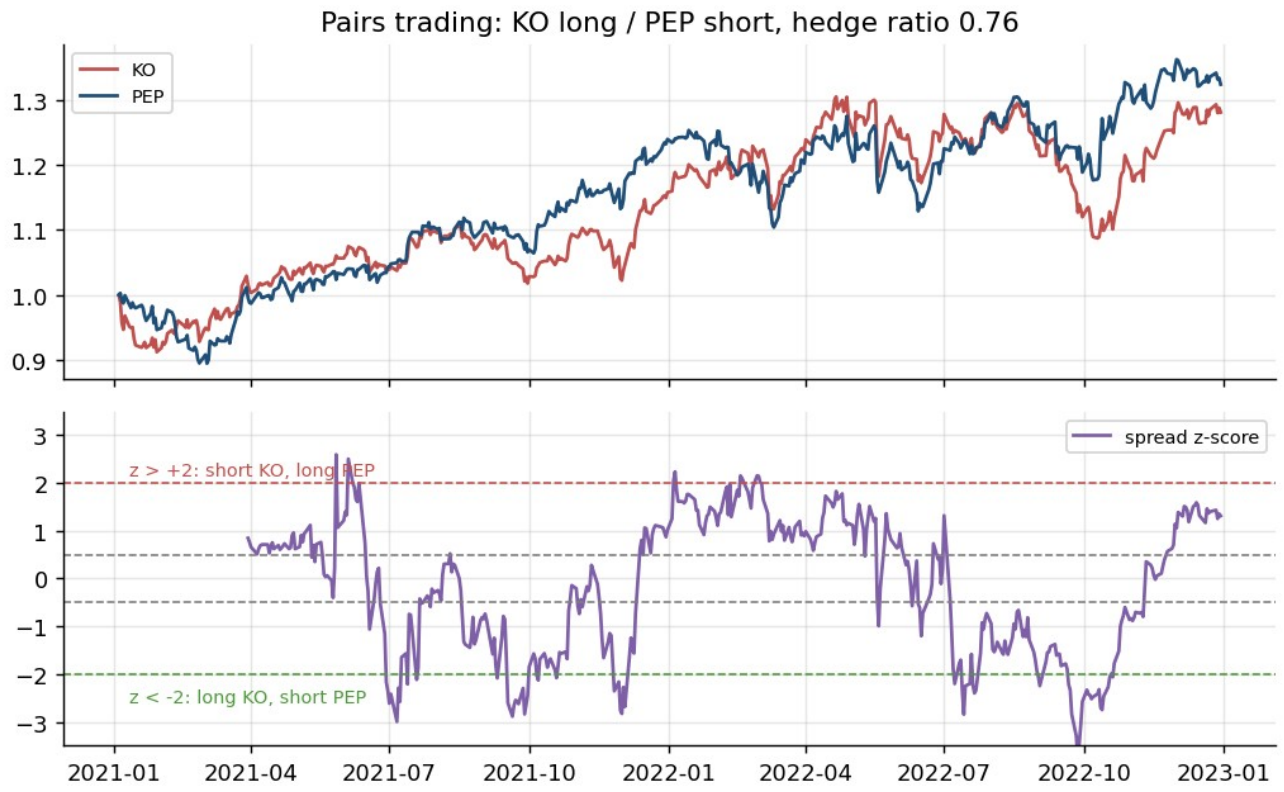


Gatev, Goetzmann and Rouwenhorst (2006): a simple pairs rule on US stocks earned average annualised excess returns of up to 11% (1962-2002); later studies find smaller and less consistent profits.

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TOPIC 3

Algorithm: Pairs Trading (Hedged Statistical Arbitrage)



HOW IT WORKS

Long one stock, short a cointegrated partner in hedge-ratio proportion: market moves cancel.

RULE

Spread z > +2: short A / long B. z < -2: long A / short B. Exit as z returns to 0.

WORKS BEST WHEN

Economically linked stocks (KO/PEP) whose spread has a stable mean.

WATCH OUT FOR

Cointegration can break; costs on both legs; short-borrow; screening many pairs.

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TOPIC 3 · HANDS-ON LAB

Strategy-Search Stage: Grid Search, Out-of-Sample Ranking and Decay

LAB 10

Grid-search SMA crossover parameters in-sample and measure how the ranking decays out of sample.

WORKFLOW

1. Simulator: Run the grid
2. Simulator: Read the decay
3. Simulator: Walk-forward

YOU'LL PRODUCE

IS and OOS Sharpe heatmaps, a rank correlation and a robust parameter region.

Duration
25 min

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LAB 10 · FOLLOW ALONG

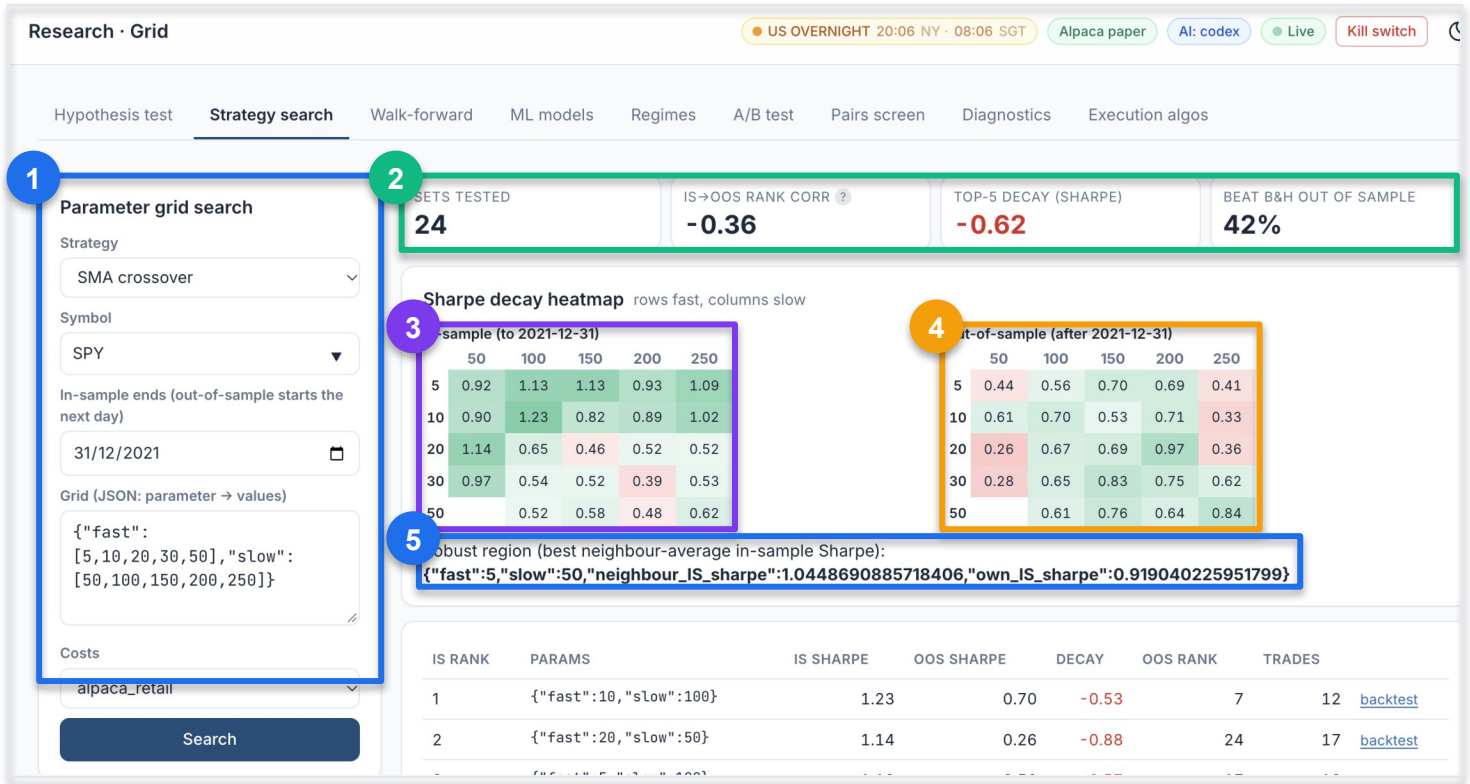
Lab 10 — Step by Step

Course Labs > Lab 10 at http://localhost:8000 - each step's Go button opens its page pre-filled. Tick each 'You should see' before moving on.

	STEP · WHERE	DO THIS	YOU SHOULD SEE
1	**Run the grid** ▸ Research > Grid SPY SMA cross	Research > Strategy search: SMA crossover on SPY, in-sample to 2021-12-31, Alpaca costs. Press Search.	✓ About 24 parameter sets; two heatmaps appear.
2	**Read the decay** ▸ Research > Strategy search	Compare the in-sample and out-of-sample heatmaps and the IS->OOS rank correlation.	✓ You can explain why the single best in-sample cell is usually not the best out of sample, and why a robust region is preferred.
3	**Walk-forward** ▸ Research > Walk-forward SPY	Research > Walk-forward: SMA crossover on SPY (3-year train, 1-year test).	✓ The stitched out-of-sample equity curve and the parameters chosen in each window.

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Lab 10 — How to Read the Screen



1 — The grid
SMA crossover on SPY: fast 5-50 x slow 50-250, in-sample to 31/12/2021.

2 — Decay summary
24 sets; IS -> OOS rank correlation -0.36: the in-sample ranking does not hold.

3 — In-sample heatmap
Greener = higher Sharpe to 2021. The best cell, 10/100, scores 1.23.

4 — Out-of-sample heatmap
The same cells after 2021: 10/100 drops to 0.70 and the winners move.

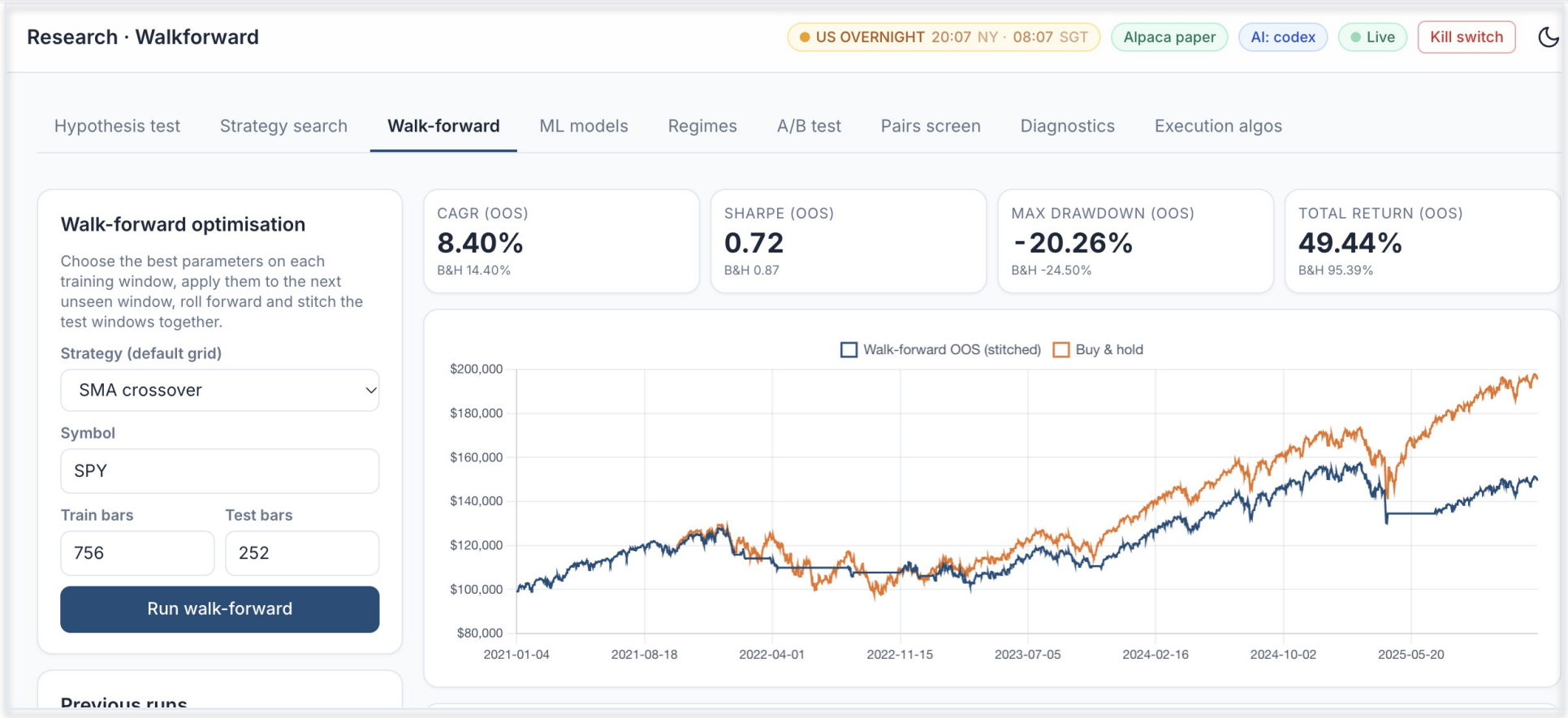
5 — Robust region
Prefer the cell whose neighbours are also good, not the single best cell.

The best in-sample setting is rarely the best out of sample: choose a robust region, then walk-forward.

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LAB 10 · SIMULATOR

Lab 10 — Walk-Forward: Re-Tune, Then Test Unseen



Research > Walk-forward: SMA crossover on SPY, 3-year train, 1-year test. Stitched out of sample: CAGR 8.40% (buy-and-hold 14.40%), Sharpe 0.72 (0.87), max drawdown -20.26% (-24.50%).

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TOPIC 3 · HANDS-ON LAB

ML-Signal Stage: a Walk-Forward Classifier from Probability to Position

LAB 11

Train logistic regression, random forest and XGBoost walk-forward and trade P(up) > threshold.

WORKFLOW

1. Simulator: Train the classifiers
2. Simulator: Backtest an ML strategy
3. Simulator: Check for leakage

YOU'LL PRODUCE

Accuracy / AUC table, feature importance and a net-of-cost backtest of the ML signal.

Duration
25 min

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LAB 11 · FOLLOW ALONG

Lab 11 — Step by Step

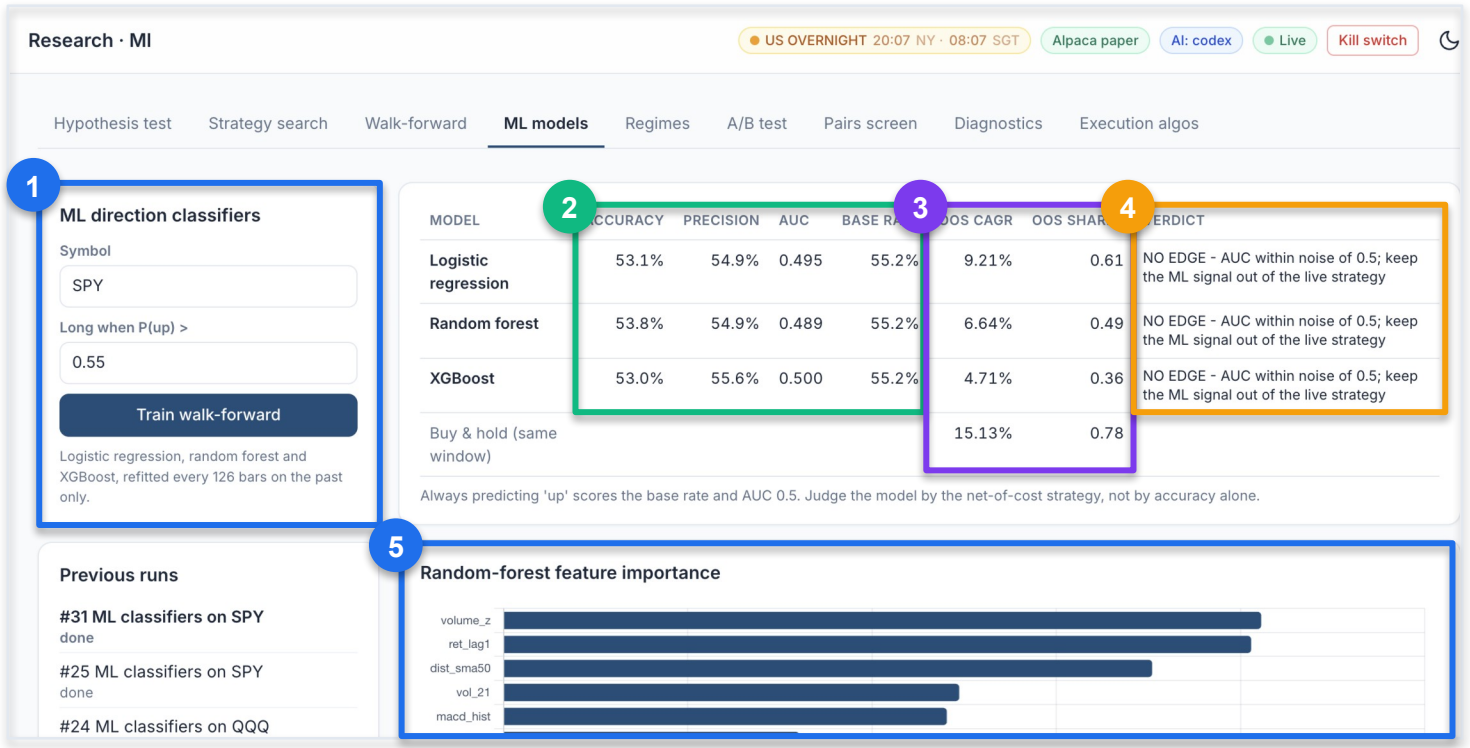
Course Labs > Lab 11 at http://localhost:8000 - each step's Go button opens its page pre-filled. Tick each 'You should see' before moving on.

	STEP · WHERE	DO THIS	YOU SHOULD SEE
1	**Train the classifiers** ▸ Research > ML SPY	Research > ML models on SPY, threshold 0.55.	✓ AUC close to 0.5 - daily direction is close to a coin toss; read the verdict.
2	**Backtest an ML strategy** ▸ Backtest SPY XGBoost	Backtest 'ML: XGBoost (walk-forward)' on SPY.	✓ Judge the model by the net-of-cost strategy, not by accuracy.
3	**Check for leakage** ▸ Research > Diagnostics	Research > Diagnostics on 'ML: logistic regression'.	✓ Look-ahead check PASS: refits use only past bars.

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LAB 11 · SIMULATOR

Lab 11 — How to Read the Screen



1 **Train walk-forward**
SPY, long when P(up) > 0.55; every model refits every 126 bars on past data only.

2 **Accuracy, AUC, base rate**
53-54% accuracy is below the 55.2% base rate; AUC about 0.5 = a coin toss.

3 **Net-of-cost result**
OOS Sharpe 0.36-0.61 vs buy-and-hold 0.78: none beats simply holding SPY.

4 **Verdict**
NO EDGE for all three models: keep the ML signal out of the live strategy.

5 **Feature importance**
The inputs the random forest leaned on: volume, lagged return, distance to SMA 50.

A model that cannot beat the base rate, with AUC near 0.5, has no edge - however clever it looks.

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TOPIC 3 · HANDS-ON LAB

Regime-Filter Stage: PCA and K-Means as a Gate on the Signals

LAB 12

Find market regimes with walk-forward K-means and gate a signal in the high-volatility regime.

WORKFLOW

1. Simulator: Analyse regimes
2. Simulator: Backtest the regime-filtered strategy

YOU'LL PRODUCE

PCA explained variance, a regime timeline and gated vs ungated performance.

Duration
20 min

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LAB 12 · FOLLOW ALONG

Lab 12 — Step by Step

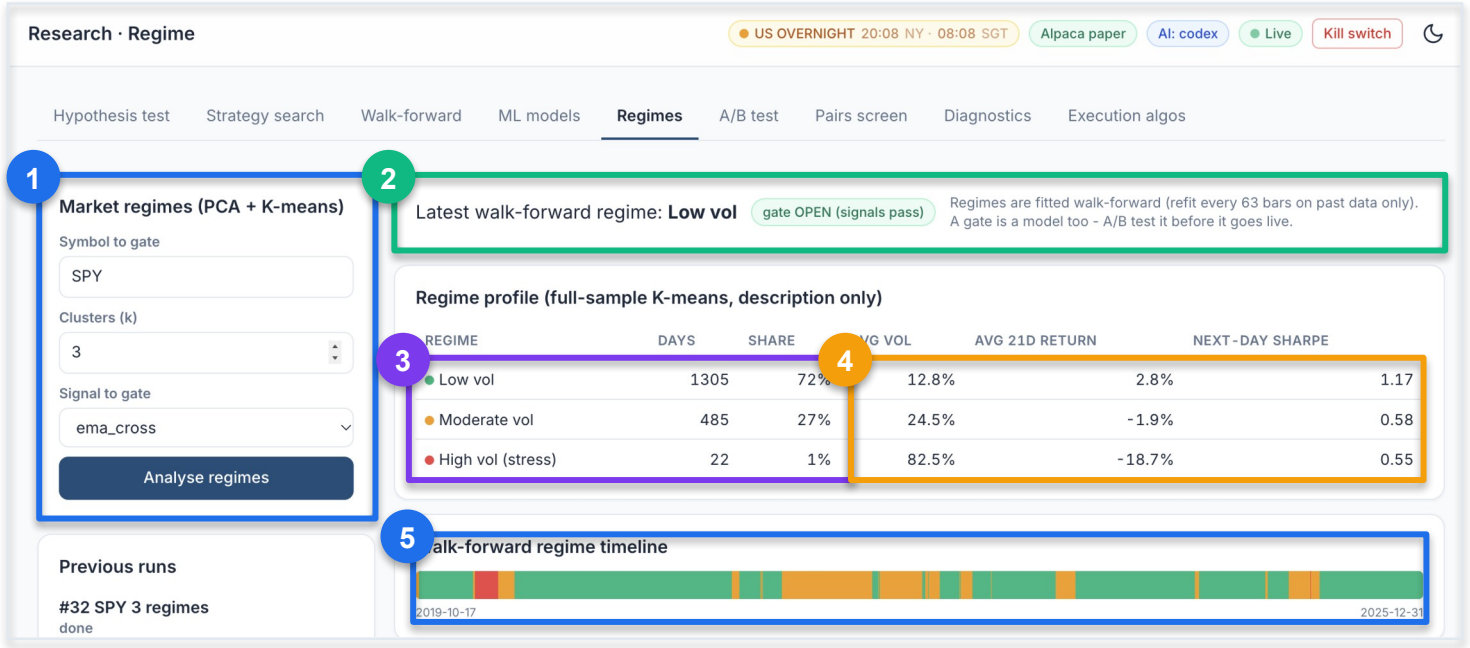
Course Labs > Lab 12 at http://localhost:8000 - each step's Go button opens its page pre-filled. Tick each 'You should see' before moving on.

	STEP · WHERE	DO THIS	YOU SHOULD SEE
1	**Analyse regimes** ▸ Research > Regimes SPY	Research > Regimes: SPY, 3 clusters, gate the EMA crossover.	✓ PC1 explains the largest share of variance (the market factor); the 2020 crash is in the stress regime.
2	**Backtest the regime-filtered strategy** ▸ Backtest SPY regime filter	Backtest 'Regime-filtered strategy (K-means gate)'.	✓ Compare max drawdown with the ungated EMA crossover.

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LAB 12 · SIMULATOR

Lab 12 — How to Read the Screen



1. **Settings**
   SPY, 3 clusters (k), gating the EMA crossover signal.

2. **Today's regime and gate**
   Low vol, gate OPEN: signals pass. Regimes are refitted walk-forward on past data only.

3. **How common each regime is**
   Low vol 72% of days, moderate vol 27%, high-vol stress 1% (22 days).

4. **How each regime behaves**
   Stress: 82.5% vol and -18.7% over 21 days; low vol has the best next-day Sharpe (1.17).

5. **Regime timeline**
   The walk-forward regime day by day, 2019-2025: red = stress (the 2020 crash), orange = moderate.

Trade the signal only while the gate is open - and A/B test the gate before it goes live.

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TOPIC 3 · HANDS-ON LAB

A/B-Test Stage: a Statistical Decision before Capital Moves

LAB 13

Decide statistically between two variants before moving capital.

WORKFLOW

1. Simulator: Run the A/B test
2. Simulator: Your own variants

YOU'LL PRODUCE

Welch and paired t-tests, a bootstrap Sharpe CI and a decision with a multiple-testing caution.

Duration
20 min

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LAB 13 · FOLLOW ALONG

Lab 13 — Step by Step

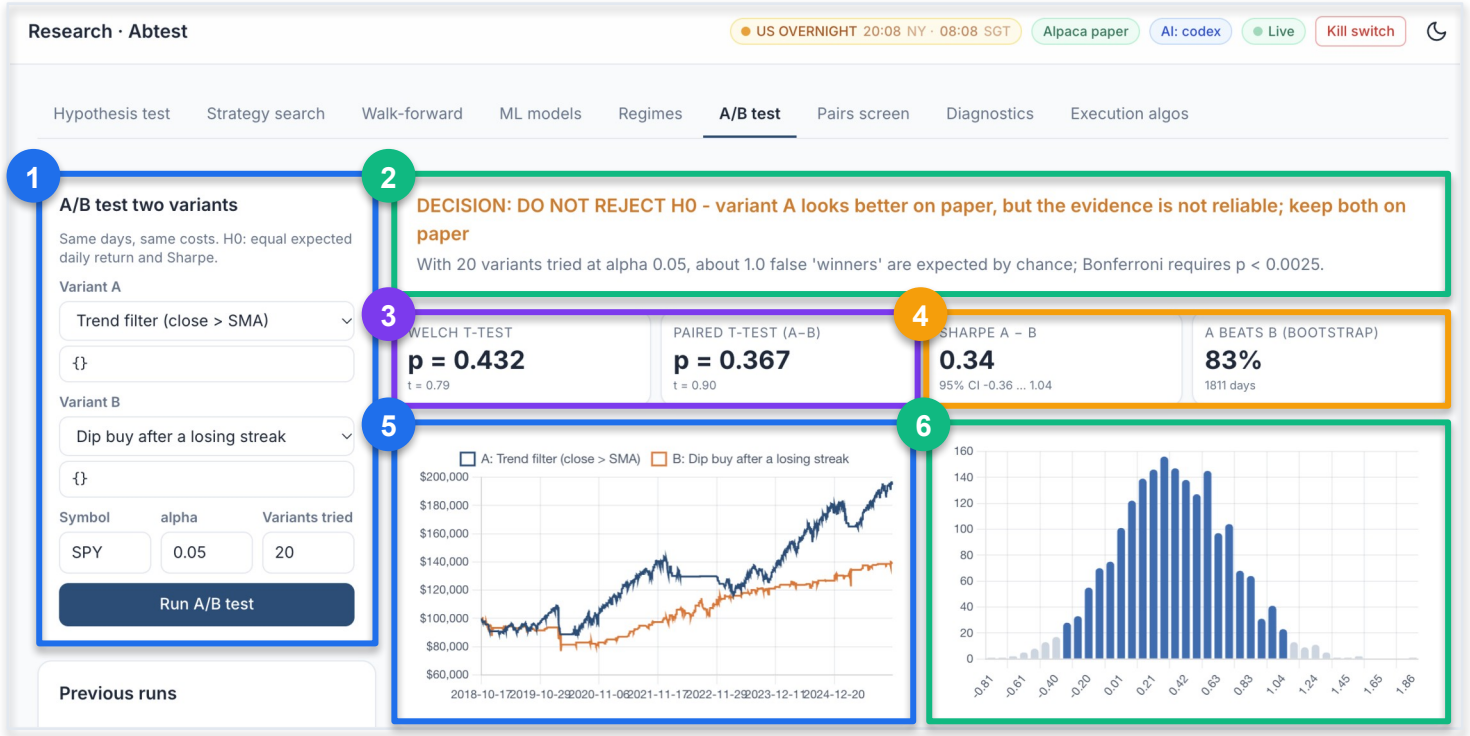
Course Labs > Lab 13 at http://localhost:8000 - each step's Go button opens its page pre-filled. Tick each 'You should see' before moving on.

	STEP · WHERE	DO THIS	YOU SHOULD SEE
1	**Run the A/B test** ▸ Research > A/B trend_filter vs dip_buy	Research > A/B test: A = Trend filter, B = Dip buy, SPY, alpha 0.05, 20 variants tried.	✓ Read the decision; if the CI contains 0, do not reject H0.
2	**Your own variants** ▸ Research > A/B test	A = EMA crossover {"fast": 10, "slow": 30}, B = EMA crossover {"fast": 20, "slow": 50}.	✓ You can explain the Bonferroni-adjusted alpha.

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LAB 13 · SIMULATOR

Lab 13 — How to Read the Screen



1 — Set up the test
A = Trend filter, B = Dip buy, SPY, alpha 0.05, 20 variants tried. H0: no difference.

2 — The decision
DO NOT REJECT H0. 20 tries at 5% expect 1 lucky winner; Bonferroni needs p < 0.0025.

3 — Welch and paired t-tests
p 0.432 and 0.367, far above 0.05. Paired is the better test: same days, same market.

4 — Sharpe gap and bootstrap
A - B = 0.34, but the 95% CI -0.36 to 1.04 contains 0. A wins 83% of resamples - not enough.

5 — Equity curves
A ends near $195k, B near $138k. Tempting - but a nicer curve is not proof.

6 — Bootstrap distribution
Resampled Sharpe gaps; dark bars are the 95% CI, and 0 falls inside them.

Keep both variants on paper: move capital only when p beats the Bonferroni alpha and the CI excludes 0.

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Lab 13 — How to Reach the A/B Decision

Read the screen in this order. H0: A and B have the same expected daily return and Sharpe.

1

Don't trust the chart. A's curve looks better, but 7 years of noisy daily returns can fool you.

2

Paired t-test on A − B (same days, same market): p = 0.367, above alpha 0.05.

3

Bootstrap Sharpe CI −0.36 … 1.04 contains 0, so 'no difference' is still plausible.

4

20 variants × 0.05 = 1 lucky 'winner' expected. Bonferroni: 0.05 ÷ 20 = 0.0025.

5

Decision: DO NOT REJECT H0. Keep both on paper and record the result.

Step 2 (EMA 10/30 vs EMA 20/50): read the same three numbers - paired p, the CI vs 0 and Bonferroni 0.0025 - and record them for the PP.

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TOPIC 3 · HANDS-ON LAB

Pairs Stage: a Hedged, Market-Neutral Statistical-Arbitrage Sleeve

LAB 14

Screen pairs for cointegration and backtest a market-neutral KO/PEP spread.

WORKFLOW

1. Simulator: Screen the pairs
2. Simulator: Backtest KO/PEP

YOU'LL PRODUCE

An Engle-Granger screen and a pairs backtest with beta near zero.

Duration
25 min

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LAB 14 · FOLLOW ALONG

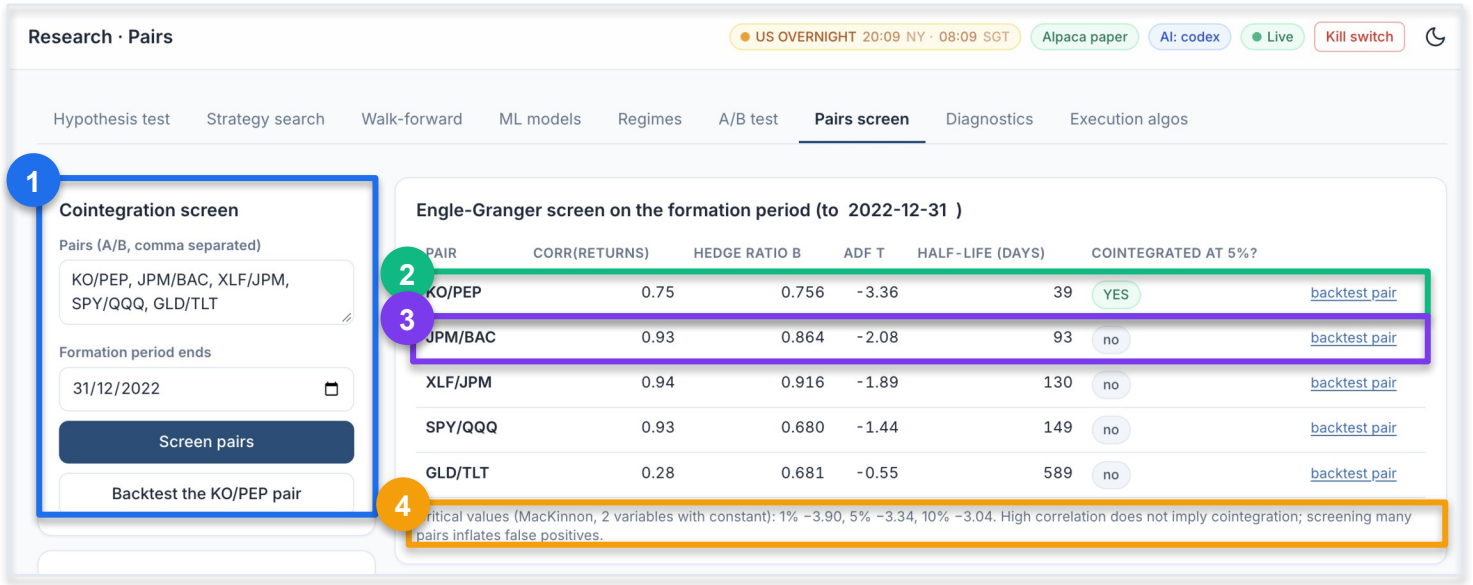
Lab 14 — Step by Step

Course Labs > Lab 14 at http://localhost:8000 - each step's Go button opens its page pre-filled. Tick each 'You should see' before moving on.

	STEP · WHERE	DO THIS	YOU SHOULD SEE
1	**Screen the pairs** ▸ Research > Pairs screen	Research > Pairs screen (formation to 2022-12-31).	✓ High correlation does not imply cointegration - compare the corr and ADF columns.
2	**Backtest KO/PEP** ▸ Backtest KO/PEP pairs	Backtest the pairs strategy on KO, PEP (both legs, short leg allowed, borrow cost charged).	✓ Beta to SPY near 0 in 'All metrics'; the trade list has long and short legs.

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Lab 14 — How to Read the Screen



1 — Pairs and formation period
Five candidate pairs, tested on data to 31/12/2022 only.

2 — KO/PEP: cointegrated
ADF t -3.36 beats the 5% critical value -3.34; the spread's half-life is 39 days.

3 — JPM/BAC: correlated, not cointegrated
0.93 correlation but ADF t -2.08: the spread can drift away for good.

4 — Critical values
1% -3.90, 5% -3.34, 10% -3.04. Screening many pairs inflates false positives.

Trade a spread only for a cointegrated pair (KO/PEP), then check its beta to SPY is near 0.

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Topic 3 Recap

1. Conduct statistical modelling of data to derive patterns and solutions
2. Perform A/B testing based on different hypotheses
3. Types of statistical analyses, data models, algorithms and advanced computational methods
4. Range and application of various types of data models
5. Statistical modelling techniques

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TOPIC 4

04

End-to-End Trading Pipeline

LEARNING OUTCOME

Perform coding and configuration of an automated trading program based on selected models or algorithms, conduct tests on its actions and outcomes, diagnose unintended outcomes, and draw insights from the results to support decisions and report accordingly.

Day 2 · Labs 15–20

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TOPIC 4

Pre-Trade Risk Controls

1. **Position and order limits**
   Maximum shares, maximum order value per symbol.

2. **Maximum daily loss**
   Stop trading for the day when breached.

3. **Market-hours and duplicate checks**
   Alpaca clock; never resend an open order.

4. **Kill switch**
   One file or flag halts all trading immediately.

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TOPIC 4

Trading Costs You Must Model

1

Commission

Zero on Alpaca US equities — not on most brokers.

2

Spread and slippage

Half-spread + slippage on every fill, in bps.

3

Regulatory fees on sells

SEC Section 31 fee and FINRA TAF.

4

Borrow and impact

Short-borrow costs; your own market impact.

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TOPIC 4

Historical Data: Backtest, Tune, Then Forward Test



BACKTEST ON HISTORY

Replay the strategy's rules over past prices - next-bar fills, trading costs - to see how it would have performed.

TUNE HYPERPARAMETERS (IN-SAMPLE)

Search the settings (MA lengths, RSI levels, model parameters) on the OLDER history only, by grid search or walk-forward.

CONFIRM OUT-OF-SAMPLE

Backtest the tuned settings ONCE on later history the search never saw. A big drop from in-sample = overfitting.

FORWARD TEST (PAPER)

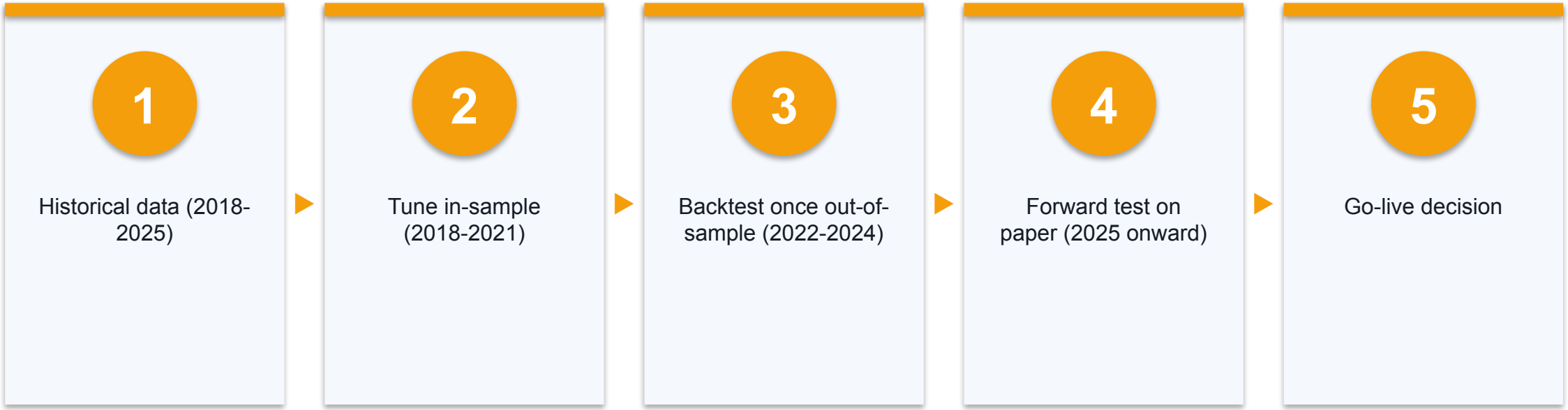
Run the tuned strategy on NEW data as it arrives - paper orders, real fills and risk checks - before any capital.

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TOPIC 4

From Historical Data to a Forward-Tested Strategy

The same rule moves through four tests; each uses data the previous step never touched.



Tuning on the test period, or re-tuning after seeing out-of-sample results, turns the test into more tuning - the result will look better than reality (Labs 10, 15, 16).

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TOPIC 4

Potential Reasons for Unintended Outcomes

1

Overfitting and data snooping

Great in-sample, collapses out of sample.

2

Look-ahead and survivorship bias

Same-bar fills; only today's survivors.

3

Cost and execution drag

Live returns trail the backtest by turnover × cost.

4

Regime change, data and code errors

Drift, bad splits, time zones, duplicate orders.

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TOPIC 4

What Goes on a Tearsheet

1. **Equity versus benchmark**
   Growth of USD 100,000, strategy and buy-and-hold.

2. **Drawdown and rolling Sharpe**
   Depth, length and stability of the edge.

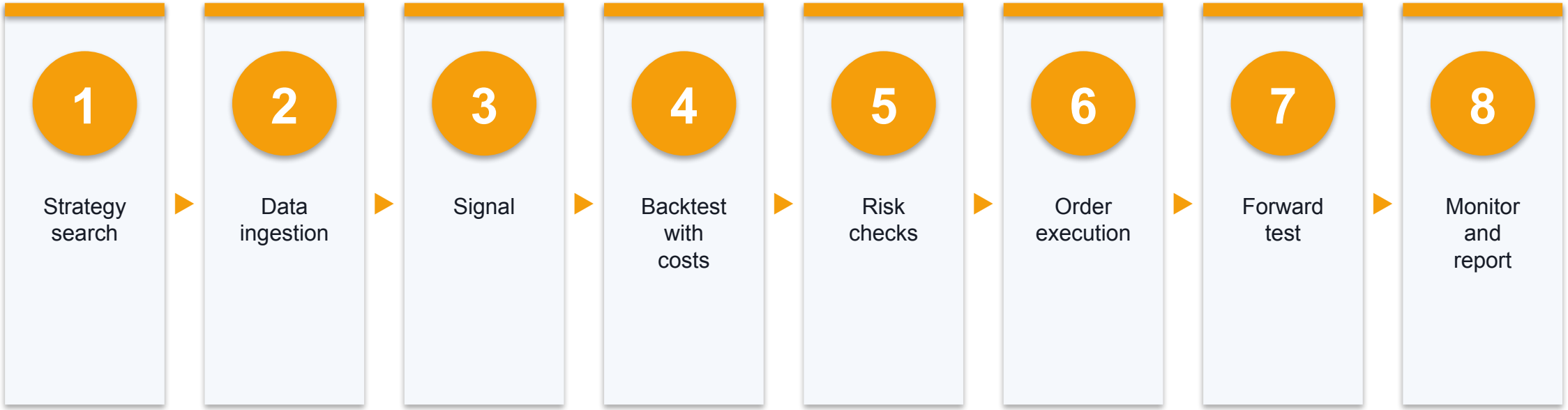
3. **Monthly returns heatmap**
   When did it make — and lose — money?

4. **Metrics and cost table**
   Gross versus net; backtest versus forward.

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TOPIC 4

The End-to-End Algorithmic Trading Pipeline



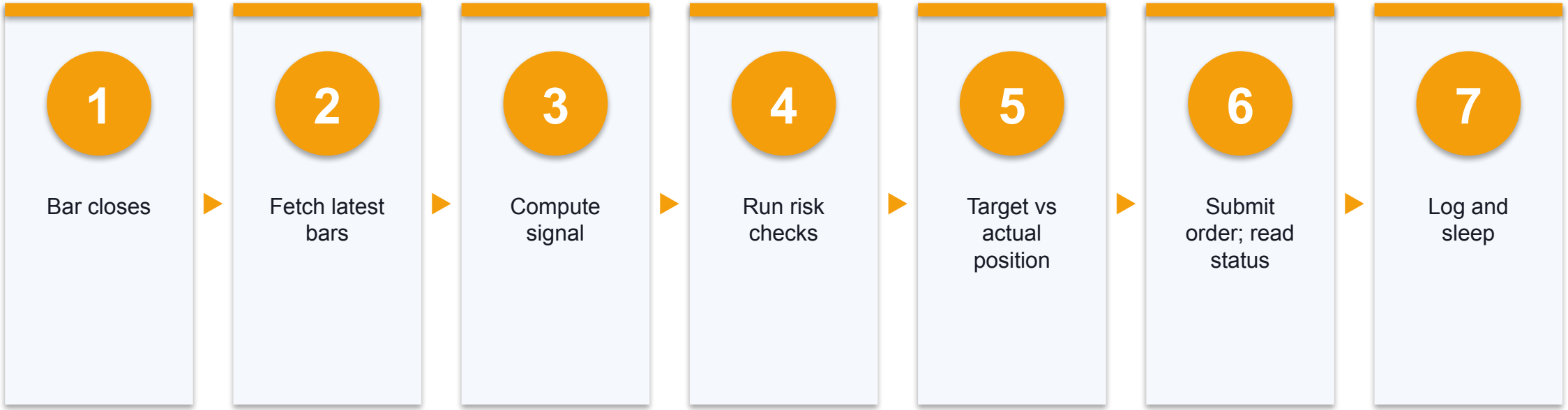
Configuration — symbols, parameters, limits, keys — lives outside the code.

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TOPIC 4

The Automated Paper-Trading Bot Loop

One cycle per closed bar.



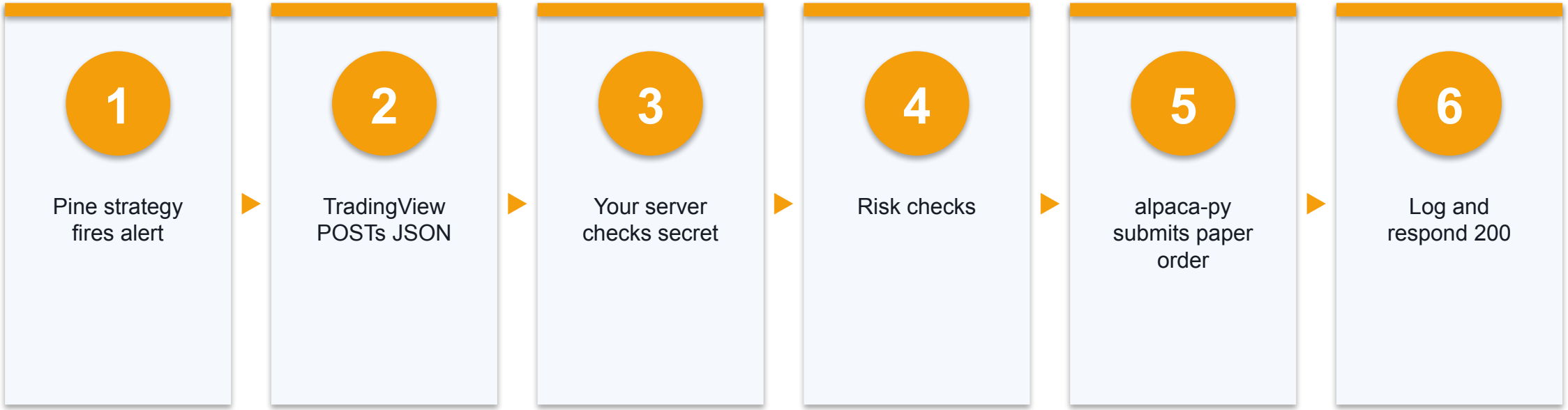
Every cycle writes to the trade log — the evidence for performance and audit.

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TOPIC 4

TradingView Alert to Alpaca Order

Webhook-driven execution.



Webhook alerts need a paid TradingView plan and a public URL (e.g. a tunnel).

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TOPIC 4

Diagnosing an Unintended Outcome

1. Reproduce the result
2. Isolate the first divergent step
3. Data · signal · order · fill?
4. Compare with a known-good run
5. Fix and re-test

Read the first failure, not the last — everything after it is a consequence.

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Lunch Break

1 hour · see you at 13:30

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TOPIC 4 · HANDS-ON LAB

Backtest-Engine Stage: Cost Models, Execution Timing and Break-Even

LAB 15

Run the same strategy under four cost models and find its break-even cost.

WORKFLOW

1. Simulator: Zero vs Alpaca vs high-cost
2. Simulator: Break-even and execution timing
3. Simulator: Stops and sizing

YOU'LL PRODUCE

Gross vs net CAGR, a cost breakdown and the break-even bps.

Duration
30 min

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LAB 15 · FOLLOW ALONG

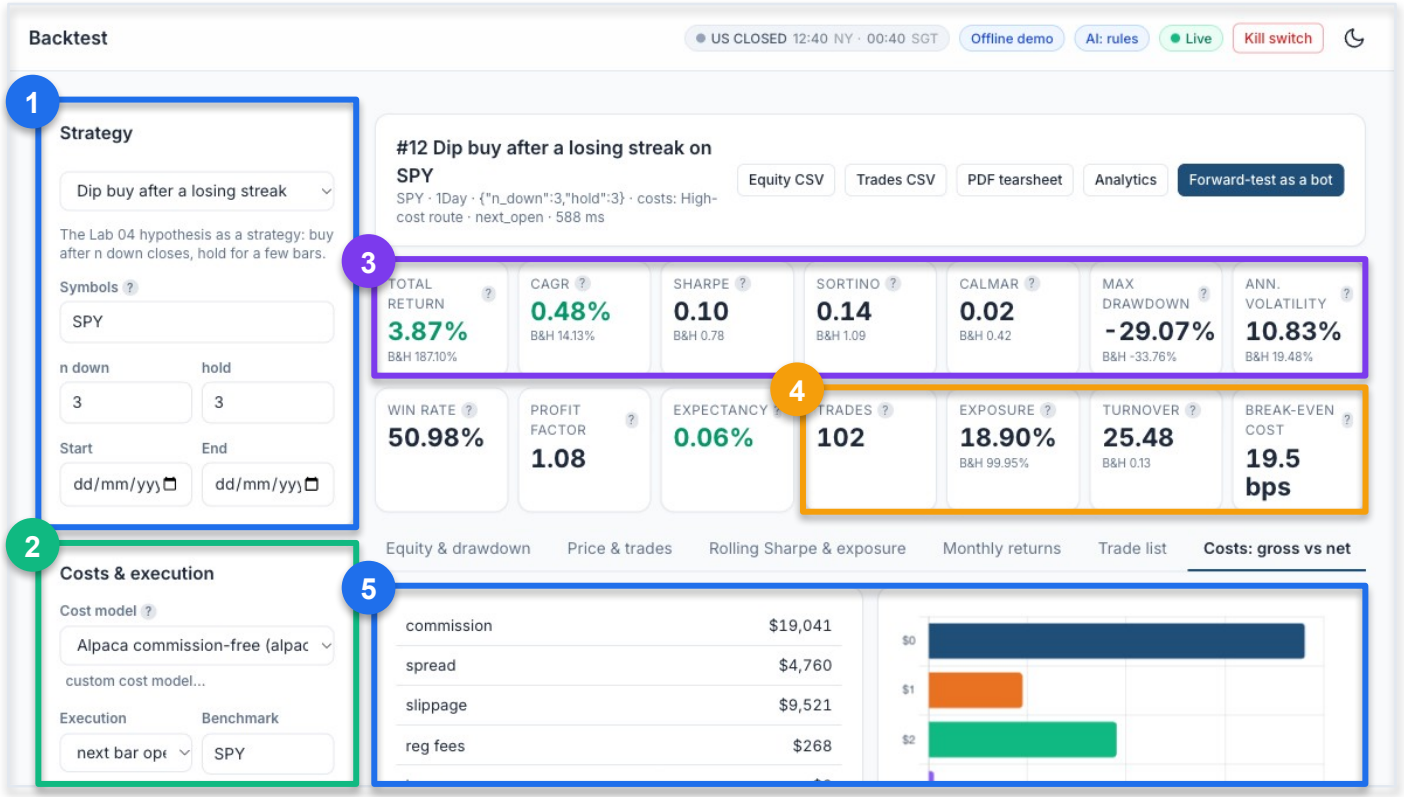
Lab 15 — Step by Step

Course Labs > Lab 15 at http://localhost:8000 - each step's Go button opens its page pre-filled. Tick each 'You should see' before moving on.

	STEP · WHERE	DO THIS	YOU SHOULD SEE
1	**Zero vs Alpaca vs high-cost** ▸ Backtest SPY dip-buy, zero cost first	Run the same strategy three times, changing ONLY the Cost model, to see how much of its profit trading costs eat.	✓ About 102 trades (203 fills) in every run.
2	**Break-even and execution timing** ▸ Backtest	Break-even cost is the one-way cost per trade at which the strategy's net CAGR falls to zero: above it the strategy loses money.	✓ Break-even is 19.5 bps. Alpaca's ~1.5 bps leaves plenty of room, but the high-cost route's ~17.5 bps (plus fees) uses almost all of it - that is why its net CAGR was only 0.48%.
3	**Stops and sizing** ▸ same page	Now add risk controls to a different strategy, EMA crossover 20/50 on SPY. The link fills in the settings below.	✓ 16 trades: 4 exit with stop_loss, the rest with signal.

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Lab 15 — How to Read the Screen



1. **Strategy**
   Dip buy after a losing streak (the Lab 04 idea): n down 3, hold 3.

2. **Cost model and execution**
   Run it with zero, alpaca_retail and high_cost; then try next-bar close.

3. **Net result (high-cost route)**
   CAGR 0.48% and Sharpe 0.10 vs SPY's 14.13% and 0.78: costs ate the edge.

4. **Trades and break-even**
   102 trades, turnover 25.5; break-even only 19.5 bps one way - fragile.

5. **Costs: gross vs net**
   The dollar drag by type: commission, spread, slippage and fees.

A frequent trader lives or dies by costs: compare the break-even bps with your broker's real cost.

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TOPIC 4 · HANDS-ON LAB

Automation Stage: a Scheduled, Risk-Checked Forward Test and Live Paper Trading

LAB 16

Create a bot, run it on a schedule, watch orders, positions and P&L live, and halt it with the kill switch.

WORKFLOW

1. Simulator: Create the bot
2. Simulator: Start it and watch
3. Simulator: Kill switch
4. Simulator: Backtest vs forward

YOU'LL PRODUCE

A bot equity curve, a blotter of its orders, a backtest-vs-forward comparison and a tested kill switch.

Duration
30 min

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LAB 16 · FOLLOW ALONG

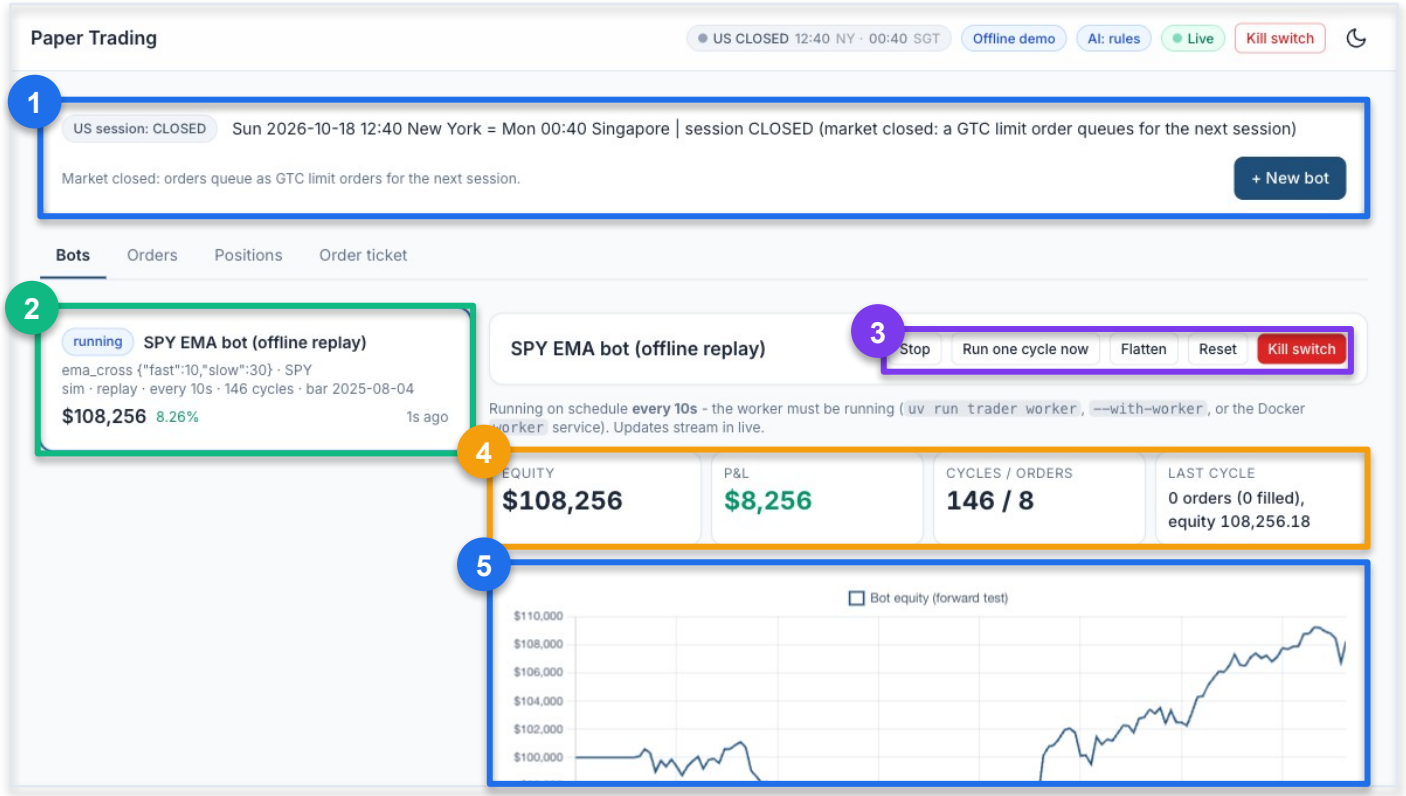
Lab 16 — Step by Step

Course Labs > Lab 16 at http://localhost:8000 - each step's Go button opens its page pre-filled. Tick each 'You should see' before moving on.

	STEP · WHERE	DO THIS	YOU SHOULD SEE
1	**Create the bot** › Paper Trading > New bot (EMA cross)	Paper Trading > + New bot: EMA crossover 10/30 on SPY, schedule 'every 10s', simulator broker, replay from 2025-01-02.	✓ The bot appears with status stopped.
2	**Start it and watch** › Paper Trading > Bots	Press Start. Every cycle replays one trading day: it sees bars to yesterday's close and its orders fill at today's open. Or press 'Run one cycle now' repeatedly.	✓ Equity curve grows cycle by cycle; the Pipeline Map path lights up during each cycle.
3	**Kill switch** › same page	Press the bot's Kill switch, then try 'Run one cycle now'. Release it. Try the global kill switch in the top bar too.	✓ Cycles report 'kill switch ON - no orders sent'; the Risk stage turns red on the Pipeline Map.
4	**Backtest vs forward** › same page	Press Compare in 'Backtest vs forward test'.	✓ The curves track closely; you can explain the remaining gap (whole shares, fills, timing).

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Lab 16 — How to Read the Screen



1 — Session and + New bot
EMA 10/30 on SPY, every 10s, simulator broker, replay from 2025-01-02.

2 — Bot card
Status (running), strategy, schedule, cycles done and the current equity.

3 — Controls
Stop, Run one cycle now, Flatten, Reset - and the red Kill switch.

4 — Live numbers
Equity $108,256, P&L $8,256, 146 cycles / 8 orders, and the last cycle's result.

5 — Forward-test equity
Grows one replayed trading day per cycle; compare it with the backtest.

With the kill switch ON, cycles send no orders - test it before you trust any bot.

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TOPIC 4 · HANDS-ON LAB

Event-Driven Automation: TradingView Webhook to Signal, Risk and Execution

LAB 17

Send TradingView alerts to the app's webhook and see them pass validation, signal confirmation and risk into paper orders.

WORKFLOW

1. Simulator: Copy the alert template
2. Simulator: Run the self-test
3. Simulator: Real alerts (paid TradingView plan)

YOU'LL PRODUCE

7 simulated alerts handled as expected and the audit log.

Duration
25 min

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LAB 17 · FOLLOW ALONG

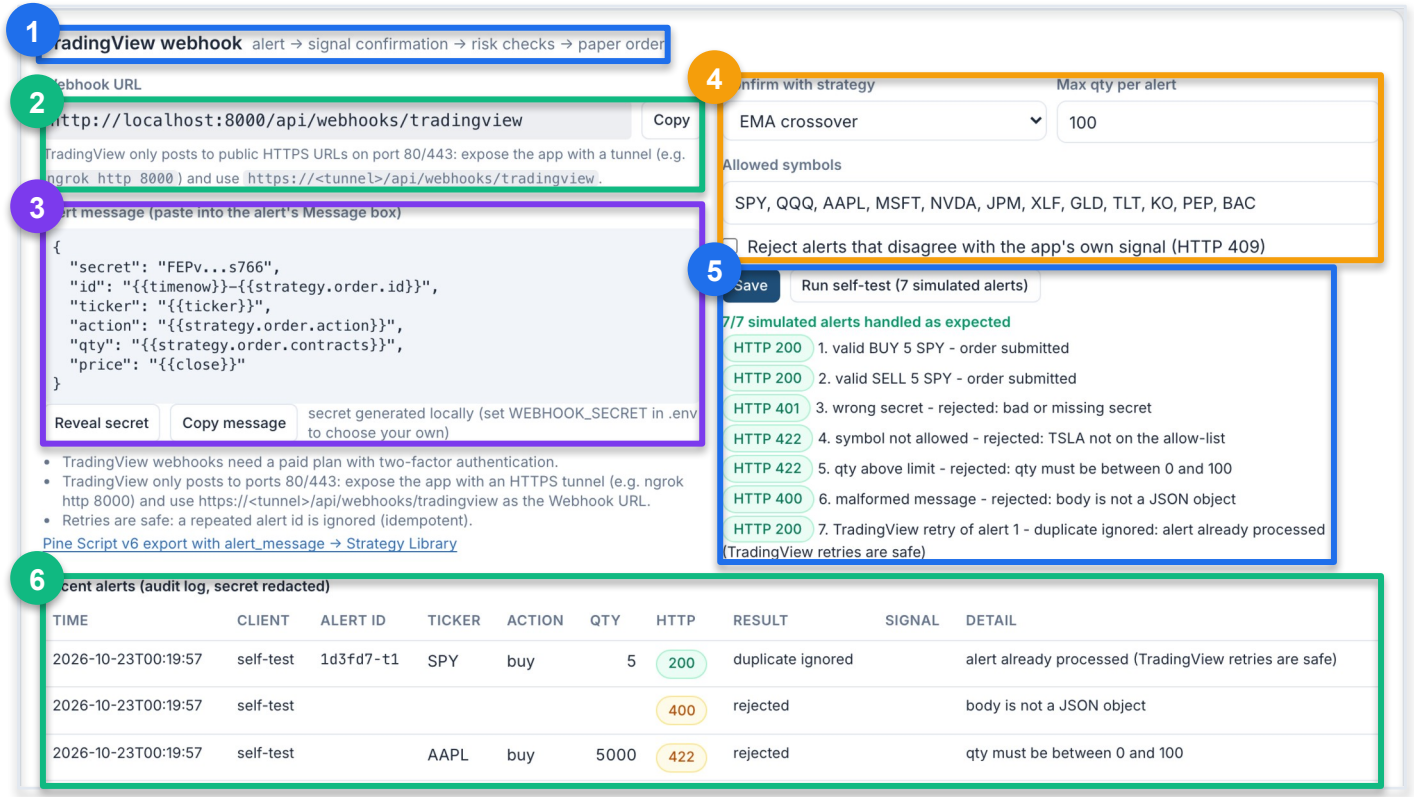
Lab 17 — Step by Step

Course Labs > Lab 17 at http://localhost:8000 - each step's Go button opens its page pre-filled. Tick each 'You should see' before moving on.

	STEP · WHERE	DO THIS	YOU SHOULD SEE
1	**Copy the alert template** ▸ Integrations > TradingView webhook	Open Integrations and scroll down to the TradingView webhook card. It holds the two things TradingView needs: where to send an alert (the Webhook URL) and what to send (the Alert message).	✓ You can say what each part of the message is for and which values TradingView fills in (the {{...}} placeholders).
2	**Run the self-test** ▸ System > Integrations	Press 'Run self-test (7 simulated alerts)'.	✓ 7/7 simulated alerts handled as expected: valid buy and sell (200), wrong secret (401), symbol not allowed and qty too large (422), malformed (400), retry ignored (200 duplicate).
3	**Real alerts (paid TradingView plan)** ▸ same page	Optional - trainer demonstration, or try it yourself if you have a paid TradingView plan. This is the same flow with a real TradingView alert.	✓ The alert appears in the audit log and its order in Paper Trading > Orders (source webhook).

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Lab 17 — How to Read the Screen



1 — The webhook path
alert -> signal confirmation -> risk checks -> paper order.

2 — Webhook URL
Copy it. TradingView can only reach it through a public HTTPS tunnel (e.g. ngrok).

3 — Alert message
The JSON for TradingView's Message box; its secret proves the alert is yours.

4 — Confirm, cap and allow
Confirm with EMA crossover, max 100 per alert, listed symbols only.

5 — Run self-test
7/7 handled: valid buy and sell 200, wrong secret 401, bad symbol or qty 422, malformed 400.

6 — Audit log
Every alert with its HTTP code and result, secret redacted; a retry is 'duplicate ignored'.

Only alerts that pass the secret, signal and risk checks become paper orders - and a retried alert never trades twice.

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TOPIC 4 · HANDS-ON LAB

Diagnostics Stage: Automated Bias Checks before Go-Live

LAB 18

Run the automated diagnostics on a deliberately biased strategy, fix it, and re-run.

WORKFLOW

1. Simulator: Diagnose the biased exhibit
2. Simulator: Fix it in the Strategy Builder
3. Simulator: Re-run diagnostics

YOU'LL PRODUCE

A FAIL on the biased exhibit, a PASS / WARN table for the fixed version and a verdict.

Duration
25 min

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LAB 18 · FOLLOW ALONG

Lab 18 — Step by Step

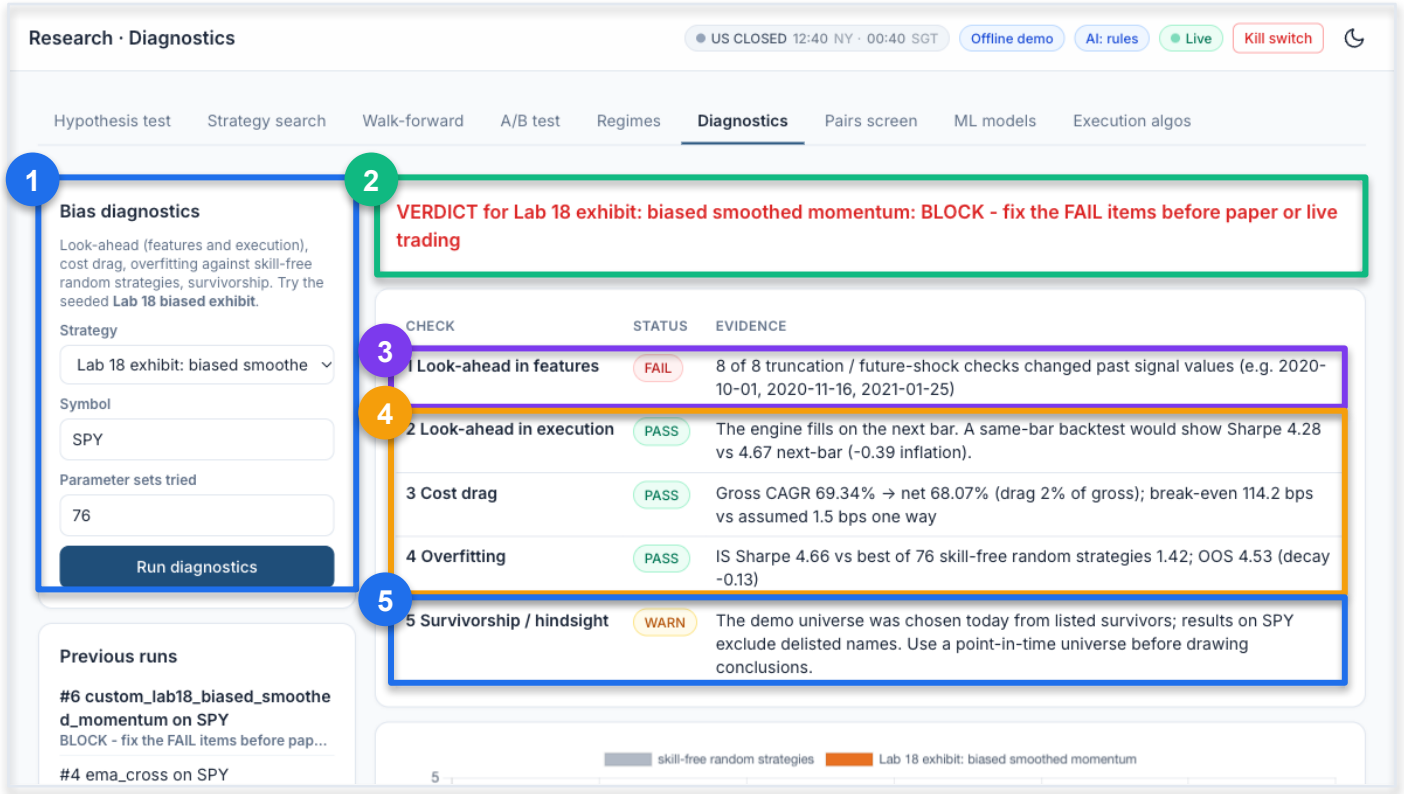
Course Labs > Lab 18 at http://localhost:8000 - each step's Go button opens its page pre-filled. Tick each 'You should see' before moving on.

	STEP · WHERE	DO THIS	YOU SHOULD SEE
1	**Diagnose the biased exhibit** ▸ Research > Diagnostics (Lab 18 exhibit)	Research > Diagnostics on 'Lab 18 exhibit: biased smoothed momentum' (seeded custom strategy).	✓ Look-ahead (features) FAIL: rolling(center=True) uses future bars. Verdict BLOCK.
2	**Fix it in the Strategy Builder** ▸ Strategy Builder (exhibit)	Open the exhibit in the Strategy Builder and change the single setting center=True to center=False (or let AI fix it: use the Lab 18 Strategy Builder prompt in the Prompt Library).	✓ The validation's look-ahead check now passes.
3	**Re-run diagnostics** ▸ Research > Diagnostics	Diagnose the fixed strategy; compare the same-bar vs next-bar Sharpe and the cost-drag row.	✓ No FAIL on look-ahead; cost drag and overfitting WARN explained.

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LAB 18 · SIMULATOR

Lab 18 — How to Read the Screen



1 **Choose the exhibit**
Lab 18 biased smoothed momentum on SPY; 76 parameter sets tried.

2 **Verdict**
BLOCK: fix every FAIL before paper or live trading.

3 **Look-ahead in features: FAIL**
rolling(center=True) peeks at future bars, so truncation changes past signals.

4 **Execution, cost drag, overfitting**
PASS: next-bar fills, small cost drag, beats 76 skill-free random strategies.

5 **Survivorship: WARN**
The universe is today's survivors - use a point-in-time universe.

Change center=True to center=False in the Strategy Builder, re-run, and the look-ahead FAIL becomes PASS.

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TOPIC 4 · HANDS-ON LAB

Monitoring and Reporting Stage: Automated Tearsheet and Backtest-vs-Forward Report

LAB 19

Produce a one-page tearsheet and explain the gap between backtest and forward test.

WORKFLOW

1. Simulator: Export the tearsheet
2. Simulator: Backtest vs forward
3. Simulator: Draw insights

YOU'LL PRODUCE

A PDF tearsheet and a written explanation of the backtest-vs-forward differences.

Duration
30 min

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LAB 19 · FOLLOW ALONG

Lab 19 — Step by Step

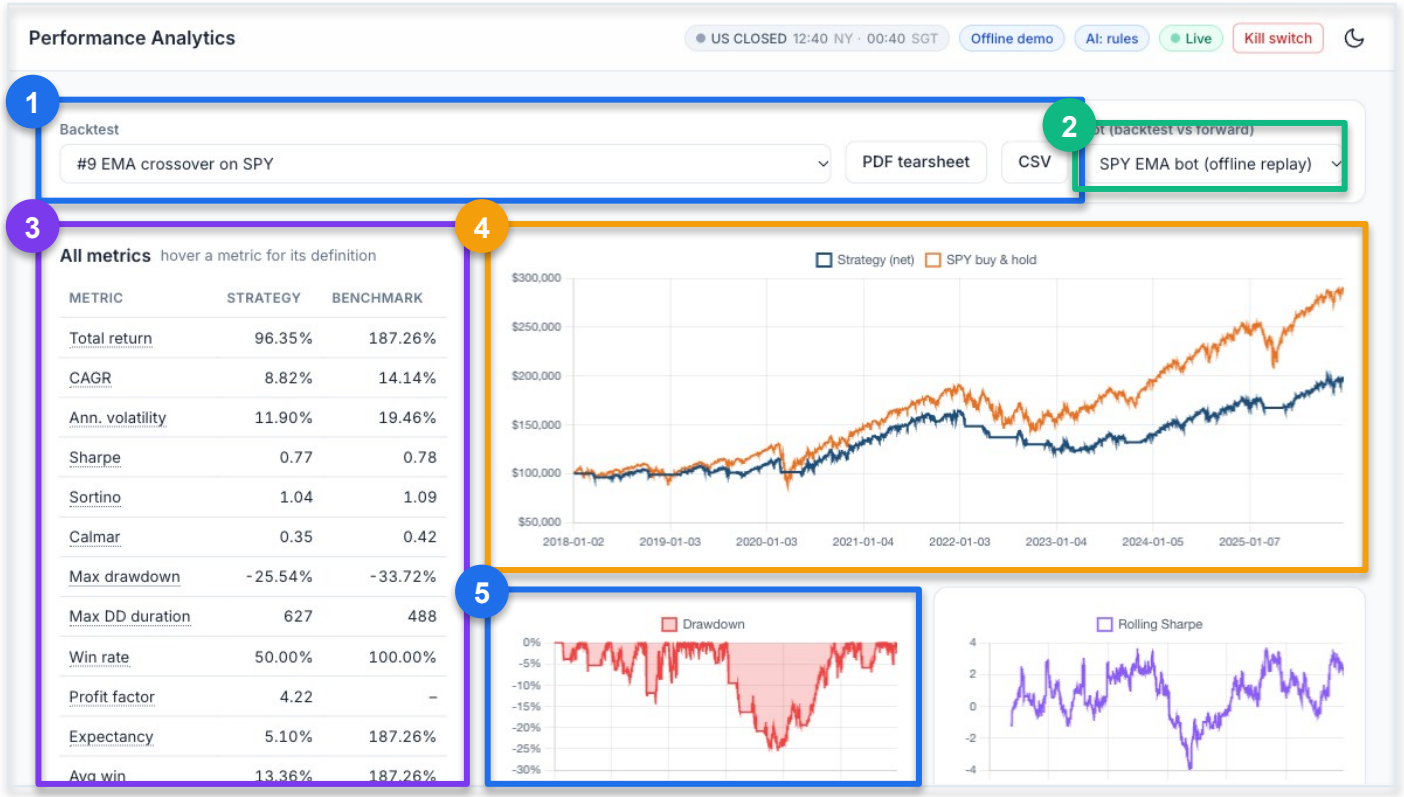
Course Labs > Lab 19 at http://localhost:8000 - each step's Go button opens its page pre-filled. Tick each 'You should see' before moving on.

	STEP · WHERE	DO THIS	YOU SHOULD SEE
1	**Export the tearsheet** ▸ Analytics > PDF tearsheet	Backtest page (or Analytics) > PDF tearsheet for your best backtest; also export the equity and trades CSV.	✓ The PDF has equity vs benchmark, drawdown, monthly heatmap, metrics and the cost table.
2	**Backtest vs forward** ▸ Analytics	Analytics: select your Lab 16 bot under 'Bot (backtest vs forward)'.	✓ Explain every material gap (fills, sizing, costs, timing).
3	**Draw insights** ▸ same page	Monthly heatmap: in which regimes did the strategy earn its returns, what drove its drawdowns?	✓ Two or three insights, not just numbers.

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LAB 19 · SIMULATOR

Lab 19 — How to Read the Screen



1 Backtest and exports
Pick your best backtest; the PDF tearsheet and CSV go into the report.

2 Bot (backtest vs forward)
Select your Lab 16 bot to compare its forward test with the backtest.

3 All metrics
CAGR 8.82% vs 14.14%, Sharpe 0.77 vs 0.78, max drawdown -25.54% vs -33.72%.

4 Equity vs SPY
Flat stretches = out of the market; it falls behind SPY from 2022.

5 Drawdown
The 2022 bear market caused the deepest loss (about -25%).

A good report explains why: which regimes earned the returns and what drove the drawdowns.

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TOPIC 4 · HANDS-ON LAB

Capstone: One Command Runs the Pipeline End to End, with an Agentic Layer and Human Approval

LAB 20

Run the whole pipeline end to end, get a human-approved proposal from the multi-agent desk, switch on the always-on agent, and create, backtest and forward-test your own strategy (from a template or written for you by AI).

WORKFLOW

1. Simulator: Run the multi-agent desk
2. Simulator: Human approval
3. Simulator: Always-on agent
4. Simulator: Create your own strategy
5. Simulator: Backtest, diagnose and forward-test it

YOU'LL PRODUCE

A Pipeline Map with every stage green, an approved agent proposal with its decision trace, and your custom strategy trading on paper.

Duration
25 min

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LAB 20 · FOLLOW ALONG

Lab 20 — Step by Step

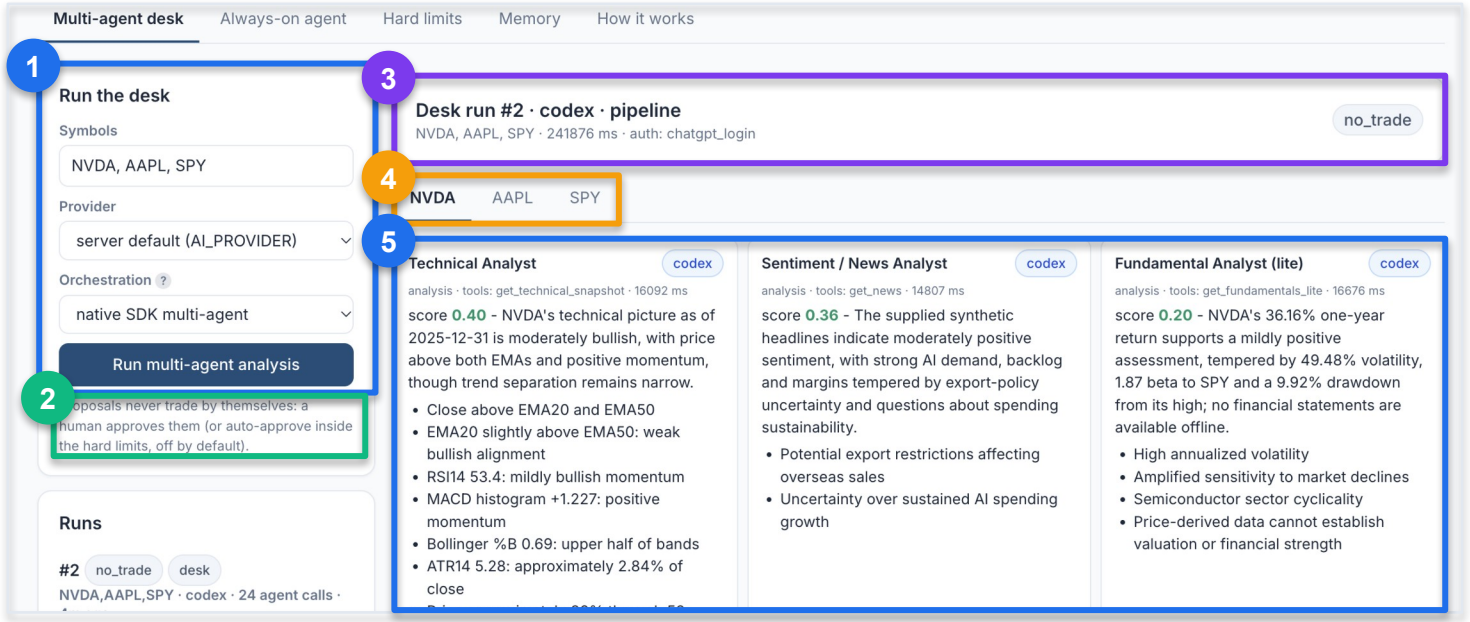
Course Labs > Lab 20 at http://localhost:8000 - each step's Go button opens its page pre-filled. Tick each 'You should see' before moving on.

	STEP · WHERE	DO THIS	YOU SHOULD SEE
1	**Run the multi-agent desk** › AI Agents > desk	AI Agents > Run multi-agent analysis on NVDA, AAPL, SPY.	✓ Eight agent entries per symbol plus the code gate; the brain badge shows which provider or rules answered.
2	**Human approval** › AI Agents	Approve (or reject) the proposals. Approved buys become paper bracket orders in the regular session or extended-hours limit orders overnight.	✓ Run status 'executed' and the orders visible in Paper Trading (source agent).
3	**Always-on agent** › same page	AI Agents > Always-on agent: enable it with a 60-minute wake-up (the worker must run), or press 'Wake it now'. Check Memory.	✓ A new always_on run and memory notes appear.
4	**Create your own strategy** › Strategy Builder (EMA + RSI template)	Strategy Builder: start from a template and change only its parameters - or describe your idea in the Lab 20 prompt of the Prompt Library and paste.	✓ Your strategy appears in the Strategy Library with an auto-generated parameter form.
5	**Backtest, diagnose and forward-test it** › Paper Trading > Bots	Backtest it, run Diagnostics, then create a paper bot for it and start it.	✓ Pipeline Map > Strategy lifecycle for your strategy shows the gates it passed.
6	**End-to-end on the Pipeline Map** › Pipeline Map	Open the Pipeline Map and press 'Run a bot cycle'. Then switch to 'Strategy lifecycle'.	✓ All 14 stages have run; you can walk a colleague through each stage.

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LAB 20 · SIMULATOR

Lab 20 — How to Read the Screen



1 — Run the desk
NVDA, AAPL, SPY; pick the AI provider and orchestration, then Run.

2 — Human approval
Proposals never trade by themselves - you approve or reject each one.

3 — Run result
Desk run #2 by codex ended no_trade: the desk chose not to trade.

4 — One tab per symbol
NVDA, AAPL and SPY each get their own analyst cards.

5 — Analyst cards
Technical, sentiment and fundamental analysts each give a score and their reasons.

Then create your own strategy, backtest, diagnose and forward-test it, and walk the Pipeline Map end to end.

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Topic 4 Recap

1. Perform coding and configuration of software agents or programs based on selected models or algorithms
2. Conduct tests on the actions taken and outcomes to assess effectiveness of the models
3. Diagnose unintended outcomes produced by statistical models
4. Draw relevant trends and insights from data analysis to support decisions and report accordingly
5. Coding languages for programming of algorithms and signals
6. Potential reasons for unintended outcomes

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SUMMARY

What You Achieved

1. **Framed testable hypotheses**
   Instrument, signal, holding period, effect, benchmark.

2. **Built indicators and strategies**
   Pine Script on TradingView; configured in the Trading Simulator.

3. **Modelled and A/B-tested**
   ML classifiers, regimes, Welch's t-test and bootstrap.

4. **Backtested net of costs**
   Commission, spread, slippage and regulatory fees.

5. **Forward-tested automatically**
   A paper-trading bot with risk checks and a trade log.

✓ **Diagnosed and reported**
Found the bias, fixed it, and wrote the recommendation.

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COURSE FEEDBACK

Before You Leave

1

Your materials

Slides, the Learner Guide and the lab manuals stay on the LMS and GitHub.

2

Course feedback

Complete the feedback form on the LMS.

3

Keep practising on paper

Your Alpaca paper account, the Trading Simulator and the lab manuals are yours to keep.

!

Not investment advice

Scenarios are fictitious; trade real money only with proper advice.

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Thank You

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